

A company is planning to migrate a commercial off-the-shelf application from its on-premises data center to AWS. The software has a software licensing model using sockets and cores with predictable capacity and uptime requirements. The company wants to use its existing licenses, which were purchased earlier this year.

Which Amazon EC2 pricing option is the MOST cost-effective?

- A. Dedicated Reserved Hosts **Most Voted**
- B. Dedicated On-Demand Hosts
- C. Dedicated Reserved Instances
- D. Dedicated On-Demand Instances

Correct Answer: A

Community vote distribution

A (92%)

8%

Comments

fk1e4 **Highly Voted** 3 years, 3 months ago

Selected Answer: A

"predictable capacity and uptime requirements" means "Reserved"

"sockets and cores" means "dedicated host"

upvoted 21 times

imvb88 **Highly Voted** 3 years, 1 month ago

Selected Answer: A

Bring custom purchased licenses to AWS -> Dedicated Host -> C,D out

Need cost effective solution -> "reserved" -> A

upvoted 8 times

imvb88 3 years, 1 month ago

<https://aws.amazon.com/ec2/dedicated-hosts/>

Amazon EC2 Dedicated Hosts allow you to use your eligible software licenses from vendors such as Microsoft and Oracle on Amazon EC2, so that you get the flexibility and cost effectiveness of using your own licenses, but with the resiliency, simplicity and elasticity of AWS.

upvoted 3 times

Hrishi_707 **Most Recent** 2 years, 3 months ago

BYOL >>> Dedicated Hosts

upvoted 5 times

Uzbekistan 2 years, 3 months ago

Selected Answer: A

A. Dedicated Reserved Hosts

Here's why:

License Flexibility: Dedicated Reserved Hosts allow the company to bring their existing licenses to AWS. This option enables them to continue using their purchased licenses without any additional cost or licensing changes.

Cost Optimization: Reserved Hosts offer significant cost savings compared to On-Demand pricing. By purchasing Reserved Hosts, the company can benefit from discounted hourly rates for the entire term of the reservation, which typically spans one or three years.

upvoted 3 times

jjcode 2 years, 3 months ago

I work with COTS applications they require a three tier architecture, its completely irrelevant and confusing to add that to the question, the key word here is licenses, since AWS wants you to use their solutions the answer to this is which of one the options solves this particular problem, in this case its dedicated hosts.

upvoted 1 times

BillaRanga 2 years, 4 months ago

Selected Answer: A

What is difference between dedicated host and reserved instance?

Dedicated Instance: The physical machine or underlying hardware is reserved for use for the whole account. You can have instances for different purposes on this hardware. Dedicated Host: The physical machine or the underlying hardware is reserved for "Single Use" only, eg. a certain application.

upvoted 2 times

BillaRanga 2 years, 4 months ago

What is the difference between a dedicated instance and a dedicated host tenancy?

Dedicated Instance (dedicated) — Your instance runs on single-tenant hardware. Dedicated Host (host) — Your instance runs on a physical server with EC2 instance capacity fully dedicated to your use, an isolated server with configurations that you can control.

upvoted 2 times

pentium75 2 years, 5 months ago

Selected Answer: A

Actually the question is a bit ambiguous because there ARE "software licensing model using sockets and cores" that accept virtual sockets are cores as the base, for which C would work. But most of these license models are based on PHYSICAL sockets, thus A.

upvoted 4 times

TariqKipkemei 2 years, 7 months ago

Selected Answer: A

Dedicated Hosts give you visibility and control over how instances are placed on a physical server and also enable you to use your existing server-bound software licenses like Windows Server

upvoted 3 times

wsdasdasdqwda 2 years, 7 months ago

Easy with one, but only 79% up to now answered correctly. It is A. Reserved because of the predictable and sockets and cores means dedicated host.

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: C

The correct answer is C. Dedicated Reserved Instances.

Dedicated Reserved Instances (DRIs) are the most cost-effective option for workloads that have predictable capacity and uptime requirements. DRIs offer a significant discount over On-Demand Instances, and they can be used to lock in a price for a period of time.

In this case, the company has predictable capacity and uptime requirements because the software has a software licensing model using sockets and cores. The company also wants to use its existing licenses, which were purchased earlier this year. Therefore, DRIs are the most cost-effective option.

upvoted 3 times

riccardoto 2 years, 10 months ago

Selected Answer: C

I don't agree with people voting "A". The question reference that the COTS Application has a licensing model based on "sockets and cores". The question does not specify if it means TCP sockets (= open connections) or hardware sockets, so I assume that "TCP sockets are intended". If this is the case, sockets and cores can also remain stable with reserved instances - which are cheaper than reserved hosts.

I would go with "A" only if the question would clearly state that the COTS application has some strong dependency on physical hardware.

upvoted 1 times

riccardoto 2 years, 10 months ago

note: instead, if by socket we mean "CPU sockets", then A would be the right one.

upvoted 2 times

pentium75 2 years, 5 months ago

Even if "sockets" mean TCP sockets there are still the cores, thus A

upvoted 3 times

JA2018 1 year, 6 months ago

Hi Riccardoto, perhaps you might to enlighten us which software licensing model will bill customer on a "per-TCP socket" basis? :-)

Cheers.

upvoted 1 times

aragon_saa 3 years, 3 months ago

A

<https://www.examttopics.com/discussions/amazon/view/35818-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 2 times

fruto123 3 years, 3 months ago

Selected Answer: A

Dedicated Host Reservations provide a billing discount compared to running On-Demand Dedicated Hosts. Reservations are available in three payment options.

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/dedicated-hosts-overview.html>

upvoted 4 times

Kenp1192 3 years, 3 months ago

A

is the most cost effective

upvoted 2 times

A company runs an application on Amazon EC2 Linux instances across multiple Availability Zones. The application needs a storage layer that is highly available and Portable Operating System Interface (POSIX)-compliant. The storage layer must provide maximum data durability and must be shareable across the EC2 instances. The data in the storage layer will be accessed frequently for the first 30 days and will be accessed infrequently after that time.

Which solution will meet these requirements MOST cost-effectively?

- A. Use the Amazon S3 Standard storage class. Create an S3 Lifecycle policy to move infrequently accessed data to S3 Glacier.
- B. Use the Amazon S3 Standard storage class. Create an S3 Lifecycle policy to move infrequently accessed data to S3 Standard-Infrequent Access (S3 Standard-IA).
- C. Use the Amazon Elastic File System (Amazon EFS) Standard storage class. Create a lifecycle management policy to move infrequently accessed data to EFS Standard-Infrequent Access (EFS Standard-IA). **Most Voted**
- D. Use the Amazon Elastic File System (Amazon EFS) One Zone storage class. Create a lifecycle management policy to move infrequently accessed data to EFS One Zone-Infrequent Access (EFS One Zone-IA).

Correct Answer: C

Community vote distribution

C (96%)

4%

Comments

TariqKipkemei **Highly Voted** 3 years ago

Selected Answer: C

Multi AZ = both EFS and S3 support
Storage classes = both EFS and S3 support
POSIX file system access = only Amazon EFS supports
upvoted 17 times

LazyTs **Highly Voted** 2 years, 9 months ago

Selected Answer: C

POSIX => EFS
<https://docs.aws.amazon.com/efs/latest/ug/whatisefs.html>
upvoted 8 times

Danilus **Most Recent** 1 year, 7 months ago

Selected Answer: C

key-Portable Operating System Interface (POSIX)-compliant.
key -shareable across the ec2 instances
key-accessed frequently for the first 30 days and will be accessed infrequently after
key-storage layer that is highly available

so the only service that work with POSIX is EFS in this case
s3 can not use posix so A and B are wrong answers
is not D because is not highly available because is One Zone so the answer is c
upvoted 2 times

zinabu 2 years, 2 months ago

Answer:B
cause there is no life cycle policy for EFS that will work in S3 only.
upvoted 2 times

jjcode 2 years, 3 months ago

"storage layer will be accessed frequently for the first 30 days and will be accessed infrequently after that time" Was the only reason they added this to trick you?

upvoted 3 times

pentium75 2 years, 5 months ago

Selected Answer: C

POSIX -> EFS, "maximum data durability" rules out One Zone

upvoted 5 times

maudsha 2 years, 7 months ago

Selected Answer: C

Both standard and one zone have same durability.

<https://docs.aws.amazon.com/efs/latest/ug/storage-classes.html>

Also EFS one zone can work with multiple EC2s in different AZs. But there will be a cost involved when you are accessing the EFS from a different AZ EC2. (EC2 data access charges)

<https://docs.aws.amazon.com/efs/latest/ug/how-it-works.html>

So if "all" EC2 instances accessing the files frequently there will be a storage cost + EC2 data access charges if you choose one zone.

So i would choose C.

upvoted 2 times

beast2091 2 years, 7 months ago

Ans: C

upvoted 2 times

baba365 2 years, 8 months ago

Ans: D, one-zone IA for 'most cost effective' .

<https://aws.amazon.com/efs/features/infrequent-access/>

upvoted 1 times

Gajendr 2 years, 5 months ago

What about " The application needs a storage layer that is highly available" and "application on Amazon EC2 Linux instances across multiple Availability Zones " ?

upvoted 2 times

AAAWrekng 2 years, 7 months ago

How does D fulfill the data durability requirement? Requirements must be met first, then consider 'most cost effective' - if you go to a tire shop, and say you want 4 new tires as cheap as possible. And they take off 4 tires and put on 2... Then they say you wanted it as cheap as possible...

upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: C

Use the Amazon Elastic File System (Amazon EFS) Standard storage class. Create a lifecycle management policy to move infrequently accessed data to EFS Standard-Infrequent Access (EFS Standard-IA).

upvoted 3 times

Yadav_Sanjay 2 years, 11 months ago

Selected Answer: D

D - It should be cost-effective

upvoted 2 times

pentium75 2 years, 5 months ago

But D does meet the durability requirement.

upvoted 1 times

Abrar2022 2 years, 12 months ago

Selected Answer: C

POSIX file system access = only Amazon EFS supports

upvoted 4 times

imvb88 3 years, 1 month ago

Selected Answer: C

POSIX + sharable across EC2 instances --> EFS --> A, B out

Instances run across multiple AZ -> C is needed.

upvoted 2 times

WherecanIstart 3 years, 2 months ago

Selected Answer: C

Linux based system points to EFS plus POSIX-compliant is also EFS related.

upvoted 3 times

fk1e4 3 years, 3 months ago

Selected Answer: C

"POSIX-compliant" means EFS.

also, file system can be shared with multiple EC2 instances means "EFS"

upvoted 5 times

KAUS2 3 years, 3 months ago

Selected Answer: C

Option C is the correct answer .

upvoted 2 times

Ruhi02 3 years, 3 months ago

Answer c : <https://aws.amazon.com/efs/features/infrequent-access/>

upvoted 1 times

A solutions architect is creating a new VPC design. There are two public subnets for the load balancer, two private subnets for web servers, and two private subnets for MySQL. The web servers use only HTTPS. The solutions architect has already created a security group for the load balancer allowing port 443 from 0.0.0.0/0. Company policy requires that each resource has the least access required to still be able to perform its tasks.

Which additional configuration strategy should the solutions architect use to meet these requirements?

- A. Create a security group for the web servers and allow port 443 from 0.0.0.0/0. Create a security group for the MySQL servers and allow port 3306 from the web servers security group.
- B. Create a network ACL for the web servers and allow port 443 from 0.0.0.0/0. Create a network ACL for the MySQL servers and allow port 3306 from the web servers security group.
- C. Create a security group for the web servers and allow port 443 from the load balancer. Create a security group for the MySQL servers and allow port 3306 from the web servers security group. **Most Voted**
- D. Create a network ACL for the web servers and allow port 443 from the load balancer. Create a network ACL for the MySQL servers and allow port 3306 from the web servers security group.

Correct Answer: C

Community vote distribution

C (100%)

Comments

WherecanIstart **Highly Voted** 2 years, 8 months ago

Selected Answer: C

Load balancer is public facing accepting all traffic coming towards the VPC (0.0.0.0/0). The web server needs to trust traffic originating from the ALB. The DB will only trust traffic originating from the Web server on port 3306 for MySQL

upvoted 6 times

TheFivePips **Most Recent** 1 year, 9 months ago

Selected Answer: C

Option C aligns with the least access principle and provides a clear and granular control over the communication between different components in the architecture.

Option D suggests using network ACLs, but security groups are more suitable for controlling access to individual instances based on their security group membership, which is why Option C is the more appropriate choice in this context

upvoted 3 times

TariqKipkemei 2 years, 1 month ago

Selected Answer: C

Create a security group for the web servers and allow port 443 from the load balancer. Create a security group for the MySQL servers and allow port 3306 from the web servers security group.

upvoted 3 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: C

C) Create a security group for the web servers and allow port 443 from the load balancer. Create a security group for the MySQL servers and allow port 3306 from the web servers security group.

This option follows the principle of least privilege by only allowing necessary access:

Web server SG allows port 443 from load balancer SG (not open to world)

MySQL SG allows port 3306 only from web server SG

upvoted 4 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: C

Create a security group for the web servers and allow port 443 from the load balancer. Create a security group for the MySQL servers and allow port 3306 from the web servers security group

upvoted 2 times

elearningtakai 2 years, 8 months ago

Selected Answer: C

Option C is the correct choice.

upvoted 2 times

fkie4 2 years, 9 months ago

Selected Answer: C

Just C. plain and simple

upvoted 2 times

aragon_saa 2 years, 9 months ago

C

<https://www.examttopics.com/discussions/amazon/view/43796-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 3 times

An ecommerce company is running a multi-tier application on AWS. The front-end and backend tiers both run on Amazon EC2, and the database runs on Amazon RDS for MySQL. The backend tier communicates with the RDS instance. There are frequent calls to return identical datasets from the database that are causing performance slowdowns.

Which action should be taken to improve the performance of the backend?

- A. Implement Amazon SNS to store the database calls.
- B. Implement Amazon ElastiCache to cache the large datasets. **Most Voted**
- C. Implement an RDS for MySQL read replica to cache database calls.
- D. Implement Amazon Kinesis Data Firehose to stream the calls to the database.

Correct Answer: B

Community vote distribution

B (100%)

Comments

elearningtakai **Highly Voted** 2 years, 8 months ago

Selected Answer: B

the best solution is to implement Amazon ElastiCache to cache the large datasets, which will store the frequently accessed data in memory, allowing for faster retrieval times. This can help to alleviate the frequent calls to the database, reduce latency, and improve the overall performance of the backend tier.

upvoted 15 times

fruto123 **Highly Voted** 2 years, 9 months ago

Selected Answer: B

Key term is identical datasets from the database it means caching can solve this issue by cached in frequently used dataset from DB

upvoted 7 times

Manimgh **Most Recent** 1 year, 1 month ago

Selected Answer: B

return identical datasets = Cache

upvoted 2 times

Ucy 1 year, 8 months ago

Selected Answer: B

Answer is B This will help reduce the frequency of calls to the database and improve overall performance by serving frequently accessed data from the cache instead of fetching it from the database every time. It's not option C as it suggests implementing an RDS for MySQL read replica to cache database calls. While read replicas can offload read operations from the primary database instance and improve read scalability, they are primarily used for read scaling and high availability rather than caching. Read replicas are intended to handle read-heavy workloads by distributing read requests across multiple instances. However, they do not inherently cache data like ElastiCache does.

upvoted 4 times

Ucy 1 year, 8 months ago

Answer is B

This will help reduce the frequency of calls to the database and improve overall performance by serving frequently accessed data from the cache instead of fetching it from the database every time.

It's not option C as it suggests implementing an RDS for MySQL read replica to cache database calls. While read replicas can offload read operations from the primary database instance and improve read scalability, they are primarily used for read scaling and high availability rather than caching.

Read replicas are intended to handle read-heavy workloads by distributing read requests across multiple instances. However, they do not inherently cache data like ElastiCache does.

upvoted 2 times

Bhanu1992 1 year, 8 months ago

Keyword is identical datasets

upvoted 2 times

thewalker 1 year, 10 months ago

Selected Answer: B

As per Amazon Q:

ElastiCache can be used to cache datasets from queries to RDS databases. Some key points:

While creating an ElastiCache cluster from the RDS console provides convenience, the application is still responsible for leveraging the cache.

Caching query results in ElastiCache can significantly improve performance by allowing high-volume read operations to be served from cache versus hitting the database.

This is especially useful for applications with high read throughput needs, as scaling the database can become more expensive compared to scaling the cache as needs increase. ElastiCache nodes can support up to 400,000 queries per second.

Cost savings are directly proportional to read throughput - higher throughput applications see greater savings.

upvoted 2 times

Murtadhaceit 2 years ago

Selected Answer: B

The best scenario to implement caching, identical calls to the same data sets.

upvoted 3 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: B

B) Implement Amazon ElastiCache to cache the large datasets.

The key issue is repeated calls to return identical datasets from the RDS database causing performance slowdowns.

Implementing Amazon ElastiCache for Redis or Memcached would allow these repeated query results to be cached, improving backend performance by reducing load on the database.

upvoted 4 times

Guru4Cloud 2 years, 3 months ago

B) Implement Amazon ElastiCache to cache the large datasets.

The key issue is repeated calls to return identical datasets from the RDS database causing performance slowdowns.

Implementing Amazon ElastiCache for Redis or Memcached would allow these repeated query results to be cached, improving backend performance by reducing load on the database.

upvoted 2 times

Abrar2022 2 years, 6 months ago

Selected Answer: B

Thanks Tariq for the simplified answer below:

frequent identical calls = ElastiCache

upvoted 4 times

TariqKipkemei 2 years, 6 months ago

frequent identical calls = ElastiCache

upvoted 2 times

Mikebonsi70 2 years, 8 months ago

Tricky question, anyway.

upvoted 2 times

Mikebonsi70 2 years, 8 months ago

Yes, caching is the solution but is ElastiCache compatible with RDS MySQL DB? So, what about the answer C with a DB read replica? For me it's C.

upvoted 1 times

aragon_saa 2 years, 9 months ago

B

<https://www.examtopycs.com/discussions/amazon/view/27874-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 2 times

A new employee has joined a company as a deployment engineer. The deployment engineer will be using AWS CloudFormation templates to create multiple AWS resources. A solutions architect wants the deployment engineer to perform job activities while following the principle of least privilege.

Which combination of actions should the solutions architect take to accomplish this goal? (Choose two.)

- A. Have the deployment engineer use AWS account root user credentials for performing AWS CloudFormation stack operations.
- B. Create a new IAM user for the deployment engineer and add the IAM user to a group that has the PowerUsers IAM policy attached.
- C. Create a new IAM user for the deployment engineer and add the IAM user to a group that has the AdministratorAccess IAM policy attached.
- D. Create a new IAM user for the deployment engineer and add the IAM user to a group that has an IAM policy that allows AWS CloudFormation actions only. **Most Voted**
- E. Create an IAM role for the deployment engineer to explicitly define the permissions specific to the AWS CloudFormation stack and launch stacks using that IAM role. **Most Voted**

Correct Answer: DE

Community vote distribution

DE (100%)

Comments

aragon_saa **Highly Voted** 2 years, 9 months ago

D, E

<https://www.examttopics.com/discussions/amazon/view/46428-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 5 times

mwwt2022 1 year, 11 months ago

thank you my friend

upvoted 2 times

truongtx8 **Highly Voted** 1 year, 10 months ago

Selected Answer: DE

The answers inside the question: CloudFormation.

A is excluded since root account is never a choice for the principle of least privilege.

D, E left are the correct ones.

upvoted 5 times

awsgeek75 **Most Recent** 1 year, 10 months ago

Selected Answer: DE

ABC are just giving too much access so CD are logical choices

upvoted 4 times

TariqKipkemei 2 years, 1 month ago

Selected Answer: DE

Create a new IAM user for the deployment engineer and add the IAM user to a group that has an IAM policy that allows AWS CloudFormation actions only.

Create an IAM role for the deployment engineer to explicitly define the permissions specific to the AWS CloudFormation stack and launch stacks using that IAM role.

upvoted 3 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: DE

The two actions that should be taken to follow the principle of least privilege are:

D) Create a new IAM user for the deployment engineer and add the IAM user to a group that has an IAM policy that allows AWS CloudFormation actions only.

E) Create an IAM role for the deployment engineer to explicitly define the permissions specific to the AWS CloudFormation stack and launch stacks using that IAM role.

The principle of least privilege states that users should only be given the minimal permissions necessary to perform their job function.

upvoted 4 times

alexandercamachop 2 years, 6 months ago

Selected Answer: DE

Option D, creating a new IAM user and adding them to a group with an IAM policy that allows AWS CloudFormation actions only, ensures that the deployment engineer has the necessary permissions to perform AWS CloudFormation operations while limiting access to other resources and actions. This aligns with the principle of least privilege by providing the minimum required permissions for their job activities.

Option E, creating an IAM role with specific permissions for AWS CloudFormation stack operations and allowing the deployment engineer to assume that role, is another valid approach. By using an IAM role, the deployment engineer can assume the role when necessary, granting them temporary permissions to perform CloudFormation actions. This provides a level of separation and limits the permissions granted to the engineer to only the required CloudFormation operations.

upvoted 3 times

Babaaaaa 2 years, 6 months ago

Selected Answer: DE

Dddd,Eeee

upvoted 2 times

elearningtakai 2 years, 8 months ago

Selected Answer: DE

D & E are a good choices

upvoted 2 times

fruto123 2 years, 9 months ago

Selected Answer: DE

I agree DE

upvoted 3 times

A company is deploying a two-tier web application in a VPC. The web tier is using an Amazon EC2 Auto Scaling group with public subnets that span multiple Availability Zones. The database tier consists of an Amazon RDS for MySQL DB instance in separate private subnets. The web tier requires access to the database to retrieve product information.

The web application is not working as intended. The web application reports that it cannot connect to the database. The database is confirmed to be up and running. All configurations for the network ACLs, security groups, and route tables are still in their default states.

What should a solutions architect recommend to fix the application?

- A. Add an explicit rule to the private subnet's network ACL to allow traffic from the web tier's EC2 instances.
- B. Add a route in the VPC route table to allow traffic between the web tier's EC2 instances and the database tier.
- C. Deploy the web tier's EC2 instances and the database tier's RDS instance into two separate VPCs, and configure VPC peering.
- D. Add an inbound rule to the security group of the database tier's RDS instance to allow traffic from the web tiers security group. **Most Voted**

Correct Answer: D

Community vote distribution

D (97%)

3%

Comments

TariqKipkemei **Highly Voted** 2 years, 6 months ago

Selected Answer: D

Security group defaults block all inbound traffic..Add an inbound rule to the security group of the database tier's RDS instance to allow traffic from the web tiers security group

upvoted 15 times

ExamGuru727 **Highly Voted** 1 year, 8 months ago

Selected Answer: D

For those questioning why the answer is not A:

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html>

Default NACLs allow all traffic, and in this question NACLs, SGs and route tables are in their default states.

upvoted 6 times

hgjdsh **Most Recent** 1 year, 8 months ago

Selected Answer: A

I think the answer should be A. Sine the services are in different subnets, the NACL would by default block all the incoming traffic to the subnet. Security group rule wouldn't be able to override NACL rule.

upvoted 1 times

njufi 1 year, 8 months ago

I selected option D as well, but I have a question regarding option A. Considering that the EC2 instances and the RDS are located in different subnets shouldn't the network ACLs for each subnet allow traffic from one another as well? Given that the default settings for network ACLs typically block all traffic, wouldn't it be necessary to explicitly permit communication between the subnets?

upvoted 2 times

smartegnine 2 years, 5 months ago

Selected Answer: D

Security Groups are tied on instance where as network ACL are tied to Subnet.

upvoted 5 times

elearningtakai 2 years, 8 months ago

Selected Answer: D

By default, all inbound traffic to an RDS instance is blocked. Therefore, an inbound rule needs to be added to the security group of the RDS instance to allow traffic from the security group of the web tier's EC2 instances.

upvoted 4 times

Russs99 2 years, 8 months ago

Selected Answer: D

D is the correct answer

upvoted 2 times

aragon_saa 2 years, 9 months ago

D

<https://www.examttopics.com/discussions/amazon/view/81445-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 2 times

KAUS2 2 years, 9 months ago

Selected Answer: D

D is correct option

upvoted 2 times

A company has a large dataset for its online advertising business stored in an Amazon RDS for MySQL DB instance in a single Availability Zone. The company wants business reporting queries to run without impacting the write operations to the production DB instance.

Which solution meets these requirements?

- A. Deploy RDS read replicas to process the business reporting queries. **Most Voted**
- B. Scale out the DB instance horizontally by placing it behind an Elastic Load Balancer.
- C. Scale up the DB instance to a larger instance type to handle write operations and queries.
- D. Deploy the DB instance in multiple Availability Zones to process the business reporting queries.

Correct Answer: A

Community vote distribution

A (100%)

Comments

mwwt2022 **Highly Voted** 1 year, 11 months ago

Selected Answer: A

reporting queries to run without impacting the write operations -> read replicas
upvoted 6 times

Guru4Cloud **Most Recent** 2 years, 3 months ago

Selected Answer: A

A) Deploy RDS read replicas to process the business reporting queries.

The key points are:

RDS read replicas allow read-only copies of the production DB instance to be created
Queries to the read replica don't affect the source DB instance performance
This isolates reporting queries from production traffic and write operations
So using RDS read replicas is the best way to meet the requirements of running reporting queries without impacting production write operations.
upvoted 4 times

james2033 2 years, 4 months ago

Selected Answer: A

"single AZ", "large dataset", "Amazon RDS for MySQL database". Want "business report queries". --> Solution "Read replicas", choose A.
upvoted 2 times

antropaws 2 years, 6 months ago

Selected Answer: A

No doubt A.
upvoted 3 times

TariqKipkemei 2 years, 6 months ago

Load balance read operations = read replicas
upvoted 2 times

TariqKipkemei 2 years, 1 month ago

reports=read replica
upvoted 2 times

KAUS2 2 years, 9 months ago

Selected Answer: A

Option "A" is the right answer . Read replica use cases - You have a production database that is taking on normal load & You want to run a reporting application to run some analytics

- You create a Read Replica to run the new workload there
- The production application is unaffected
- Read replicas are used for SELECT (=read) only kind of statements (not INSERT, UPDATE, DELETE)

upvoted 3 times

cegama543 2 years, 9 months ago

Selected Answer: A

option A is the best solution for ensuring that business reporting queries can run without impacting write operations to the production DB instance.

upvoted 4 times

A company hosts a three-tier ecommerce application on a fleet of Amazon EC2 instances. The instances run in an Auto Scaling group behind an Application Load Balancer (ALB). All ecommerce data is stored in an Amazon RDS for MariaDB Multi-AZ DB instance.

The company wants to optimize customer session management during transactions. The application must store session data durably.

Which solutions will meet these requirements? (Choose two.)

- A. Turn on the sticky sessions feature (session affinity) on the ALB. **Most Voted**
- B. Use an Amazon DynamoDB table to store customer session information.
- C. Deploy an Amazon Cognito user pool to manage user session information.
- D. Deploy an Amazon ElastiCache for Redis cluster to store customer session information. **Most Voted**
- E. Use AWS Systems Manager Application Manager in the application to manage user session information.

Correct Answer: AD

Community vote distribution



Comments

fruto123 **Highly Voted** 3 years, 3 months ago

Selected Answer: AD

It is A and D. Proof is in link below.

<https://aws.amazon.com/caching/session-management/>

upvoted 28 times

pentium75 2 years, 5 months ago

This doesn't say anything about durability

upvoted 4 times

babayomi 1 year, 7 months ago

First consider the durability requirement of a session. It's never meant to be kept after the session terminates, tha5 said. You wouldn't need a permanent database for a session. You would only require a cache, so that it can expire after a while when the session has already completed. That is why D is correct. A is required in the first place in order to initialize the use of the cache.

upvoted 2 times

babayomi 1 year, 7 months ago

Thanks,

First consider the durability requirement of a session. It's never meant to be kept after the session terminates, tha5 said. You wouldn't need a permanent database for a session. You would only require a cache, so that it can expire after a while when the session has already completed. That is why D is correct. A is required in the first place in order to initialize the use of the cache.

upvoted 2 times

Marco_St **Highly Voted** 2 years, 5 months ago

Selected Answer: BD

I did not get why A is most voted? The question did not mention anything about fixed routing target so the ALB should route traffic randomly to each server. Then we just need to provide cache session management to avoid session lost issue instead of using sticky session.

upvoted 15 times

babayomi 1 year, 7 months ago

AD

First consider the durability requirement of a session. It's never meant to be kept after the session terminates, tha5 said. You wouldn't need a pamanent database for a session. You would only require a cache, so that it can expire after a while when the session has already completed. That is why D is correct. A is required in the first place in order to initialize the use of the cache.

upvoted 1 times

mahesuma Most Recent 1 week, 5 days ago

Selected Answer: BD

Both are specialised managing session data.

upvoted 1 times

SnaxAttax 6 months, 2 weeks ago

Selected Answer: BD

BD. Breaking down why:

A: ALB Sticky Sessions: ties a particular session to ONE instance. Not durable if that instance fails.

B: DynamoDB DOES durably store session data.

C: Cognito is about authentication and user pools.

D: Elasticache is used for caching OR as session stores/session management in distributed use cases.

E: AWS Systems Manager app manager is more about visualizing resources: management and monitoring, not data store.

B & D are best options given the above.

upvoted 1 times

JustEugen 8 months, 4 weeks ago

Selected Answer: BD

The question specifies that session storage should be durable, which is why A is not the case, because EC2 storage cannot be considered durable.

In addition to that, it is common practice to store user sessions in databases. In addition to that, using DynamoDB, you can enable TTL for your data, which comes with no deletion cost.

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/TTL.html>

upvoted 1 times

In2718 9 months, 3 weeks ago

Selected Answer: BD

Chat AI says B and D

B. DynamoDB table → DynamoDB is fully managed, highly available, durable, and scales seamlessly. Perfect for storing session state durably.

D. ElastiCache for Redis → Redis is commonly used as a session store. If deployed with replication + persistence (AOF or snapshotting), it can provide durability as well as fast access.

upvoted 1 times

chest_jd 1 year, 2 months ago

Selected Answer: BD

First of all - application MUST STORE session DATA. We should get some storage on backend side. Just Band D can do it.

upvoted 1 times

bigjft 1 year, 3 months ago

Selected Answer: BD

Avoid sticky sessions:

While using sticky sessions on an Application Load Balancer (ALB) might seem like a solution, it can hinder scalability as it ties a user to a specific instance, which could become a bottleneck if traffic spikes.

Dynamodb and Elastic cache are best i think

upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: BD

A is not right, the question is to where to save user session, the sticky session is used for ALB to send the request of the same client to the same target.

upvoted 1 times

FlyingHawk 1 year, 6 months ago

Selected Answer: BD

People select AD because they think it is combine A and D to make a final solution. if the expectation is that each selected option should independently meet the requirements, then B and D. I think "Which solutions will meet these requirements?" should mean two independent solutions.

upvoted 2 times

Arad 1 year, 6 months ago

Selected Answer: BD

session management during transaction ---> DynamoDB and ElastiCache

upvoted 1 times

Sergantus 1 year, 7 months ago

Selected Answer: BD

We are not looking here for a combination of options, just two independent solutions.

Because of the "durability" requirement -

B - DynamoDB is durable

D - ElastiCache for Redis although mostly used as in-memory cache, ElastiCache for Redis provides options for data persistence - RDB (Redis Database Backup) and AOF (Append-Only File). RDB periodically saves a snapshot of the dataset to disk, while AOF logs every write operation and replays it in case of a restart. ElastiCache for Redis also allows configuring replication groups with Multi-AZ and read replicas, providing additional redundancy and failover options.

upvoted 3 times

FlyingHawk 1 year, 6 months ago

I agree with your comments

upvoted 1 times

babayomi 1 year, 7 months ago

AD

First consider the durability requirement of a session. It's never meant to be kept after the session terminates, tha5 said. You wouldn't need a pamanent database for a session. You would only require a cache, so that it can expire after a while when the session has already completed. That is why D is correct. A is required in the first place in order to initialize the use of the cache.

upvoted 1 times

Uzbekistan 2 years, 3 months ago

Selected Answer: BD

Option A suggests using sticky sessions (session affinity) on the Application Load Balancer (ALB). While sticky sessions can help route requests from the same client to the same backend server, it doesn't directly address the requirement for durable storage of session data. Sticky sessions are typically used to maintain session state at the load balancer level, but they do not provide data durability in case of server failures or restarts. Option A - is not correct !!!

So answer is option B and D !!!

upvoted 3 times

babayomi 1 year, 7 months ago

First consider the durability requirement of a session. It's never meant to be kept after the session terminates, tha5 said. You wouldn't need a pamanent database for a session. You would only require a cache, so that it can expire after a while when the session has already completed. That is why D is correct. A is required in the first place in order to initialize the use of the cache.

upvoted 1 times

jjcode 2 years, 3 months ago

why does it matter to store user sessions durably? they EXPIRE, why would a company care about storing user sessions, thats not something thats done in the real world, those things are usually data dumped, or overwritten with new session tokens LOL, this whole question is &^%&*&^\$#@%^

upvoted 4 times

tuso 2 years, 4 months ago

I think the question is intended to mean "Combination of services", as some answers say "to store" or "to manage". So i am going for A+B, as sticky sessions are intended to manage the sessions and DynamoDB to store durably.

upvoted 1 times

pentium75 2 years, 5 months ago

Selected Answer: AB

Going for AB. Sticky Sessions to "optimize customer session management during transactions" and DynamoDB to "store session data durably".

D, ElastiCache does NOT allow "durable" storage. Just because there's an article that contains both words "ElastiCache" and "durable" does not prove the contrary.

C and E, Cognito and Systems Manager, have nothing to do with the issue.

upvoted 5 times

avdxeqtr 2 years, 5 months ago

<https://aws.amazon.com/blogs/developer/elasticache-as-an-asp-net-session-store/>

Amazon ElastiCache for Redis is highly suited as a session store to manage session information such as user authentication tokens, session state, and more. Simply use ElastiCache for Redis as a fast key-value store with appropriate TTL on session keys to manage your session information.

Session management is commonly required for online applications, including games, e-commerce websites, and social media platforms.

upvoted 2 times

avdxeqtr 2 years, 5 months ago

Correct link: <https://aws.amazon.com/elasticache/redis/>
upvoted 3 times

dkw2342 2 years, 3 months ago

I agree that ElastiCache for Redis is not a durable KV store.

But what about the phrasing?

"Which solutions will meet these requirements? (Choose two.)" Solutions (plural) implies two ways to *independently* fulfill the requirements. If you're supposed to select a combination of options, it's usually phrased like this: "Which combination of solutions ..."

upvoted 4 times

A company needs a backup strategy for its three-tier stateless web application. The web application runs on Amazon EC2 instances in an Auto Scaling group with a dynamic scaling policy that is configured to respond to scaling events. The database tier runs on Amazon RDS for PostgreSQL. The web application does not require temporary local storage on the EC2 instances. The company's recovery point objective (RPO) is 2 hours.

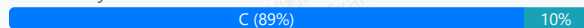
The backup strategy must maximize scalability and optimize resource utilization for this environment.

Which solution will meet these requirements?

- A. Take snapshots of Amazon Elastic Block Store (Amazon EBS) volumes of the EC2 instances and database every 2 hours to meet the RPO.
- B. Configure a snapshot lifecycle policy to take Amazon Elastic Block Store (Amazon EBS) snapshots. Enable automated backups in Amazon RDS to meet the RPO.
- C. Retain the latest Amazon Machine Images (AMIs) of the web and application tiers. Enable automated backups in Amazon RDS and use point-in-time recovery to meet the RPO. **Most Voted**
- D. Take snapshots of Amazon Elastic Block Store (Amazon EBS) volumes of the EC2 instances every 2 hours. Enable automated backups in Amazon RDS and use point-in-time recovery to meet the RPO.

Correct Answer: C

Community vote distribution



Comments

elearningtakai **Highly Voted** 3 years, 2 months ago

Selected Answer: C

that if there is no temporary local storage on the EC2 instances, then snapshots of EBS volumes are not necessary. Therefore, if your application does not require temporary storage on EC2 instances, using AMIs to back up the web and application tiers is sufficient to restore the system after a failure

Snapshots of EBS volumes would be necessary if you want to back up the entire EC2 instance, including any applications and temporary data stored on the EBS volumes attached to the instances. When you take a snapshot of an EBS volume, it backs up the entire contents of that volume. This ensures that you can restore the entire EC2 instance to a specific point in time more quickly. However, if there is no temporary data stored on the EBS volumes, then snapshots of EBS volumes are not necessary.

upvoted 37 times

MssP 3 years, 2 months ago

I think "temporal local storage" refers to "instance store", no instance store is required. EBS is durable storage, not temporal.

upvoted 4 times

MssP 3 years, 2 months ago

Look at the first paragraph. <https://repost.aws/knowledge-center/instance-store-vs-ebs>

upvoted 2 times

MatAlves 1 year, 9 months ago

Considering it's a "stateless web application", that would still be no reason to back up the EBS volumes.

upvoted 2 times

CloudForFun **Highly Voted** 3 years, 2 months ago

Selected Answer: C

The web application does not require temporary local storage on the EC2 instances => No EBS snapshot is required, retaining the latest AMI is enough.

upvoted 16 times

JustEugen Most Recent 8 months, 4 weeks ago

Selected Answer: C

In question, it says that the EC2 instances are stateless, which means that we do not care about the data on the attached EBS, so no reason to make EBS snapshots

upvoted 1 times

Mikado211 2 years, 6 months ago

Selected Answer: C

The web application does not require temporary local storage on the EC2 instances so we do not care about ECS. We only need two things here, the image of the instance (AMI) and a database backup.

C

upvoted 4 times

TariqKipkemei 2 years, 7 months ago

Selected Answer: C

"The web application does not require temporary local storage on the EC2 instances" rules out any option to back up the EC2 EBS volumes.

upvoted 3 times

darekw 2 years, 10 months ago

Question says: ...stateless web application.. that means application doesn't store any data, so no EBS required

upvoted 2 times

kruasan 3 years, 1 month ago

Selected Answer: C

Since the application has no local data on instances, AMIs alone can meet the RPO by restoring instances from the most recent AMI backup. When combined with automated RDS backups for the database, this provides a complete backup solution for this environment. The other options involving EBS snapshots would be unnecessary given the stateless nature of the instances. AMIs provide all the backup needed for the app tier.

This uses native, automated AWS backup features that require minimal ongoing management:

- AMI automated backups provide point-in-time recovery for the stateless app tier.
- RDS automated backups provide point-in-time recovery for the database.

upvoted 4 times

neosis91 3 years, 1 month ago

Selected Answer: B

BBBBBBBBBB

upvoted 1 times

pentium75 2 years, 5 months ago

Why back up EBS volumes of the autoscaled instances?

upvoted 2 times

Rob1L 3 years, 2 months ago

Selected Answer: D

I vote for D

upvoted 1 times

pentium75 2 years, 5 months ago

Why back up EBS volumes of the autoscaled instances?

upvoted 2 times

CapJackSparrow 3 years, 2 months ago

Selected Answer: C

makes more sense.

upvoted 3 times

nileshlg 3 years, 2 months ago

Selected Answer: C

Answer is C. Keyword to notice "Stateless"

upvoted 3 times

cra2yk 3 years, 2 months ago

Selected Answer: C

why B? I mean "stateless" and "does not require temporary local storage" have indicate that we don't need to take snapshot for ec2 volume.

upvoted 4 times

ktulu2602 3 years, 3 months ago

Selected Answer: B

Option B is the most appropriate solution for the given requirements.

With this solution, a snapshot lifecycle policy can be created to take Amazon Elastic Block Store (Amazon EBS) snapshots periodically, which will ensure that EC2 instances can be restored in the event of an outage. Additionally, automated backups can be enabled in Amazon RDS for PostgreSQL to take frequent backups of the database tier. This will help to minimize the RPO to 2 hours.

Taking snapshots of Amazon EBS volumes of the EC2 instances and database every 2 hours (Option A) may not be cost-effective and efficient, as this approach would require taking regular backups of all the instances and volumes, regardless of whether any changes have occurred or not. Retaining the latest Amazon Machine Images (AMIs) of the web and application tiers (Option C) would provide only an image backup and not a data backup, which is required for the database tier. Taking snapshots of Amazon EBS volumes of the EC2 instances every 2 hours and enabling automated backups in Amazon RDS and using point-in-time recovery (Option D) would result in higher costs and may not be necessary to meet the RPO requirement of 2 hours.

upvoted 4 times

pentium75 2 years, 5 months ago

Why back up EBS volumes of the autoscaled instances?

"Retaining the latest Amazon Machine Images (AMIs) of the web and application tiers (Option C) would provide only an image backup and not a data backup, which is required for the database tier." False because option C also includes "automated backups in Amazon RDS".

upvoted 2 times

cegama543 3 years, 3 months ago

Selected Answer: B

B. Configure a snapshot lifecycle policy to take Amazon Elastic Block Store (Amazon EBS) snapshots. Enable automated backups in Amazon RDS to meet the RPO.

The best solution is to configure a snapshot lifecycle policy to take Amazon Elastic Block Store (Amazon EBS) snapshots, and enable automated backups in Amazon RDS to meet the RPO. An RPO of 2 hours means that the company needs to ensure that the backup is taken every 2 hours to minimize data loss in case of a disaster. Using a snapshot lifecycle policy to take Amazon EBS snapshots will ensure that the web and application tier can be restored quickly and efficiently in case of a disaster. Additionally, enabling automated backups in Amazon RDS will ensure that the database tier can be restored quickly and efficiently in case of a disaster. This solution maximizes scalability and optimizes resource utilization because it uses automated backup solutions built into AWS.

upvoted 3 times

pentium75 2 years, 5 months ago

No need to back up the EBS volumes of autoscaled instances.

upvoted 3 times

A company wants to deploy a new public web application on AWS. The application includes a web server tier that uses Amazon EC2 instances. The application also includes a database tier that uses an Amazon RDS for MySQL DB instance.

The application must be secure and accessible for global customers that have dynamic IP addresses.

How should a solutions architect configure the security groups to meet these requirements?

- A. Configure the security group for the web servers to allow inbound traffic on port 443 from 0.0.0.0/0. Configure the security group for the DB instance to allow inbound traffic on port 3306 from the security group of the web servers. **Most Voted**
- B. Configure the security group for the web servers to allow inbound traffic on port 443 from the IP addresses of the customers. Configure the security group for the DB instance to allow inbound traffic on port 3306 from the security group of the web servers.
- C. Configure the security group for the web servers to allow inbound traffic on port 443 from the IP addresses of the customers. Configure the security group for the DB instance to allow inbound traffic on port 3306 from the IP addresses of the customers.
- D. Configure the security group for the web servers to allow inbound traffic on port 443 from 0.0.0.0/0. Configure the security group for the DB instance to allow inbound traffic on port 3306 from 0.0.0.0/0.

Correct Answer: A

Community vote distribution

A (87%)

13%

Comments

awsgeek75 **Highly Voted** 1 year, 10 months ago

Selected Answer: A

"The application must be secure and accessible for global customers that have dynamic IP addresses." This just means "anyone" so BC are wrong as you cannot know in advance about the dynamic IP addresses. D is just opening the DB to the internet.

A is most secure as web is open to internet and db is open to web only.

upvoted 7 times

Bhanu1992 **Highly Voted** 1 year, 8 months ago

The keyword is dynamic IPs from the customer, then B, C out, D out due to 0.0.0.0/0

upvoted 6 times

bora4motion **Most Recent** 1 year, 1 month ago

Selected Answer: A

a is 100% correct.

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: A

It allows HTTPS access from any public IP address, meeting the requirement for global customer access.

HTTPS provides encryption for secure communication.

And for the database security group, only allowing inbound port 3306 from the web server security group properly restricts access to only the resources that need it.

upvoted 4 times

jayce5 2 years, 6 months ago

Selected Answer: A

Should be A since the customer IPs are dynamically.

upvoted 2 times

antropaws 2 years, 6 months ago

Selected Answer: A

A no doubt.

upvoted 3 times

omoakin 2 years, 6 months ago

BBBBBBBBBBBBBBBBBBBBBBB

from customers IPs

upvoted 1 times

MostafaWardany 2 years, 6 months ago

Correct answer A, customer dynamic IPs ==> > 443 from 0.0.0.0/0

upvoted 3 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: A

dynamic source ips = allow all traffic - Configure the security group for the web servers to allow inbound traffic on port 443 from 0.0.0.0/0. Configure the security group for the DB instance to allow inbound traffic on port 3306 from the security group of the web servers.

upvoted 2 times

elarningtakai 2 years, 8 months ago

Selected Answer: A

If the customers have dynamic IP addresses, option A would be the most appropriate solution for allowing global access while maintaining security.

upvoted 4 times

Kenzo 2 years, 8 months ago

Correct answer is A.

B and C are out.

D is out because it is accepting traffic from every where instead of from web servers only

upvoted 4 times

Grace83 2 years, 8 months ago

A is correct

upvoted 3 times

WheracanIstart 2 years, 8 months ago

Selected Answer: B

Keyword dynamic ...A is the right answer. If the IP were static and specific, B would be the right answer

upvoted 4 times

pentium75 1 year, 11 months ago

Then why voted B?

upvoted 2 times

boxu03 2 years, 8 months ago

Selected Answer: A

aaaaaaa

upvoted 1 times

kprakashbehera 2 years, 9 months ago

Selected Answer: A

Ans - A

upvoted 1 times

A payment processing company records all voice communication with its customers and stores the audio files in an Amazon S3 bucket. The company needs to capture the text from the audio files. The company must remove from the text any personally identifiable information (PII) that belongs to customers.

What should a solutions architect do to meet these requirements?

- A. Process the audio files by using Amazon Kinesis Video Streams. Use an AWS Lambda function to scan for known PII patterns.
- B. When an audio file is uploaded to the S3 bucket, invoke an AWS Lambda function to start an Amazon Textract task to analyze the call recordings.
- C. Configure an Amazon Transcribe transcription job with PII redaction turned on. When an audio file is uploaded to the S3 bucket, invoke an AWS Lambda function to start the transcription job. Store the output in a separate S3 bucket. **Most Voted**
- D. Create an Amazon Connect contact flow that ingests the audio files with transcription turned on. Embed an AWS Lambda function to scan for known PII patterns. Use Amazon EventBridge to start the contact flow when an audio file is uploaded to the S3 bucket.

Correct Answer: C

Community vote distribution

C (100%)

Comments

TariqKipkemei **Highly Voted** 1 year, 7 months ago

Selected Answer: C

speech to text = Amazon Transcribe
upvoted 6 times

Guru4Cloud **Highly Voted** 1 year, 9 months ago

Selected Answer: C

Amazon Transcribe is a service provided by Amazon Web Services (AWS) that converts speech to text using automatic speech recognition (ASR) technology
upvoted 6 times

james2033 **Most Recent** 1 year, 10 months ago

Selected Answer: C

AWS Transcribe <https://aws.amazon.com/transcribe/> . Redacting or identifying (Personally identifiable instance) PII in real-time stream
<https://docs.aws.amazon.com/transcribe/latest/dg/pii-redaction-stream.html> .
upvoted 2 times

SimiTik 2 years, 1 month ago

C
Amazon Transcribe is a service provided by Amazon Web Services (AWS) that converts speech to text using automatic speech recognition (ASR) technology. gtp
upvoted 4 times

elearningtakai 2 years, 2 months ago

Selected Answer: C

Option C is the most suitable solution as it suggests using Amazon Transcribe with PII redaction turned on. When an audio file is uploaded to the S3 bucket, an AWS Lambda function can be used to start the transcription job. The output can be stored in a separate S3 bucket to ensure that the PII redaction is applied to the transcript. Amazon Transcribe can redact PII such as credit card numbers, social security numbers, and phone numbers.
upvoted 4 times

WherecanIstart 2 years, 2 months ago

Selected Answer: C

C for sure.....

upvoted 2 times

WherecanIstart 2 years, 2 months ago

C for sure

upvoted 2 times

boxu03 2 years, 2 months ago

Selected Answer: C

cccccccc

upvoted 2 times

Ruhi02 2 years, 3 months ago

answer c

upvoted 2 times

KAUS2 2 years, 3 months ago

Selected Answer: C

Option C is correct..

upvoted 2 times

A company is running a multi-tier ecommerce web application in the AWS Cloud. The application runs on Amazon EC2 instances with an Amazon RDS for MySQL Multi-AZ DB instance. Amazon RDS is configured with the latest generation DB instance with 2,000 GB of storage in a General Purpose SSD (gp3) Amazon Elastic Block Store (Amazon EBS) volume. The database performance affects the application during periods of high demand.

A database administrator analyzes the logs in Amazon CloudWatch Logs and discovers that the application performance always degrades when the number of read and write IOPS is higher than 20,000.

What should a solutions architect do to improve the application performance?

- A. Replace the volume with a magnetic volume.
- B. Increase the number of IOPS on the gp3 volume.
- C. Replace the volume with a Provisioned IOPS SSD (io2) volume. **Most Voted**
- D. Replace the 2,000 GB gp3 volume with two 1,000 GB gp3 volumes.

Correct Answer: C

Community vote distribution



Comments

Bezha **Highly Voted** 3 years, 2 months ago

Selected Answer: D

A - Magnetic Max IOPS 200 - Wrong
B - gp3 Max IOPS 16000 per volume - Wrong
C - RDS not supported io2 - Wrong
D - Correct; 2 gp3 volume with 16 000 each $2 \times 16000 = 32\ 000$ IOPS
upvoted 37 times

joechen2023 2 years, 11 months ago

<https://repost.aws/knowledge-center/ebs-volume-type-differences>
RDS does support io2
upvoted 3 times

wRhIH 2 years, 11 months ago

that Link is to EBS instead of RDS
upvoted 6 times

allanm 10 months ago

Still doesn't change the fact that RDS now supports io2 https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html
upvoted 1 times

FlyingHawk 1 year, 6 months ago

gp3 Max IOPS 16000 per volume is for SQL server, for MySQL server it is 64000 per volume, so B should work.
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html
upvoted 1 times

baba365 2 years, 8 months ago

'the application performance always degrades when the number of read and write IOPS is higher than 20,000' ... question didn't say read and write IOPs can't be higher than 32,000. Answer: C if it's based on performance and not cost related.

'Amazon RDS provides three storage types: General Purpose SSD (also known as gp2 and gp3), Provisioned IOPS SSD (also known as io1), and magnetic (also known as standard). They differ in performance characteristics and price, which means that you can tailor your storage performance

and cost to the needs of your database workload.'
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html

upvoted 1 times

dkw2342 2 years, 3 months ago

I really wonder how this answer can be the top answer. How would it even be possible to provision multiple gp3 volumes for RDS? RDS manages the storage, we have no influence on the number of volumes.

Striping is something that RDS does automatically depending on storage class and volume size: "When you select General Purpose SSD or Provisioned IOPS SSD, depending on the engine selected and the amount of storage requested, Amazon RDS automatically stripes across multiple volumes to enhance performance (...)"

For MariaDB with 400 to 64,000 GiB of gp3 storage, RDS automatically provisions 4 volumes. This gives us 12,000 IOPS *baseline* and can be increased up to 64,000 *provisioned* IOPS.

RDS does not support io2.

Therefore: Option B

upvoted 6 times

dkw2342 2 years, 3 months ago

PS: https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html

upvoted 2 times

zits88 2 years, 2 months ago

It must be that io2 was originally not supported by RDS, because I see this untruth reposted everywhere. It totally is.

upvoted 4 times

Michal_L_95 Highly Voted 3 years, 2 months ago

Selected Answer: B

It can not be option C as RDS does not support io2 storage type (only io1).

Here is a link to the RDS storage documentation: https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html

Also it is not the best option to take Magnetic storage as it supports max 1000 IOPS.

I vote for option B as gp3 storage type supports up to 64 000 IOPS where question mentioned with problem at level of 20 000.

upvoted 15 times

GalileoEC2 3 years, 2 months ago

is this true? Amazon RDS (Relational Database Service) supports the Provisioned IOPS SSD (io2) storage type for its database instances. The io2 storage type is designed to deliver predictable performance for critical and highly demanding database workloads. It provides higher durability, higher IOPS, and lower latency compared to other Amazon EBS (Elastic Block Store) storage types. RDS offers the option to choose between the General Purpose SSD (gp3) and Provisioned IOPS SSD (io2) storage types for database instances.

upvoted 3 times

1rob 2 years, 5 months ago

Please add a reference where it states that io2 is supported by RDS.

upvoted 1 times

zits88 2 years, 2 months ago

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html - Right there in the first paragraph: "Amazon RDS provides three storage types: General Purpose SSD (also known as gp2 and gp3), Provisioned IOPS SSD (also known as io1 and io2 Block Express), and magnetic (also known as standard). They differ in performance characteristics and price, which means that you can tailor your storage performance and cost to the needs of your database workload. You can create Db2, MySQL, MariaDB, Oracle, and PostgreSQL RDS DB instances with up to 64 terabytes (TiB) of storage. "

upvoted 7 times

joechen2023 2 years, 11 months ago

check the link below <https://repost.aws/knowledge-center/ebs-volume-type-differences>
it states:

General Purpose SSD volumes are good for a wide variety of transactional workloads that require less than the following:

16,000 IOPS
1,000 MiB/s of throughput
160-TiB volume size

upvoted 2 times

MultipleWorkerMirroredStrategy Most Recent 9 months, 1 week ago

Selected Answer: C

Since ~ March 24, C is arguably the right answer:

<https://aws.amazon.com/blogs/aws/amazon-rds-now-supports-io2-block-express-volumes-for-mission-critical-database-workloads/>

upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: C

RDS now support io2
upvoted 3 times

EllenLiu 1 year, 5 months ago

Selected Answer: C

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html#Concepts.Storage.GeneralSSD
upvoted 1 times

FlyingHawk 1 year, 6 months ago

Selected Answer: B

I selected C initially, but after read the doc https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html, noticed maximum IOPS for gp3 volume is 64K (16K for SQL server), For an RDS for MySQL workload, increasing the provisioned IOPS on the gp3 volume to exceed the application's peak demand (e.g., 30,000 IOPS) is a cost-effective and scalable solution.
upvoted 1 times

LeonSauveterre 1 year, 6 months ago

Selected Answer: C

How on earth would the answer be D? Splitting the gp3 volume into two does not solve the IOPS limitation because each volume would still be subject to the gp3 performance cap. The core of the crux is still there!

C - io2 volumes are exactly designed for workloads requiring consistent, high-performance storage.
upvoted 1 times

Mish 1 year, 7 months ago

Selected Answer: C

Provisioned IOPS SSDs (io2) are specifically designed to deliver sustained high performance and low latency (RDS is supported in IO2). They can handle more than 20,000 IOPS.
upvoted 2 times

studydue 1 year, 7 months ago

Updated 2024

Answer: C

upvoted 4 times

ashishs174 1 year, 8 months ago

Answer is C, io2 volumes are supported

<https://aws.amazon.com/blogs/aws/amazon-rds-now-supports-io2-block-express-volumes-for-mission-critical-database-workloads/>
upvoted 2 times

MatAlves 1 year, 9 months ago

Nice to see that everyone just picked a different answer...

upvoted 3 times

ChymKuBoy 1 year, 10 months ago

Selected Answer: B

B for sure

upvoted 1 times

example_ 1 year, 10 months ago

Selected Answer: C

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html
upvoted 5 times

FrozenCarrot 1 year, 11 months ago

Selected Answer: C

Now EBS support io2.

upvoted 6 times

theamachine 1 year, 11 months ago

Selected Answer: C

Provisioned IOPS SSDs (io2) are specifically designed to deliver sustained high performance and low latency (RDS is supported in IO2). They can handle more than 20,000 IOPS.
upvoted 7 times

FlyingHawk 1 year, 6 months ago

C is more expensive and overkill.

upvoted 1 times

Lin878 1 year, 12 months ago

Selected Answer: C

It should be "C" right, now.

<https://aws.amazon.com/blogs/aws/amazon-rds-now-supports-io2-block-express-volumes-for-mission-critical-database-workloads/>

upvoted 5 times

Scheldon 2 years ago

Selected Answer: C

Per the newest info it should be C right now

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html

upvoted 4 times

An IAM user made several configuration changes to AWS resources in their company's account during a production deployment last week. A solutions architect learned that a couple of security group rules are not configured as desired. The solutions architect wants to confirm which IAM user was responsible for making changes.

Which service should the solutions architect use to find the desired information?

- A. Amazon GuardDuty
- B. Amazon Inspector
- C. AWS CloudTrail **Most Voted**
- D. AWS Config

Correct Answer: C

Community vote distribution

C (100%)

Comments

cegama543 **Highly Voted** 2 years, 9 months ago

Selected Answer: C

C. AWS CloudTrail

The best option is to use AWS CloudTrail to find the desired information. AWS CloudTrail is a service that enables governance, compliance, operational auditing, and risk auditing of AWS account activities. CloudTrail can be used to log all changes made to resources in an AWS account, including changes made by IAM users, EC2 instances, AWS management console, and other AWS services. By using CloudTrail, the solutions architect can identify the IAM user who made the configuration changes to the security group rules.

upvoted 15 times

Manimgh **Most Recent** 1 year, 1 month ago

Selected Answer: C

which IAM user was responsible = CloudTrail

upvoted 2 times

BatVanyo 1 year, 7 months ago

Selected Answer: C

I was initially a bit confused on what Config and CloudTrail actually do, as both can be used to track configuration changes. However, this explanation is probably the best one I have come across so far:
"Config reports on what has changed, whereas CloudTrail reports on who made the change, when, and from which location"

Since the question is which IAM user was responsible for making the changes, the answer is CloudTrail.

upvoted 4 times

d401c0d 1 year, 8 months ago

Selected Answer: C

CloudTrail = which user made which api calls. This is used for audit purpose.

upvoted 4 times

sheq 2 years ago

This question is the same with the question 388, isn't it?

upvoted 1 times

kambarami 2 years, 2 months ago

This is how you know not to trust the moderators with their answers.

upvoted 1 times

Wayne23Fang 2 years, 3 months ago

There is an article "How to use AWS Config and CloudTrail to find who made changes to a resource" in aws blog. Given CloudTrail provided AWS config original info, it seems for this particular one, C is better than AWS config.

upvoted 3 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: C

AWS CloudTrail is the correct service to use here to identify which user was responsible for the security group configuration changes

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: C

AWS CloudTrail

upvoted 2 times

Bezha 2 years, 8 months ago

Selected Answer: C

AWS CloudTrail

upvoted 2 times

kprakashbehera 2 years, 9 months ago

Selected Answer: C

CloudTrail logs will tell who did that

upvoted 4 times

KAUS2 2 years, 9 months ago

Selected Answer: C

Option "C" AWS CloudTrail is correct.

upvoted 3 times

Nithin1119 2 years, 9 months ago

ccccc

upvoted 2 times

A company has implemented a self-managed DNS service on AWS. The solution consists of the following:

- Amazon EC2 instances in different AWS Regions
- Endpoints of a standard accelerator in AWS Global Accelerator

The company wants to protect the solution against DDoS attacks.

What should a solutions architect do to meet this requirement?

- A. Subscribe to AWS Shield Advanced. Add the accelerator as a resource to protect. **Most Voted**
- B. Subscribe to AWS Shield Advanced. Add the EC2 instances as resources to protect.
- C. Create an AWS WAF web ACL that includes a rate-based rule. Associate the web ACL with the accelerator.
- D. Create an AWS WAF web ACL that includes a rate-based rule. Associate the web ACL with the EC2 instances.

Correct Answer: A

Community vote distribution

A (97%) 3%

Comments

WherecanIstart **Highly Voted** 2 years, 8 months ago

Selected Answer: A

DDoS attacks = AWS Shield Advance
Shield Advance protects Global Accelerator, NLB, ALB, etc
upvoted 14 times

pentium75 **Highly Voted** 1 year, 11 months ago

Selected Answer: A

Global Accelerator is what is exposed to the Internet = where DDoS attacks could land = what must be protected by Shield Advanced
upvoted 6 times

Guru4Cloud **Most Recent** 2 years, 3 months ago

Selected Answer: B

So, the correct option is:

B. Subscribe to AWS Shield Advanced. Add the EC2 instances as resources to protect.

Here's why this option is the most appropriate:

A. While you can add the accelerator as a resource to protect with AWS Shield Advanced, it's generally more effective to protect the individual resources (in this case, the EC2 instances) because AWS Shield Advanced will automatically protect resources associated with Global Accelerator
upvoted 1 times

awsgeek75 1 year, 10 months ago

Which EC2 instance? Global Accelerator works by providing anycast IP addresses for the underlying resource (our EC2 in this case) so every end use trying to reach the EC2 server HAS to go through the Global Accelerator which is why the Global Accelerator needs to be protected and not the EC2.

upvoted 6 times

Abrar2022 2 years, 6 months ago

Selected Answer: A

DDoS attacks = AWS Shield Advance
resource as Global Acc

upvoted 5 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: A

DDoS attacks = AWS Shield Advanced

upvoted 3 times

nileshlg 2 years, 8 months ago

Selected Answer: A

Answer is A

<https://docs.aws.amazon.com/waf/latest/developerguide/ddos-event-mitigation-logic-gax.html>

upvoted 2 times

ktulu2602 2 years, 9 months ago

Selected Answer: A

AWS Shield is a managed service that provides protection against Distributed Denial of Service (DDoS) attacks for applications running on AWS. AWS Shield Standard is automatically enabled to all AWS customers at no additional cost. AWS Shield Advanced is an optional paid service. AWS Shield Advanced provides additional protections against more sophisticated and larger attacks for your applications running on Amazon Elastic Compute Cloud (EC2), Elastic Load Balancing (ELB), Amazon CloudFront, AWS Global Accelerator, and Route 53.

upvoted 4 times

[Removed] 2 years, 9 months ago

Selected Answer: A

aaaaa

accelator can not be attached to shield

upvoted 2 times

ktulu2602 2 years, 9 months ago

Yes it can:

AWS Shield is a managed service that provides protection against Distributed Denial of Service (DDoS) attacks for applications running on AWS. AWS Shield Standard is automatically enabled to all AWS customers at no additional cost. AWS Shield Advanced is an optional paid service. AWS Shield Advanced provides additional protections against more sophisticated and larger attacks for your applications running on Amazon Elastic Compute Cloud (EC2), Elastic Load Balancing (ELB), Amazon CloudFront, AWS Global Accelerator, and Route 53.

upvoted 2 times

An ecommerce company needs to run a scheduled daily job to aggregate and filter sales records for analytics. The company stores the sales records in an Amazon S3 bucket. Each object can be up to 10 GB in size. Based on the number of sales events, the job can take up to an hour to complete. The CPU and memory usage of the job are constant and are known in advance.

A solutions architect needs to minimize the amount of operational effort that is needed for the job to run.

Which solution meets these requirements?

- A. Create an AWS Lambda function that has an Amazon EventBridge notification. Schedule the EventBridge event to run once a day.
- B. Create an AWS Lambda function. Create an Amazon API Gateway HTTP API, and integrate the API with the function. Create an Amazon EventBridge scheduled event that calls the API and invokes the function.
- C. Create an Amazon Elastic Container Service (Amazon ECS) cluster with an AWS Fargate launch type. Create an Amazon EventBridge scheduled event that launches an ECS task on the cluster to run the job. **Most Voted**
- D. Create an Amazon Elastic Container Service (Amazon ECS) cluster with an Amazon EC2 launch type and an Auto Scaling group with at least one EC2 instance. Create an Amazon EventBridge scheduled event that launches an ECS task on the cluster to run the job.

Correct Answer: C

Community vote distribution

C (100%)

Comments

ktulu2602 **Highly Voted** 2 years, 9 months ago

Selected Answer: C

The requirement is to run a daily scheduled job to aggregate and filter sales records for analytics in the most efficient way possible. Based on the requirement, we can eliminate option A and B since they use AWS Lambda which has a limit of 15 minutes of execution time, which may not be sufficient for a job that can take up to an hour to complete.

Between options C and D, option C is the better choice since it uses AWS Fargate which is a serverless compute engine for containers that eliminates the need to manage the underlying EC2 instances, making it a low operational effort solution. Additionally, Fargate also provides instant scale-up and scale-down capabilities to run the scheduled job as per the requirement.

Therefore, the correct answer is:

C. Create an Amazon Elastic Container Service (Amazon ECS) cluster with an AWS Fargate launch type. Create an Amazon EventBridge scheduled event that launches an ECS task on the cluster to run the job.

upvoted 27 times

imvb88 **Highly Voted** 2 years, 7 months ago

Selected Answer: C

"1-hour job" -> A, B out since max duration for Lambda is 15 min

Between C and D, "minimize operational effort" means Fargate -> C

upvoted 6 times

awsgeek75 **Most Recent** 1 year, 11 months ago

Selected Answer: C

A&B are out due to Lambda 15 min limits

C is less operationally complex than D so C is the right answer. Fargate is managed ECS cluster whereas EC2 based ECS will require more config overhead.

upvoted 5 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: C

C. Create an Amazon Elastic Container Service (Amazon ECS) cluster with an AWS Fargate launch type. Create an Amazon EventBridge scheduled event that launches an ECS task on the cluster to run the job

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: C

The best option is C.

'The job can take up to an hour to complete' rules out lambda functions as they only execute up to 15 mins. Hence option A and B are out.

'The CPU and memory usage of the job are constant and are known in advance' rules out the need for autoscaling. Hence option D is out.

upvoted 4 times

klayytech 2 years, 8 months ago

Selected Answer: C

The solution that meets the requirements with the least operational overhead is to create a **Regional AWS WAF web ACL with a rate-based rule** and associate the web ACL with the API Gateway stage. This solution will protect the application from HTTP flood attacks by monitoring incoming requests and blocking requests from IP addresses that exceed the predefined rate.

Amazon CloudFront distribution with Lambda@Edge in front of the API Gateway Regional API endpoint is also a good solution but it requires more operational overhead than the previous solution.

Using Amazon CloudWatch metrics to monitor the Count metric and alerting the security team when the predefined rate is reached is not a solution that can protect against HTTP flood attacks.

Creating an Amazon CloudFront distribution in front of the API Gateway Regional API endpoint with a maximum TTL of 24 hours is not a solution that can protect against HTTP flood attacks.

upvoted 1 times

klayytech 2 years, 8 months ago

Selected Answer: C

The solution that meets these requirements is C. Create an Amazon Elastic Container Service (Amazon ECS) cluster with an AWS Fargate launch type. Create an Amazon EventBridge scheduled event that launches an ECS task on the cluster to run the job. This solution will minimize the amount of operational effort that is needed for the job to run.

AWS Lambda which has a limit of 15 minutes of execution time,

upvoted 2 times

A company needs to transfer 600 TB of data from its on-premises network-attached storage (NAS) system to the AWS Cloud. The data transfer must be complete within 2 weeks. The data is sensitive and must be encrypted in transit. The company's internet connection can support an upload speed of 100 Mbps.

Which solution meets these requirements MOST cost-effectively?

- A. Use Amazon S3 multi-part upload functionality to transfer the files over HTTPS.
- B. Create a VPN connection between the on-premises NAS system and the nearest AWS Region. Transfer the data over the VPN connection.
- C. Use the AWS Snow Family console to order several AWS Snowball Edge Storage Optimized devices. Use the devices to transfer the data to Amazon S3. **Most Voted**
- D. Set up a 10 Gbps AWS Direct Connect connection between the company location and the nearest AWS Region. Transfer the data over a VPN connection into the Region to store the data in Amazon S3.

Correct Answer: C

Community vote distribution

C (97%)

3%

Comments

shanwford **Highly Voted** 2 years, 2 months ago

Selected Answer: C

With the existing data link the transfer takes ~ 600 days in the best case. Thus, (A) and (B) are not applicable. Solution (D) could meet the target with a transfer time of 6 days, but the lead time for the direct connect deployment can take weeks! Thus, (C) is the only valid solution.

upvoted 14 times

3916aef **Most Recent** 3 days, 20 hours ago

Selected Answer: D

The question asked for "Most cost-effectively."
C) would cost around 5000\$ - per Device. And it says "to order several" devices.
D) would cost 22,50\$ per hour. in 6 Days that would sum around 6000\$.
The question did not ask for any time-frame.

upvoted 1 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: C

Use the AWS Snow Family console to order several AWS Snowball Edge Storage Optimized devices. Use the devices to transfer the data to Amazon S3.

upvoted 2 times

TariqKipkemei 2 years ago

Selected Answer: C

C is the best option considering the time and bandwidth limitations

upvoted 2 times

pbpally 2 years, 1 month ago

Selected Answer: C

We need the admin in here to tell us how they plan on this being achieved over connection with such a slow connection lol.
It's C, folks.

upvoted 3 times

KAUS2 2 years, 3 months ago

Selected Answer: C

Best option is to use multiple AWS Snowball Edge Storage Optimized devices. Option "C" is the correct one.
upvoted 2 times

ktulu2602 2 years, 3 months ago

Selected Answer: C

All others are limited by the bandwidth limit
upvoted 1 times

ktulu2602 2 years, 3 months ago

Or provisioning time in the D case
upvoted 2 times

KZM 2 years, 3 months ago

It is C. Snowball (from Snow Family).
upvoted 2 times

cegama543 2 years, 3 months ago

Selected Answer: C

C. Use the AWS Snow Family console to order several AWS Snowball Edge Storage Optimized devices. Use the devices to transfer the data to Amazon S3.

The best option is to use the AWS Snow Family console to order several AWS Snowball Edge Storage Optimized devices and use the devices to transfer the data to Amazon S3. Snowball Edge is a petabyte-scale data transfer device that can help transfer large amounts of data securely and quickly. Using Snowball Edge can be the most cost-effective solution for transferring large amounts of data over long distances and can help meet the requirement of transferring 600 TB of data within two weeks.

upvoted 4 times

A financial company hosts a web application on AWS. The application uses an Amazon API Gateway Regional API endpoint to give users the ability to retrieve current stock prices. The company's security team has noticed an increase in the number of API requests. The security team is concerned that HTTP flood attacks might take the application offline.

A solutions architect must design a solution to protect the application from this type of attack.

Which solution meets these requirements with the LEAST operational overhead?

- A. Create an Amazon CloudFront distribution in front of the API Gateway Regional API endpoint with a maximum TTL of 24 hours.
- B. Create a Regional AWS WAF web ACL with a rate-based rule. Associate the web ACL with the API Gateway stage. **Most Voted**
- C. Use Amazon CloudWatch metrics to monitor the Count metric and alert the security team when the predefined rate is reached.
- D. Create an Amazon CloudFront distribution with Lambda@Edge in front of the API Gateway Regional API endpoint. Create an AWS Lambda function to block requests from IP addresses that exceed the predefined rate.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Guru4Cloud **Highly Voted** 1 year, 9 months ago

Selected Answer: B

Regional AWS WAF web ACL is a managed web application firewall that can be used to protect your API Gateway API from a variety of attacks, including HTTP flood attacks.

Rate-based rule is a type of rule that can be used to limit the number of requests that can be made from a single IP address within a specified period of time.

API Gateway stage is a logical grouping of API resources that can be used to control access to your API.

upvoted 10 times

elearningtakai **Highly Voted** 2 years, 2 months ago

Selected Answer: B

A rate-based rule in AWS WAF allows the security team to configure thresholds that trigger rate-based rules, which enable AWS WAF to track the rate of requests for a specified time period and then block them automatically when the threshold is exceeded. This provides the ability to prevent HTTP flood attacks with minimal operational overhead.

upvoted 6 times

TariqKipkemei **Most Recent** 2 years ago

Selected Answer: B

Answer is B

upvoted 2 times

maxicalypse 2 years, 2 months ago

B is correct

upvoted 2 times

kampatra 2 years, 2 months ago

Selected Answer: B

<https://docs.aws.amazon.com/waf/latest/developerguide/web-acl.html>

upvoted 2 times

A meteorological startup company has a custom web application to sell weather data to its users online. The company uses Amazon DynamoDB to store its data and wants to build a new service that sends an alert to the managers of four internal teams every time a new weather event is recorded. The company does not want this new service to affect the performance of the current application.

What should a solutions architect do to meet these requirements with the LEAST amount of operational overhead?

- A. Use DynamoDB transactions to write new event data to the table. Configure the transactions to notify internal teams.
- B. Have the current application publish a message to four Amazon Simple Notification Service (Amazon SNS) topics. Have each team subscribe to one topic.
- C. Enable Amazon DynamoDB Streams on the table. Use triggers to write to a single Amazon Simple Notification Service (Amazon SNS) topic to which the teams can subscribe. **Most Voted**
- D. Add a custom attribute to each record to flag new items. Write a cron job that scans the table every minute for items that are new and notifies an Amazon Simple Queue Service (Amazon SQS) queue to which the teams can subscribe.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 2 years, 2 months ago

Selected Answer: C

The best solution to meet these requirements with the least amount of operational overhead is to enable Amazon DynamoDB Streams on the table and use triggers to write to a single Amazon Simple Notification Service (Amazon SNS) topic to which the teams can subscribe. This solution requires minimal configuration and infrastructure setup, and Amazon DynamoDB Streams provide a low-latency way to capture changes to the DynamoDB table. The triggers automatically capture the changes and publish them to the SNS topic, which notifies the internal teams.

upvoted 15 times

Buruguduystunstugudunstuy 2 years, 2 months ago

Answer A is not a suitable solution because it requires additional configuration to notify the internal teams, and it could add operational overhead to the application.

Answer B is not the best solution because it requires changes to the current application, which may affect its performance, and it creates additional work for the teams to subscribe to multiple topics.

Answer D is not a good solution because it requires a cron job to scan the table every minute, which adds additional operational overhead to the system.

Therefore, the correct answer is C. Enable Amazon DynamoDB Streams on the table. Use triggers to write to a single Amazon SNS topic to which the teams can subscribe.

upvoted 6 times

Manimgh **Most Recent** 1 year, 1 month ago

Selected Answer: C

Sends an alert = SNS

upvoted 1 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: C

Enable Amazon DynamoDB Streams on the table. Use triggers to write to a single Amazon Simple Notification Service (Amazon SNS) topic to which the teams can subscribe

upvoted 4 times

james2033 1 year, 10 months ago

Selected Answer: C

Question keyword: "sends an alert", a new weather event is recorded". Answer keyword C "Amazon DynamoDB Streams on the table", "Amazon Simple Notification Service" (Amazon SNS). Choose C. Easy question.

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/Streams.html>

<https://aws.amazon.com/blogs/database/dynamodb-streams-use-cases-and-design-patterns/>

upvoted 4 times

TariqKipkemei 2 years ago

Selected Answer: C

Best answer is C

upvoted 3 times

TariqKipkemei 1 year, 7 months ago

DynamoDB Streams captures a time-ordered sequence of item-level modifications in any DynamoDB table and stores this information in a log for up to 24 hours. This capture activity can also invoke triggers to write the event to a single Amazon Simple Notification Service (Amazon SNS) topic to which the teams can subscribe to.

upvoted 7 times

Hemanthgowda1932 2 years, 2 months ago

C is correct

upvoted 2 times

Santosh43 2 years, 2 months ago

definitely C

upvoted 2 times

Bezha 2 years, 2 months ago

Selected Answer: C

DynamoDB Streams

upvoted 4 times

sitha 2 years, 3 months ago

Selected Answer: C

Answer : C

upvoted 2 times

A company wants to use the AWS Cloud to make an existing application highly available and resilient. The current version of the application resides in the company's data center. The application recently experienced data loss after a database server crashed because of an unexpected power outage.

The company needs a solution that avoids any single points of failure. The solution must give the application the ability to scale to meet user demand.

Which solution will meet these requirements?

A. Deploy the application servers by using Amazon EC2 instances in an Auto Scaling group across multiple Availability Zones. Use an Amazon RDS DB instance in a Multi-AZ configuration. **Most Voted**

B. Deploy the application servers by using Amazon EC2 instances in an Auto Scaling group in a single Availability Zone. Deploy the database on an EC2 instance. Enable EC2 Auto Recovery.

C. Deploy the application servers by using Amazon EC2 instances in an Auto Scaling group across multiple Availability Zones. Use an Amazon RDS DB instance with a read replica in a single Availability Zone. Promote the read replica to replace the primary DB instance if the primary DB instance fails.

D. Deploy the application servers by using Amazon EC2 instances in an Auto Scaling group across multiple Availability Zones. Deploy the primary and secondary database servers on EC2 instances across multiple Availability Zones. Use Amazon Elastic Block Store (Amazon EBS) Multi-Attach to create shared storage between the instances.

Correct Answer: A

Community vote distribution

A (93%)

3%

Comments

pentium75 **Highly Voted** 1 year, 11 months ago

Selected Answer: A

B has app servers in a single AZ and a database on a single instance

C has both DB replicas in a single AZ

D does not work (EBS Multi-Attach requires EC2 instances in same AZ), and if it would work then the EBS volume would be an SPOF

upvoted 7 times

Guru4Cloud **Most Recent** 2 years, 3 months ago

Selected Answer: A

Deploy the application servers by using Amazon EC2 instances in an Auto Scaling group across multiple Availability Zones. Use an Amazon RDS DB instance in a Multi-AZ configuration

upvoted 2 times

czyboi 2 years, 4 months ago

Why is C incorrect ?

upvoted 1 times

Guru4Cloud 2 years, 3 months ago

C is incorrect because the read replica also resides in a single AZ

upvoted 4 times

antropaws 2 years, 6 months ago

Selected Answer: A

A most def.

upvoted 3 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: A

Deploy the application servers by using Amazon EC2 instances in an Auto Scaling group across multiple Availability Zones. Use an Amazon RDS DB instance in a Multi-AZ configuration.

upvoted 3 times

Buruguduystunstugudunstuy 2 years, 8 months ago

Selected Answer: A

The correct answer is A. Deploy the application servers by using Amazon EC2 instances in an Auto Scaling group across multiple Availability Zones. Use an Amazon RDS DB instance in a Multi-AZ configuration.

To make an existing application highly available and resilient while avoiding any single points of failure and giving the application the ability to scale to meet user demand, the best solution would be to deploy the application servers using Amazon EC2 instances in an Auto Scaling group across multiple Availability Zones and use an Amazon RDS DB instance in a Multi-AZ configuration.

By using an Amazon RDS DB instance in a Multi-AZ configuration, the database is automatically replicated across multiple Availability Zones, ensuring that the database is highly available and can withstand the failure of a single Availability Zone. This provides fault tolerance and avoids any single points of failure.

upvoted 4 times

Thief 2 years, 8 months ago

Selected Answer: D

Why not D?

upvoted 1 times

Guru4Cloud 2 years, 3 months ago

D is incorrect because using Multi-Attach EBS adds complexity and doesn't provide automatic DB failover

upvoted 2 times

pentium75 1 year, 11 months ago

Multi-Attach does not work across Availability Zones.

upvoted 2 times

Buruguduystunstugudunstuy 2 years, 8 months ago

Answer D, deploying the primary and secondary database servers on EC2 instances across multiple Availability Zones and using Amazon Elastic Block Store (Amazon EBS) Multi-Attach to create shared storage between the instances, may provide high availability for the database but may introduce additional complexity, and management overhead, and potential performance issues.

upvoted 2 times

WheracanIstart 2 years, 8 months ago

Selected Answer: A

Highly available = Multi-AZ approach

upvoted 3 times

nileshlg 2 years, 8 months ago

Selected Answer: A

Answers is A

upvoted 1 times

quentin17 2 years, 9 months ago

Selected Answer: C

Highly available

upvoted 1 times

pentium75 1 year, 11 months ago

No because instance and read replica "in a single Availability Zone"

upvoted 2 times

KAUS2 2 years, 9 months ago

Selected Answer: A

Yes , agree with A

upvoted 2 times

cegama543 2 years, 9 months ago

Selected Answer: A

agree with that

upvoted 2 times

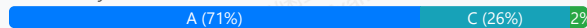
A company needs to ingest and handle large amounts of streaming data that its application generates. The application runs on Amazon EC2 instances and sends data to Amazon Kinesis Data Streams, which is configured with default settings. Every other day, the application consumes the data and writes the data to an Amazon S3 bucket for business intelligence (BI) processing. The company observes that Amazon S3 is not receiving all the data that the application sends to Kinesis Data Streams.

What should a solutions architect do to resolve this issue?

- A. Update the Kinesis Data Streams default settings by modifying the data retention period. **Most Voted**
- B. Update the application to use the Kinesis Producer Library (KPL) to send the data to Kinesis Data Streams.
- C. Update the number of Kinesis shards to handle the throughput of the data that is sent to Kinesis Data Streams.
- D. Turn on S3 Versioning within the S3 bucket to preserve every version of every object that is ingested in the S3 bucket.

Correct Answer: A

Community vote distribution



Comments

WherecanIstart **Highly Voted** 3 years, 2 months ago

Selected Answer: A

"A Kinesis data stream stores records from 24 hours by default, up to 8760 hours (365 days)."
<https://docs.aws.amazon.com/streams/latest/dev/kinesis-extended-retention.html>

The question mentioned Kinesis data stream default settings and "every other day". After 24hrs, the data isn't in the Data stream if the default settings is not modified to store data more than 24hrs.

upvoted 34 times

babayomi 1 year, 7 months ago

Thank you for the link

upvoted 2 times

cegama543 **Highly Voted** 3 years, 3 months ago

Selected Answer: C

C. Update the number of Kinesis shards to handle the throughput of the data that is sent to Kinesis Data Streams.

The best option is to update the number of Kinesis shards to handle the throughput of the data that is sent to Kinesis Data Streams. Kinesis Data Streams scales horizontally by increasing or decreasing the number of shards, which controls the throughput capacity of the stream. By increasing the number of shards, the application will be able to send more data to Kinesis Data Streams, which can help ensure that S3 receives all the data.

upvoted 17 times

Buruguduystunstugudunstuy 3 years, 2 months ago

Answer C:

C. Update the number of Kinesis shards to handle the throughput of the data that is sent to Kinesis Data Streams.

- Answer C updates the number of Kinesis shards to handle the throughput of the data that is sent to Kinesis Data Streams. By increasing the number of shards, the data is distributed across multiple shards, which allows for increased throughput and ensures that all data is ingested and processed by Kinesis Data Streams.

- Monitoring the Kinesis Data Streams and adjusting the number of shards as needed to handle changes in data throughput can ensure that the application can handle large amounts of streaming data.

upvoted 2 times

Buruguduystunstugudunstuy 3 years, 2 months ago

@cegama543, my apologies. Moderator if you can disapprove of the post above? I made a mistake. It is supposed to be intended on the post that I submitted.

Thanks.

upvoted 2 times

CapJackSparrow 3 years, 2 months ago

lets say you had infinity shards... if the retention period is 24 hours and you get the data every 48 hours, you will lose 24 hours of data no matter the amount of shards no?

upvoted 16 times

enzomv 3 years, 2 months ago

Amazon Kinesis Data Streams supports changes to the data record retention period of your data stream. A Kinesis data stream is an ordered sequence of data records meant to be written to and read from in real time. Data records are therefore stored in shards in your stream temporarily. The time period from when a record is added to when it is no longer accessible is called the retention period. A Kinesis data stream stores records from 24 hours by default, up to 8760 hours (365 days).

upvoted 6 times

lalaraj **Most Recent** 10 months, 3 weeks ago

Selected Answer: C

Why?

✓ Problem Root Cause:

Kinesis Data Streams with default settings may have insufficient shards to handle the incoming data rate.

Each shard supports 1 MB/s input and 2 MB/s output. If exceeded, data is throttled or lost.

✓ Solution:

Increase shard count to match the data throughput.

Calculate required shards:

text

Number of shards = (Total data input rate in MB/s) ÷ 1

Example: If the app sends 5 MB/s, use at least 5 shards.

upvoted 1 times

harrura 11 months, 3 weeks ago

Selected Answer: A

question is malformed, answer A is right if the scenario is one application (application1) is running on EC2 instances and sends its data to Kinesis data streams. Another application consumes the data every alternate day from Kinesis data streams of application 1 and writes data to S3. Answer A is correct only if two applications exist else the answer is C

upvoted 1 times

Rcosmos 1 year, 4 months ago

Selected Answer: U

C. Atualizar o número de fragmentos do Kinesis:

Aumentar o número de fragmentos melhora a capacidade de processamento e a taxa de transferência do stream. No entanto, este não é o problema aqui, pois a perda de dados ocorre devido ao período de retenção insuficiente, e não devido a limitações de capacidade.

upvoted 1 times

abriggy 1 year, 10 months ago

Selected Answer: C

Answer is C

Issue with A) Update the Kinesis Data Streams default settings by modifying the data retention period. is below

Limitation: Modifying the data retention period affects how long data is kept in the stream, but it does not address the issue of the stream's capacity to ingest data. If the stream is unable to handle the incoming data volume, extending the retention period will not resolve the data loss issue.

upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: A

Every other day, = 48 hours

Default settings = 24 hours

B: Development library so won't help

C: More shards may retain more data but they will have same limitation of 24 hours retention

D: Irrelevant

A: Increase the default limit from 24 hours to 48 hours

upvoted 6 times

pentium75 2 years, 5 months ago

Selected Answer: A

"Default settings" = 24 hour retention

upvoted 5 times

Murtadhaceit 2 years, 6 months ago

Selected Answer: A

KDS has two modes:

1. Provisioned Mode: Answer C would be correct if KDS runs in this mode. We need to increase the number of shards.
2. On-Demand: Scales automatically, which means it doesn't need to adjust the number of shards based on observed throughput.

And since the question does not mention which type, I would go with On-demand. Therefore, A is the correct answer.

upvoted 3 times

TariqKipkemei 2 years, 7 months ago

Selected Answer: A

Data records are stored in shards in a Kinesis data stream temporarily. The time period from when a record is added, to when it is no longer accessible is called the retention period. This time period is 24 hours by default, but could be adjusted to 365 days.

Kinesis Data Streams automatically scales the number of shards in response to changes in data volume and traffic, so this rules out option C.

<https://docs.aws.amazon.com/streams/latest/dev/service-sizes-and-limits.html#:~:text=the%20number%20of%20shards,-in%20response%20to>

upvoted 2 times

Ramdi1 2 years, 8 months ago

Selected Answer: A

I have only voted A because it mentions the default setting in Kinesis, if it did not mention that then I would look to increase the Shards. By default it is 24 hours and can go to 365 days. I think the question should be rephrased slightly. I had trouble deciding between A & C. Also apparently the most voted answer is the correct answer as per some advice I was given.

upvoted 3 times

BrijMohan08 2 years, 9 months ago

Selected Answer: A

Default retention is 24 hrs, but the data read is every other day, so the S3 will never receive the data, Change the default retention period to 48 hours.

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: C

By default, a Kinesis data stream is created with one shard. If the data throughput to the stream is higher than the capacity of the single shard, the data stream may not be able to handle all the incoming data, and some data may be lost.

Therefore, to handle the high volume of data that the application sends to Kinesis Data Streams, the number of Kinesis shards should be increased to handle the required throughput.

Kinesis Data Streams shards are the basic units of scalability and availability. Each shard can process up to 1,000 records per second with a maximum of 1 MB of data per second. If the application is sending more data to Kinesis Data Streams than the shards can handle, then some of the data will be dropped.

upvoted 1 times

Guru4Cloud 2 years, 9 months ago

If you have doubts, Please read about Kinesis Data Streams shards.

Ans: A is not the correct answer here

upvoted 1 times

Amycert 2 years, 10 months ago

Selected Answer: A

the default retention period is 24 hours "The default retention period of 24 hours covers scenarios where intermittent lags in processing require catch-up with the real-time data."

so we should increment this

upvoted 2 times

hsinchang 2 years, 10 months ago

Selected Answer: A

As "Default settings" is mentioned here, I vote for A.

upvoted 2 times

jaydesai8 2 years, 11 months ago

Selected Answer: A

keyword here is - default settings and every other day and since "A Kinesis data stream stores records from 24 hours by default, up to 8760 hours (365 days)."

<https://docs.aws.amazon.com/streams/latest/dev/kinesis-extended-retention.html>

Will go with A

upvoted 2 times

jayce5 3 years ago

Selected Answer: A

C is wrong because even if you update the number of Kinesis shards, you still need to change the default data retention period first. Otherwise, you would lose data after 24 hours.

upvoted 3 times

A developer has an application that uses an AWS Lambda function to upload files to Amazon S3 and needs the required permissions to perform the task. The developer already has an IAM user with valid IAM credentials required for Amazon S3.

What should a solutions architect do to grant the permissions?

- A. Add required IAM permissions in the resource policy of the Lambda function.
- B. Create a signed request using the existing IAM credentials in the Lambda function.
- C. Create a new IAM user and use the existing IAM credentials in the Lambda function.
- D. Create an IAM execution role with the required permissions and attach the IAM role to the Lambda function. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 3 years, 2 months ago

To grant the necessary permissions to an AWS Lambda function to upload files to Amazon S3, a solutions architect should create an IAM execution role with the required permissions and attach the IAM role to the Lambda function. This approach follows the principle of least privilege and ensures that the Lambda function can only access the resources it needs to perform its specific task.

Therefore, the correct answer is D. Create an IAM execution role with the required permissions and attach the IAM role to the Lambda function.
upvoted 7 times

AWSSURI 1 year, 9 months ago

Oh you're here

upvoted 3 times

Guru4Cloud **Most Recent** 2 years, 9 months ago

Selected Answer: D

Create Lambda execution role and attach existing S3 IAM role to the lambda function

upvoted 4 times

Bilalglg93350 3 years, 2 months ago

D. Créez un rôle d'exécution IAM avec les autorisations requises et attachez le rôle IAM à la fonction Lambda.

L'architecte de solutions doit créer un rôle d'exécution IAM ayant les autorisations nécessaires pour accéder à Amazon S3 et effectuer les opérations requises (par exemple, charger des fichiers). Ensuite, le rôle doit être associé à la fonction Lambda, de sorte que la fonction puisse assumer ce rôle et avoir les autorisations nécessaires pour interagir avec Amazon S3.

upvoted 4 times

nileshlg 3 years, 2 months ago

Selected Answer: D

Answer is D

upvoted 3 times

kampatra 3 years, 2 months ago

Selected Answer: D

D - correct ans

upvoted 3 times

sitha 3 years, 3 months ago

Selected Answer: D

Create Lambda execution role and attach existing S3 IAM role to the lambda function

upvoted 3 times

ktulu2602 3 years, 3 months ago

Selected Answer: D

Definitely D

upvoted 3 times

Nithin1119 3 years, 3 months ago

Selected Answer: D

ddddddd

upvoted 2 times

A company has deployed a serverless application that invokes an AWS Lambda function when new documents are uploaded to an Amazon S3 bucket. The application uses the Lambda function to process the documents. After a recent marketing campaign, the company noticed that the application did not process many of the documents.

What should a solutions architect do to improve the architecture of this application?

- A. Set the Lambda function's runtime timeout value to 15 minutes.
- B. Configure an S3 bucket replication policy. Stage the documents in the S3 bucket for later processing.
- C. Deploy an additional Lambda function. Load balance the processing of the documents across the two Lambda functions.
- D. Create an Amazon Simple Queue Service (Amazon SQS) queue. Send the requests to the queue. Configure the queue as an event source for Lambda. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 3 years, 2 months ago

Selected Answer: D

To improve the architecture of this application, the best solution would be to use Amazon Simple Queue Service (Amazon SQS) to buffer the request and decouple the S3 bucket from the Lambda function. This will ensure that the documents are not lost and can be processed at a later time if the Lambda function is not available.

This will ensure that the documents are not lost and can be processed at a later time if the Lambda function is not available. By using Amazon SQS, the architecture is decoupled and the Lambda function can process the documents in a scalable and fault-tolerant manner.

upvoted 5 times

Guru4Cloud **Most Recent** 2 years, 9 months ago

Selected Answer: D

D. Create an Amazon Simple Queue Service (Amazon SQS) queue. Send the requests to the queue. Configure the queue as an event source for Lambda

upvoted 3 times

TariqKipkemei 3 years ago

Selected Answer: D

D is the best approach

upvoted 3 times

Russ99 3 years, 2 months ago

Selected Answer: D

D is the correct answer

upvoted 3 times

Bilalgl93350 3 years, 2 months ago

D. Créez une file d'attente Amazon Simple Queue Service (Amazon SQS). Envoyez les demandes à la file d'attente. Configurez la file d'attente en tant que source d'événement pour Lambda.

Cette solution permet de gérer efficacement les pics de charge et d'éviter la perte de documents en cas d'augmentation soudaine du trafic. Lorsque de nouveaux documents sont chargés dans le compartiment Amazon S3, les demandes sont envoyées à la file d'attente Amazon SQS, qui agit comme un tampon. La fonction Lambda est déclenchée en fonction des événements dans la file d'attente, ce qui permet un traitement équilibré et évite que l'application ne soit submergée par un grand nombre de documents simultanés.

upvoted 2 times

Russs99 3 years, 2 months ago

exactement. si je pouvais explique come cela en Francais aussi
upvoted 1 times

JA2018 1 year, 6 months ago

Frrom Google Translate:

Create an Amazon Simple Queue Service (Amazon SQS) queue. Send requests to the queue. Configure the queue as an event source for Lambda.

This solution makes it possible to effectively manage load peaks and avoid the loss of documents in the event of a sudden increase in traffic. When new documents are uploaded to the Amazon S3 bucket, requests are sent to the Amazon SQS queue, which acts as a buffer. The Lambda function is triggered based on events in the queue, which provides balanced processing and prevents the application from being overwhelmed by a large number of concurrent documents.

upvoted 1 times

WherecanIstart 3 years, 2 months ago

Selected Answer: D

D is the correct answer.

upvoted 1 times

kampatra 3 years, 2 months ago

Selected Answer: D

D is correct

upvoted 1 times

A solutions architect is designing the architecture for a software demonstration environment. The environment will run on Amazon EC2 instances in an Auto Scaling group behind an Application Load Balancer (ALB). The system will experience significant increases in traffic during working hours but is not required to operate on weekends.

Which combination of actions should the solutions architect take to ensure that the system can scale to meet demand? (Choose two.)

- A. Use AWS Auto Scaling to adjust the ALB capacity based on request rate.
- B. Use AWS Auto Scaling to scale the capacity of the VPC internet gateway.
- C. Launch the EC2 instances in multiple AWS Regions to distribute the load across Regions.
- D. Use a target tracking scaling policy to scale the Auto Scaling group based on instance CPU utilization. **Most Voted**
- E. Use scheduled scaling to change the Auto Scaling group minimum, maximum, and desired capacity to zero for weekends. Revert to the default values at the start of the week. **Most Voted**

Correct Answer: DE

Community vote distribution

DE (69%) AD (15%) Other

Comments

pentium75 **Highly Voted** 2 years, 5 months ago

Selected Answer: DE

Not A - "AWS Auto Scaling" cannot adjust "ALB capacity" (<https://aws.amazon.com/autoscaling/faqs/>)

Not B - VPC internet gateway has nothing to do with this

Not C - Regions have nothing to do with scaling

"The system will experience significant increases in traffic during working hours" -> addressed by D

"But is not required to operate on weekends" -> addressed by E

upvoted 16 times

foha2012 2 years, 4 months ago

Good explanation!

upvoted 2 times

cd93 **Highly Voted** 2 years, 9 months ago

What does "ALB capacity" even means anyway? It should be "Target Group capacity" no?

Answer should be DE, as D is a more comprehensive answer (and more practical in real life)

upvoted 13 times

bora4motion **Most Recent** 1 year, 1 month ago

Selected Answer: DE

who on this earth voted a?

upvoted 1 times

BigHammer 2 years, 9 months ago

AD

E - the question doesn't ask about cost. Also, shutting it down during the weekend does nothing to improve scaling during the week. It doesn't address the requirements.

upvoted 2 times

JA2018 1 year, 6 months ago

Key in STEM:

Nobody will use the system during weekend

upvoted 1 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: DE

The solutions architect should take actions D and E:

D) Use a target tracking scaling policy to scale the Auto Scaling group based on instance CPU utilization. This will allow the Auto Scaling group to dynamically scale in and out based on demand.

E) Use scheduled scaling to change the Auto Scaling group capacity to zero on weekends when traffic is expected to be low. This will minimize costs by terminating unused instances.

upvoted 7 times

fuzzycr 2 years, 11 months ago

Selected Answer: AE

Basado en los requerimientos la opción que se requiere para optimizar los costos de 0 operaciones en los fines de semana

upvoted 2 times

jaydesai8 2 years, 11 months ago

Selected Answer: DE

DE - This seems more close for the auto scaling -

A - Its says auto scaling on ALB, but it should always be on EC2 instances and not ELB

upvoted 7 times

XaviL 2 years, 11 months ago

Hi guys, very simple

* A. because the question are asking about request rate!!!! This is a requirement!

* E. The weekend is not necessary to execute anything!

A&D. Is not possible, way you can put an ALB capacity based in cpu and in request rate???? You need to select one or another option (and this is for all questions here guys!)

upvoted 3 times

[Removed] 2 years, 11 months ago

Selected Answer: AE

ALBRequestCountPerTarget—Average Application Load Balancer request count per target.

<https://docs.aws.amazon.com/autoscaling/ec2/userguide/as-scaling-target-tracking.html#target-tracking-choose-metrics>

It is possible to set to zero. "is not required to operate on weekends" means the instances are not required during the weekends.

<https://docs.aws.amazon.com/autoscaling/ec2/userguide/asg-capacity-limits.html>

upvoted 3 times

pentium75 2 years, 5 months ago

A says to scale "ALB capacity", not number of EC2 instances. But "AWS Auto Scaling" cannot scale ALB capacity.

upvoted 1 times

Uzi_m 3 years ago

Option E is incorrect because the question specifically mentions an increase in traffic during working hours. Therefore, it is not advisable to schedule the instances for 24 hours using default settings throughout the entire week.

E. Use scheduled scaling to change the Auto Scaling group minimum, maximum, and desired capacity to zero for weekends. Revert to the default values at the start of the week.

upvoted 1 times

omoakin 3 years ago

AD are the correct answws

upvoted 3 times

TariqKipkemei 3 years ago

Selected Answer: ADE

Either one or two or all of these combinations will meet the need:

Use AWS Auto Scaling to adjust the ALB capacity based on request rate.

Use a target tracking scaling policy to scale the Auto Scaling group based on instance CPU utilization.

Use scheduled scaling to change the Auto Scaling group minimum, maximum, and desired capacity to zero for weekends. Revert to the default values at the start of the week.

upvoted 2 times

TariqKipkemei 2 years, 7 months ago

Scheduled scaling was specifically designed to handle these kind of requirements.

I therefore take out target scaling.

<https://docs.aws.amazon.com/autoscaling/ec2/userguide/ec2-auto-scaling-scheduled-scaling.html#:~:text=RSS-,Scheduled%20scaling,-helps%20you%20to>

upvoted 2 times

Joe94KR 3 years, 1 month ago

Selected Answer: DE

<https://docs.aws.amazon.com/autoscaling/ec2/userguide/as-scaling-target-tracking.html#target-tracking-choose-metrics>

Based on docs, ASG can't track ALB's request rate, so the answer is D&E
meanwhile ASG can track CPU rates.

upvoted 5 times

[Removed] 3 years ago

The link shows:

ALBRequestCountPerTarget—Average Application Load Balancer request count per target.

upvoted 2 times

kraken21 3 years, 2 months ago

Selected Answer: DE

Scaling should be at the ASG not ALB. So, not sure about "Use AWS Auto Scaling to adjust the ALB capacity based on request rate"

upvoted 6 times

channn 3 years, 2 months ago

Selected Answer: AD

A. Use AWS Auto Scaling to adjust the ALB capacity based on request rate: This will allow the system to scale up or down based on incoming traffic demand. The solutions architect should use AWS Auto Scaling to monitor the request rate and adjust the ALB capacity as needed.

D. Use a target tracking scaling policy to scale the Auto Scaling group based on instance CPU utilization: This will allow the system to scale up or down based on the CPU utilization of the EC2 instances in the Auto Scaling group. The solutions architect should use a target tracking scaling policy to maintain a specific CPU utilization target and adjust the number of EC2 instances in the Auto Scaling group accordingly.

upvoted 9 times

pentium75 2 years, 5 months ago

Auto scaling for ALB capacity?

upvoted 3 times

neosis91 3 years, 2 months ago

Selected Answer: AD

A. Use a target tracking scaling policy to scale the Auto Scaling group based on instance CPU utilization. This approach allows the Auto Scaling group to automatically adjust the number of instances based on the specified metric, ensuring that the system can scale to meet demand during working hours.

D. Use scheduled scaling to change the Auto Scaling group minimum, maximum, and desired capacity to zero for weekends. Revert to the default values at the start of the week. This approach allows the Auto Scaling group to reduce the number of instances to zero during weekends when traffic is expected to be low. It will help the organization to save costs by not paying for instances that are not needed during weekends.

Therefore, options A and D are the correct answers. Options B and C are not relevant to the scenario, and option E is not a scalable solution as it would require manual intervention to adjust the group capacity every week.

upvoted 1 times

zooba72 3 years, 2 months ago

Selected Answer: DE

This is why I don't believe A is correct use auto scaling to adjust the ALB D&E

upvoted 4 times

pentium75 2 years, 5 months ago

Autoscaling can't scale the ALB

upvoted 2 times

A solutions architect is designing a two-tiered architecture that includes a public subnet and a database subnet. The web servers in the public subnet must be open to the internet on port 443. The Amazon RDS for MySQL DB instance in the database subnet must be accessible only to the web servers on port 3306.

Which combination of steps should the solutions architect take to meet these requirements? (Choose two.)

- A. Create a network ACL for the public subnet. Add a rule to deny outbound traffic to 0.0.0.0/0 on port 3306.
- B. Create a security group for the DB instance. Add a rule to allow traffic from the public subnet CIDR block on port 3306.
- C. Create a security group for the web servers in the public subnet. Add a rule to allow traffic from 0.0.0.0/0 on port 443.
Most Voted
- D. Create a security group for the DB instance. Add a rule to allow traffic from the web servers' security group on port 3306.
Most Voted
- E. Create a security group for the DB instance. Add a rule to deny all traffic except traffic from the web servers' security group on port 3306.

Correct Answer: CD

Community vote distribution

CD (100%)

Comments

Guru4Cloud **Highly Voted** 2 years, 3 months ago

Selected Answer: CD

Remember guys that SG is not used for Deny action, just Allow
upvoted 11 times

datmd77 **Highly Voted** 2 years, 7 months ago

Selected Answer: CD

Remember guys that SG is not used for Deny action, just Allow
upvoted 5 times

Tendu **Most Recent** 1 year, 3 months ago

Selected Answer: CD

C and D
upvoted 1 times

waldirlsantos 1 year, 8 months ago

Selected Answer: CD

The following are the default rules for a security group that you create:

Allows no inbound traffic

Allows all outbound traffic

upvoted 3 times

TariqKipkemei 2 years, 1 month ago

Selected Answer: CD

'must be accessible only to the web servers' is the key here.
Option B almost threw me off, but with this then all that exists in the public subnet would be able to access the DB security group.
Therefore C,D well applies the principle of least privilege.
upvoted 5 times

Buruguduystunstugudunstuy 2 years, 8 months ago

Selected Answer: CD

To meet the requirements of allowing access to the web servers in the public subnet on port 443 and the Amazon RDS for MySQL DB instance in the database subnet on port 3306, the best solution would be to create a security group for the web servers and another security group for the DB instance, and then define the appropriate inbound and outbound rules for each security group.

1. Create a security group for the web servers in the public subnet. Add a rule to allow traffic from 0.0.0.0/0 on port 443.
2. Create a security group for the DB instance. Add a rule to allow traffic from the web servers' security group on port 3306.

This will allow the web servers in the public subnet to receive traffic from the internet on port 443, and the Amazon RDS for MySQL DB instance in the database subnet to receive traffic only from the web servers on port 3306.

upvoted 3 times

kampatra 2 years, 8 months ago

Selected Answer: CD

CD - Correct ans.

upvoted 3 times

Eden 2 years, 9 months ago

I choose CE

upvoted 1 times

lili_9 2 years, 9 months ago

CE support @sitha

upvoted 1 times

sitha 2 years, 9 months ago

Answer: CE . The solution is to deny accessing DB from Internet and allow only access from webserver.

upvoted 1 times

KAUS2 2 years, 9 months ago

Selected Answer: CD

C & D are the right choices. correct

upvoted 2 times

KS2020 2 years, 9 months ago

why not CE?

upvoted 2 times

kampatra 2 years, 8 months ago

By default Security Group deny all traffic and we need to configure to enable.

upvoted 5 times

A company is implementing a shared storage solution for a gaming application that is hosted in the AWS Cloud. The company needs the ability to use Lustre clients to access data. The solution must be fully managed.

Which solution meets these requirements?

- A. Create an AWS DataSync task that shares the data as a mountable file system. Mount the file system to the application server.
- B. Create an AWS Storage Gateway file gateway. Create a file share that uses the required client protocol. Connect the application server to the file share.
- C. Create an Amazon Elastic File System (Amazon EFS) file system, and configure it to support Lustre. Attach the file system to the origin server. Connect the application server to the file system.
- D. Create an Amazon FSx for Lustre file system. Attach the file system to the origin server. Connect the application server to the file system. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 2 years, 2 months ago

Selected Answer: D

To meet the requirements of a shared storage solution for a gaming application that can be accessed using Lustre clients and is fully managed, the best solution would be to use Amazon FSx for Lustre.

Amazon FSx for Lustre is a fully managed file system that is optimized for compute-intensive workloads, such as high-performance computing, machine learning, and gaming. It provides a POSIX-compliant file system that can be accessed using Lustre clients and offers high performance, scalability, and data durability.

This solution provides a highly available, scalable, and fully managed shared storage solution that can be accessed using Lustre clients. Amazon FSx for Lustre is optimized for compute-intensive workloads and provides high performance and durability.

upvoted 6 times

Buruguduystunstugudunstuy 2 years, 2 months ago

Answer A, creating an AWS DataSync task that shares the data as a mountable file system and mounting the file system to the application server, may not provide the required performance and scalability for a gaming application.

Answer B, creating an AWS Storage Gateway file gateway and connecting the application server to the file share, may not provide the required performance and scalability for a gaming application.

Answer C, creating an Amazon Elastic File System (Amazon EFS) file system and configuring it to support Lustre, may not provide the required performance and scalability for a gaming application and may require additional configuration and management overhead.

upvoted 3 times

Manimgh **Most Recent** 1 year, 1 month ago

Selected Answer: D

Lustre = FSx

upvoted 1 times

Tendu 1 year, 3 months ago

Selected Answer: D

D is correct answer.

upvoted 1 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: D

Lustre clients = Amazon FSx for Lustre file system
upvoted 4 times

TariqKipkemei 2 years ago

Selected Answer: D

Lustre clients = Amazon FSx for Lustre file system
upvoted 4 times

kampatra 2 years, 2 months ago

Selected Answer: D

D - correct ans
upvoted 2 times

kprakashbehera 2 years, 3 months ago

Selected Answer: D

FSx for Lustre
DDDDDD
upvoted 2 times

KAUS2 2 years, 3 months ago

Selected Answer: D

Amazon FSx for Lustre is the right answer
• Lustre is a type of parallel distributed file system, for large-scale computing, Machine Learning, High Performance Computing (HPC)
• Video Processing, Financial Modeling, Electronic Design Automatio
upvoted 2 times

cegama543 2 years, 3 months ago

Selected Answer: D

Option D is the best solution because Amazon FSx for Lustre is a fully managed, high-performance file system that is designed to support compute-intensive workloads, such as those required by gaming applications. FSx for Lustre provides sub-millisecond access to petabyte-scale file systems, and supports Lustre clients natively. This means that the gaming application can access the shared data directly from the FSx for Lustre file system without the need for additional configuration or setup.

Additionally, FSx for Lustre is a fully managed service, meaning that AWS takes care of all maintenance, updates, and patches for the file system, which reduces the operational overhead required by the company.
upvoted 2 times

A company runs an application that receives data from thousands of geographically dispersed remote devices that use UDP. The application processes the data immediately and sends a message back to the device if necessary. No data is stored.

The company needs a solution that minimizes latency for the data transmission from the devices. The solution also must provide rapid failover to another AWS Region.

Which solution will meet these requirements?

A. Configure an Amazon Route 53 failover routing policy. Create a Network Load Balancer (NLB) in each of the two Regions. Configure the NLB to invoke an AWS Lambda function to process the data.

B. Use AWS Global Accelerator. Create a Network Load Balancer (NLB) in each of the two Regions as an endpoint. Create an Amazon Elastic Container Service (Amazon ECS) cluster with the Fargate launch type. Create an ECS service on the cluster. Set the ECS service as the target for the NLB. Process the data in Amazon ECS. **Most Voted**

C. Use AWS Global Accelerator. Create an Application Load Balancer (ALB) in each of the two Regions as an endpoint. Create an Amazon Elastic Container Service (Amazon ECS) cluster with the Fargate launch type. Create an ECS service on the cluster. Set the ECS service as the target for the ALB. Process the data in Amazon ECS.

D. Configure an Amazon Route 53 failover routing policy. Create an Application Load Balancer (ALB) in each of the two Regions. Create an Amazon Elastic Container Service (Amazon ECS) cluster with the Fargate launch type. Create an ECS service on the cluster. Set the ECS service as the target for the ALB. Process the data in Amazon ECS.

Correct Answer: B

Community vote distribution

B (100%)

Comments

UnluckyDucky **Highly Voted** 2 years, 9 months ago

Selected Answer: B

Key words: geographically dispersed, UDP.

Geographically dispersed (related to UDP) - Global Accelerator - multiple entrances worldwide to the AWS network to provide better transfer rates.
UDP - NLB (Network Load Balancer).

upvoted 15 times

Ruhi02 **Highly Voted** 2 years, 9 months ago

Answer should be B.. there is typo mistake in B. Correct Answer is : Use AWS Global Accelerator. Create a Network Load Balancer (NLB) in each of the two Regions as an endpoint. Create an Amazon Elastic Container Service (Amazon ECS) cluster with the Fargate launch type. Create an ECS service on the cluster. Set the ECS service as the target for the NLB. Process the data in Amazon ECS.

upvoted 5 times

wizcloudifa **Most Recent** 1 year, 7 months ago

Selected Answer: B

if its UDP it has to be Global Accelerator + NLB package, plus it has the provision for rapid failover as well, piece of cake.

upvoted 3 times

sandordini 1 year, 7 months ago

Selected Answer: B

UDP: NLB + AWS Global Accelerator

upvoted 4 times

zinabu 1 year, 8 months ago

UDP/TCP=NLB
rapid failover= AWS global accelerator
upvoted 3 times

ferdzcruz 1 year, 11 months ago
devices that use UDP = NLB
upvoted 3 times

ferdzcruz 1 year, 11 months ago
minimizes latency = AWS Global Accelerator
upvoted 3 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: B

This option meets the requirements:

Global Accelerator provides UDP support and minimizes latency using the AWS global network.
Using NLBs allows the UDP traffic to be load balanced across Availability Zones.
ECS Fargate provides rapid scaling and failover across Regions.
NLB endpoints allow rapid failover if one Region goes down.

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: B

UDP = AWS Global Accelerator and Network Load Balancer
upvoted 2 times

kraken21 2 years, 8 months ago

Selected Answer: B

Global accelerator for multi region automatic failover. NLB for UDP.
upvoted 3 times

MaxMa 2 years, 8 months ago

why not A?
upvoted 1 times

kraken21 2 years, 8 months ago

NLBs do not support lambda target type. Tricky!!! <https://docs.aws.amazon.com/elasticloadbalancing/latest/network/load-balancer-target-groups.html>
upvoted 10 times

Buruguduystunstugudunstuy 2 years, 8 months ago

Selected Answer: B

To meet the requirements of minimizing latency for data transmission from the devices and providing rapid failover to another AWS Region, the best solution would be to use AWS Global Accelerator in combination with a Network Load Balancer (NLB) and Amazon Elastic Container Service (Amazon ECS).

AWS Global Accelerator is a service that improves the availability and performance of applications by using static IP addresses (Anycast) to route traffic to optimal AWS endpoints. With Global Accelerator, you can direct traffic to multiple Regions and endpoints, and provide automatic failover to another AWS Region.
upvoted 4 times

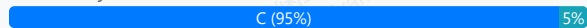
A solutions architect must migrate a Windows Internet Information Services (IIS) web application to AWS. The application currently relies on a file share hosted in the user's on-premises network-attached storage (NAS). The solutions architect has proposed migrating the IIS web servers to Amazon EC2 instances in multiple Availability Zones that are connected to the storage solution, and configuring an Elastic Load Balancer attached to the instances.

Which replacement to the on-premises file share is MOST resilient and durable?

- A. Migrate the file share to Amazon RDS.
- B. Migrate the file share to AWS Storage Gateway.
- C. Migrate the file share to Amazon FSx for Windows File Server. **Most Voted**
- D. Migrate the file share to Amazon Elastic File System (Amazon EFS).

Correct Answer: C

Community vote distribution



Comments

channn **Highly Voted** 2 years, 8 months ago

Selected Answer: C

- A) RDS is a database service
 - B) Storage Gateway is a hybrid cloud storage service that connects on-premises applications to AWS storage services.
 - D) provides shared file storage for Linux-based workloads, but it does not natively support Windows-based workloads.
- upvoted 9 times

UnluckyDucky **Highly Voted** 2 years, 9 months ago

Selected Answer: C

FSx for Windows - Windows.
EFS - Linux.

upvoted 7 times

mwwt2022 1 year, 11 months ago

good summary
upvoted 3 times

Manimgh **Most Recent** 1 year, 1 month ago

Selected Answer: C

IIS = FSx
upvoted 1 times

com7 2 years ago

Selected Answer: C

Windows Server to FSx For Windows
upvoted 3 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: C

Windows client = Amazon FSx for Windows File Serve
upvoted 3 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: C

Windows client = Amazon FSx for Windows File Server

upvoted 4 times

Buruguduystunstugudunstuy 2 years, 8 months ago

Selected Answer: C

The most resilient and durable replacement for the on-premises file share in this scenario would be Amazon FSx for Windows File Server.

Amazon FSx is a fully managed Windows file system service that is built on Windows Server and provides native support for the SMB protocol. It is designed to be highly available and durable, with built-in backup and restore capabilities. It is also fully integrated with AWS security services, providing encryption at rest and in transit, and it can be configured to meet compliance standards.

upvoted 7 times

Buruguduystunstugudunstuy 2 years, 8 months ago

Migrating the file share to Amazon RDS or AWS Storage Gateway is not appropriate as these services are designed for database workloads and block storage respectively, and do not provide native support for the SMB protocol.

Migrating the file share to Amazon EFS (Linux ONLY) could be an option, but Amazon FSx for Windows File Server would be more appropriate in this case because it is specifically designed for Windows file shares and provides better performance for Windows applications.

upvoted 6 times

Grace83 2 years, 8 months ago

Obviously C is the correct answer - FSx for Windows - Windows

upvoted 5 times

elearningtakai 2 years, 9 months ago

Selected Answer: D

Amazon EFS is a scalable and fully-managed file storage service that is designed to provide high availability and durability. It can be accessed by multiple EC2 instances across multiple Availability Zones simultaneously. Additionally, it offers automatic and instantaneous data replication across different availability zones within a region, which makes it resilient to failures.

upvoted 2 times

asoli 2 years, 8 months ago

EFS is a wrong choice because it can only work with Linux instances. That application has a Windows web server, so its OS is Windows and EFS cannot connect to it

upvoted 5 times

sitha 2 years, 9 months ago

Amazon FSx makes it easy and cost effective to launch, run, and scale feature-rich, high-performance file systems in the cloud.

Answer : C

upvoted 2 times

KAUS2 2 years, 9 months ago

Selected Answer: C

FSx for Windows is a fully managed Windows file system share drive. Hence C is the correct answer.

upvoted 3 times

Ruhi02 2 years, 9 months ago

FSx for Windows is ideal in this case. So answer is C.

upvoted 2 times

A company is deploying a new application on Amazon EC2 instances. The application writes data to Amazon Elastic Block Store (Amazon EBS) volumes. The company needs to ensure that all data that is written to the EBS volumes is encrypted at rest.

Which solution will meet this requirement?

- A. Create an IAM role that specifies EBS encryption. Attach the role to the EC2 instances.
- B. Create the EBS volumes as encrypted volumes. Attach the EBS volumes to the EC2 instances. **Most Voted**
- C. Create an EC2 instance tag that has a key of Encrypt and a value of True. Tag all instances that require encryption at the EBS level.
- D. Create an AWS Key Management Service (AWS KMS) key policy that enforces EBS encryption in the account. Ensure that the key policy is active.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 2 years, 2 months ago

Selected Answer: B

The solution that will meet the requirement of ensuring that all data that is written to the EBS volumes is encrypted at rest is B. Create the EBS volumes as encrypted volumes and attach the encrypted EBS volumes to the EC2 instances.

When you create an EBS volume, you can specify whether to encrypt the volume. If you choose to encrypt the volume, all data written to the volume is automatically encrypted at rest using AWS-managed keys. You can also use customer-managed keys (CMKs) stored in AWS KMS to encrypt and protect your EBS volumes. You can create encrypted EBS volumes and attach them to EC2 instances to ensure that all data written to the volumes is encrypted at rest.

Answer A is incorrect because attaching an IAM role to the EC2 instances does not automatically encrypt the EBS volumes.

Answer C is incorrect because adding an EC2 instance tag does not ensure that the EBS volumes are encrypted.

upvoted 12 times

Kds53829 **Most Recent** 1 year, 7 months ago

B is the answer

upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: B

B. Create the EBS volumes as encrypted volumes. Attach the EBS volumes to the EC2 instances.

upvoted 2 times

TariqKipkemei 2 years ago

Selected Answer: B

Windows client = Amazon FSx for Windows File Server

upvoted 2 times

TariqKipkemei 1 year, 7 months ago

ignore this, mind stuck on last question hhhhhh.

Just create the EBS volumes as encrypted volumes then attach the EBS volumes to the EC2 instances.

upvoted 5 times

elearningtakai 2 years, 2 months ago

Selected Answer: B

The other options either do not meet the requirement of encrypting data at rest (A and C) or do so in a more complex or less efficient manner (D).
upvoted 2 times

Bofi 2 years, 2 months ago

Why not D, EBS encryption require the use of KMS key
upvoted 1 times

Buruguduystunstugudunstuy 2 years, 2 months ago

Answer D is incorrect because creating a KMS key policy that enforces EBS encryption does not automatically encrypt EBS volumes. You need to create encrypted EBS volumes and attach them to EC2 instances to ensure that all data written to the volumes are encrypted at rest.
upvoted 10 times

WheracanIstart 2 years, 2 months ago

Selected Answer: B

Create encrypted EBS volumes and attach encrypted EBS volumes to EC2 instances..
upvoted 3 times

sitha 2 years, 3 months ago

Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances. Select KMS Keys either default or custom
upvoted 2 times

Ruhi02 2 years, 3 months ago

Answer B. You can enable encryption for EBS volumes while creating them.
upvoted 2 times

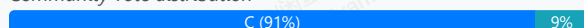
A company has a web application with sporadic usage patterns. There is heavy usage at the beginning of each month, moderate usage at the start of each week, and unpredictable usage during the week. The application consists of a web server and a MySQL database server running inside the data center. The company would like to move the application to the AWS Cloud, and needs to select a cost-effective database platform that will not require database modifications.

Which solution will meet these requirements?

- A. Amazon DynamoDB
- B. Amazon RDS for MySQL
- C. MySQL-compatible Amazon Aurora Serverless **Most Voted**
- D. MySQL deployed on Amazon EC2 in an Auto Scaling group

Correct Answer: C

Community vote distribution



Comments

channn **Highly Voted** 2 years, 8 months ago

Selected Answer: C

C: Aurora Serverless is a MySQL-compatible relational database engine that automatically scales compute and memory resources based on application usage. no upfront costs or commitments required.

A: DynamoDB is a NoSQL

B: Fixed cost on RDS class

D: More operation requires

upvoted 12 times

Buruguduystunstugudunstuy **Highly Voted** 2 years, 8 months ago

Selected Answer: C

Answer C, MySQL-compatible Amazon Aurora Serverless, would be the best solution to meet the company's requirements.

Aurora Serverless can be a cost-effective option for databases with sporadic or unpredictable usage patterns since it automatically scales up or down based on the current workload. Additionally, Aurora Serverless is compatible with MySQL, so it does not require any modifications to the application's database code.

upvoted 6 times

TariqKipkemei **Most Recent** 2 years, 1 month ago

Selected Answer: C

There is a huge demand for auto-scaling which Amazon RDS cannot do. This contributes to the cost savings as Aurora serverless would scale down in low peak times, this contributes to low costs.

upvoted 4 times

JKevin778 2 years, 2 months ago

Selected Answer: B

RDS is cheaper than Aurora.

upvoted 1 times

pentium75 1 year, 11 months ago

RDS is cheaper than Aurora if you have a fixed instance size, but NOT if you have "unpredictable" usage patterns, then Aurora Serverless (!) is cheaper.

upvoted 7 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: C

Answer C, MySQL-compatible Amazon Aurora Serverless, would be the best solution to meet the company's requirements.

upvoted 2 times

MrAWSAssociate 2 years, 5 months ago

Selected Answer: C

Since we have sporadic & unpredictable usage for DB, Aurora Serverless would be fit more cost-efficient for this case scenario than RDS MySQL.

<https://www.techtarget.com/searchcloudcomputing/answer/When-should-I-use-Amazon-RDS-vs-Aurora-Serverless>

upvoted 1 times

antropaws 2 years, 6 months ago

Selected Answer: C

C for sure.

upvoted 3 times

klayytech 2 years, 8 months ago

Selected Answer: B

Amazon RDS for MySQL is a cost-effective database platform that will not require database modifications. It makes it easier to set up, operate, and scale MySQL deployments in the cloud. With Amazon RDS, you can deploy scalable MySQL servers in minutes with cost-efficient and resizable hardware capacity².

Amazon DynamoDB is a fully managed NoSQL database service that provides fast and predictable performance with seamless scalability. DynamoDB is a good choice for applications that require low-latency data access¹.

MySQL-compatible Amazon Aurora Serverless is an on-demand, auto-scaling configuration for Amazon Aurora (MySQL-compatible edition), where the database will automatically start up, shut down, and scale capacity up or down based on your application's needs³.

So, Amazon RDS for MySQL is the best option for your requirements.

upvoted 2 times

klayytech 2 years, 8 months ago

sorry i will change to C , because

Amazon RDS for MySQL is a fully-managed relational database service that makes it easy to set up, operate, and scale MySQL deployments in the cloud. Amazon Aurora Serverless is an on-demand, auto-scaling configuration for Amazon Aurora (MySQL-compatible edition), where the database will automatically start up, shut down, and scale capacity up or down based on your application's needs. It is a simple, cost-effective option for infrequent, intermittent, or unpredictable workloads.

upvoted 3 times

boxu03 2 years, 8 months ago

Selected Answer: C

Amazon Aurora Serverless : a simple, cost-effective option for infrequent, intermittent, or unpredictable workloads

upvoted 4 times

An image-hosting company stores its objects in Amazon S3 buckets. The company wants to avoid accidental exposure of the objects in the S3 buckets to the public. All S3 objects in the entire AWS account need to remain private.

Which solution will meet these requirements?

- A. Use Amazon GuardDuty to monitor S3 bucket policies. Create an automatic remediation action rule that uses an AWS Lambda function to remediate any change that makes the objects public.
- B. Use AWS Trusted Advisor to find publicly accessible S3 buckets. Configure email notifications in Trusted Advisor when a change is detected. Manually change the S3 bucket policy if it allows public access.
- C. Use AWS Resource Access Manager to find publicly accessible S3 buckets. Use Amazon Simple Notification Service (Amazon SNS) to invoke an AWS Lambda function when a change is detected. Deploy a Lambda function that programmatically remediates the change.
- D. Use the S3 Block Public Access feature on the account level. Use AWS Organizations to create a service control policy (SCP) that prevents IAM users from changing the setting. Apply the SCP to the account. **Most Voted**

Correct Answer: D

Community vote distribution

D (96%)

4%

Comments

Ruhi02 **Highly Voted** 3 years, 3 months ago

Answer is D ladies and gentlemen. While guard duty helps to monitor s3 for potential threats its a reactive action. We should always be proactive and not reactive in our solutions so D, block public access to avoid any possibility of the info becoming publicly accessible
upvoted 23 times

Buruguduystunstugudunstuy **Highly Voted** 3 years, 2 months ago

Selected Answer: D

Answer D is the correct solution that meets the requirements. The S3 Block Public Access feature allows you to restrict public access to S3 buckets and objects within the account. You can enable this feature at the account level to prevent any S3 bucket from being made public, regardless of the bucket policy settings. AWS Organizations can be used to apply a Service Control Policy (SCP) to the account to prevent IAM users from changing this setting, ensuring that all S3 objects remain private. This is a straightforward and effective solution that requires minimal operational overhead.
upvoted 11 times

noircesar25 **Most Recent** 2 years, 3 months ago

its 1 aws account, how could D be the answer?
upvoted 1 times

JA2018 1 year, 6 months ago

make no difference, 1 AWS account, 10 AWS accounts or > 100 AWS accounts, SCP enforces the same guard rails
upvoted 2 times

TariqKipkemei 2 years, 7 months ago

Selected Answer: D

Use the S3 Block Public Access feature on the account level. Use AWS Organizations to create a service control policy (SCP) that prevents IAM users from changing the setting. Apply the SCP to the account
upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

Use the S3 Block Public Access feature on the account level. Use AWS Organizations to create a service control policy (SCP) that prevents IAM users from changing the setting. Apply the SCP to the account
upvoted 2 times

MrAWSAssociate 2 years, 11 months ago

Selected Answer: A

A is correct!

upvoted 1 times

pentium75 2 years, 5 months ago

No, first it would not remove any existing public access (only detect changes), second it would just detect and then remediate, but in the meantime someone could access the objects. It's clearly D.

upvoted 3 times

Yadav_Sanjay 3 years ago

Selected Answer: D

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/access-control-block-public-access.html>

upvoted 4 times

elearningtakai 3 years, 2 months ago

Selected Answer: D

This is the most effective solution to meet the requirements.

upvoted 3 times

Bofi 3 years, 2 months ago

Selected Answer: D

Option D provided real solution by using bucket policy to restrict public access. Other options were focus on detection which wasn't what was been asked

upvoted 3 times

An ecommerce company is experiencing an increase in user traffic. The company's store is deployed on Amazon EC2 instances as a two-tier web application consisting of a web tier and a separate database tier. As traffic increases, the company notices that the architecture is causing significant delays in sending timely marketing and order confirmation email to users. The company wants to reduce the time it spends resolving complex email delivery issues and minimize operational overhead.

What should a solutions architect do to meet these requirements?

- A. Create a separate application tier using EC2 instances dedicated to email processing.
- B. Configure the web instance to send email through Amazon Simple Email Service (Amazon SES). **Most Voted**
- C. Configure the web instance to send email through Amazon Simple Notification Service (Amazon SNS).
- D. Create a separate application tier using EC2 instances dedicated to email processing. Place the instances in an Auto Scaling group.

Correct Answer: B

Community vote distribution

B (100%)

Comments

elearningtakai **Highly Voted** 2 years, 8 months ago

Selected Answer: B

Amazon SES is a cost-effective and scalable email service that enables businesses to send and receive email using their own email addresses and domains. Configuring the web instance to send email through Amazon SES is a simple and effective solution that can reduce the time spent resolving complex email delivery issues and minimize operational overhead.

upvoted 11 times

Buruguduystunstugudunstuy **Highly Voted** 2 years, 8 months ago

Selected Answer: B

The best option for addressing the company's needs of minimizing operational overhead and reducing time spent resolving email delivery issues is to use Amazon Simple Email Service (Amazon SES).

Answer A of creating a separate application tier for email processing may add additional complexity to the architecture and require more operational overhead.

Answer C of using Amazon Simple Notification Service (Amazon SNS) is not an appropriate solution for sending marketing and order confirmation emails since Amazon SNS is a messaging service that is designed to send messages to subscribed endpoints or clients.

Answer D of creating a separate application tier using EC2 instances dedicated to email processing placed in an Auto Scaling group is a more complex solution than necessary and may result in additional operational overhead.

upvoted 6 times

waldirIsantos **Most Recent** 1 year, 8 months ago

Selected Answer: B

B meet these requirements

upvoted 2 times

TariqKipkemei 2 years, 1 month ago

Selected Answer: B

Amazon Simple Email Service (Amazon SES) lets you reach customers confidently without an on-premises Simple Mail Transfer Protocol (SMTP) email server using the Amazon SES API or SMTP interface.

upvoted 3 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: B

B. Configure the web instance to send email through Amazon Simple Email Service (Amazon SES)

upvoted 2 times

nileshlg 2 years, 8 months ago

Answer is B

upvoted 2 times

Ruhi02 2 years, 9 months ago

Answer B.. SES is meant for sending high volume e-mail efficiently and securely.

SNS is meant as a channel publisher/subscriber service

upvoted 5 times

A company has a business system that generates hundreds of reports each day. The business system saves the reports to a network share in CSV format. The company needs to store this data in the AWS Cloud in near-real time for analysis.

Which solution will meet these requirements with the LEAST administrative overhead?

- A. Use AWS DataSync to transfer the files to Amazon S3. Create a scheduled task that runs at the end of each day.
- B. Create an Amazon S3 File Gateway. Update the business system to use a new network share from the S3 File Gateway.
Most Voted
- C. Use AWS DataSync to transfer the files to Amazon S3. Create an application that uses the DataSync API in the automation workflow.
- D. Deploy an AWS Transfer for SFTP endpoint. Create a script that checks for new files on the network share and uploads the new files by using SFTP.

Correct Answer: B

Community vote distribution

B (87%) 11%

Comments

TariqKipkemei **Highly Voted** 2 years, 1 month ago

Selected Answer: B

Both Amazon S3 File Gateway and AWS DataSync are suitable for this scenario.

But there is a requirement for 'LEAST administrative overhead'.

Option C involves the creation of an entirely new application to consume the DataSync API, this rules out this option.

upvoted 16 times

channn **Highly Voted** 2 years, 8 months ago

Selected Answer: B

Key words:

1. near-real-time (A is out)

2. LEAST administrative (C n D is out)

upvoted 11 times

zdi561 **Most Recent** 1 year, 4 months ago

Selected Answer: D

Firstly the requirement is not about hybrid cloud e.g. use the data migrated to aws from on-premise, so B is gone. Secondly it is near real time, A is gone. Thirdly less administratively, C is one since an application is a headache. D is the best, it only involves a script. SFTP is good for light file transfer. Here there are only hundreds of files in a day.

upvoted 1 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: C

This option has the least administrative overhead because:

Using DataSync avoids having to rewrite the business system to use a new file gateway or SFTP endpoint.

Calling the DataSync API from an application allows automating the data transfer instead of running scheduled tasks or scripts.

DataSync directly transfers files from the network share to S3 without needing an intermediate server

upvoted 2 times

pentium75 1 year, 11 months ago

"Create an application" hell no, the application must run somewhere etc., this is massive "administrative overhead".

upvoted 5 times

antropaws 2 years, 6 months ago

Selected Answer: B

B. Data Sync is better for one time migrations.
upvoted 3 times

kruasan 2 years, 7 months ago

Selected Answer: B

The correct solution here is:

B. Create an Amazon S3 File Gateway. Update the business system to use a new network share from the S3 File Gateway.

This option requires the least administrative overhead because:

- It presents a simple network file share interface that the business system can write to, just like a standard network share. This requires minimal changes to the business system.
 - The S3 File Gateway automatically uploads all files written to the share to an S3 bucket in the background. This handles the transfer and upload to S3 without requiring any scheduled tasks, scripts or automation.
 - All ongoing management like monitoring, scaling, patching etc. is handled by AWS for the S3 File Gateway.
- upvoted 5 times

kruasan 2 years, 7 months ago

The other options would require more ongoing administrative effort:

- A) AWS DataSync would require creating and managing scheduled tasks and monitoring them.
- C) Using the DataSync API would require developing an application and then managing and monitoring it.
- D) The SFTP option would require creating scripts, managing SFTP access and keys, and monitoring the file transfer process.

So overall, the S3 File Gateway requires the least amount of ongoing management and administration as it presents a simple file share interface but handles the upload to S3 in a fully managed fashion. The business system can continue writing to a network share as is, while the files are transparently uploaded to S3.

The S3 File Gateway is the most hands-off, low-maintenance solution in this scenario.
upvoted 4 times

elarningtakai 2 years, 8 months ago

Selected Answer: B

- A - creating a scheduled task is not near-real time.
 - B - The S3 File Gateway caches frequently accessed data locally and automatically uploads it to Amazon S3, providing near-real-time access to the data.
 - C - creating an application that uses the DataSync API in the automation workflow may provide near-real-time data access, but it requires additional development effort.
 - D - it requires additional development effort.
- upvoted 5 times

zooba72 2 years, 8 months ago

Selected Answer: B

It's B. DataSync has a scheduler and it runs on hour intervals, it cannot be used real-time
upvoted 2 times

Buruguduystunstugudunstuy 2 years, 8 months ago

Selected Answer: C

The correct answer is C. Use AWS DataSync to transfer the files to Amazon S3. Create an application that uses the DataSync API in the automation workflow.

To store the CSV reports generated by the business system in the AWS Cloud in near-real time for analysis, the best solution with the least administrative overhead would be to use AWS DataSync to transfer the files to Amazon S3 and create an application that uses the DataSync API in the automation workflow.

AWS DataSync is a fully managed service that makes it easy to automate and accelerate data transfer between on-premises storage systems and AWS Cloud storage, such as Amazon S3. With DataSync, you can quickly and securely transfer large amounts of data to the AWS Cloud, and you can automate the transfer process using the DataSync API.
upvoted 4 times

Buruguduystunstugudunstuy 2 years, 8 months ago

Answer A, using AWS DataSync to transfer the files to Amazon S3 and creating a scheduled task that runs at the end of each day, is not the best solution because it does not meet the requirement of storing the CSV reports in near-real time for analysis.

Answer B, creating an Amazon S3 File Gateway and updating the business system to use a new network share from the S3 File Gateway, is not the best solution because it requires additional configuration and management overhead.

Answer D, deploying an AWS Transfer for the SFTP endpoint and creating a script to check for new files on the network share and upload the new files using SFTP, is not the best solution because it requires additional scripting and management overhead

upvoted 3 times

COTIT 2 years, 8 months ago

Selected Answer: B

I think B is the better answer, "LEAST administrative overhead"

https://aws.amazon.com/storagegateway/file/?nc1=h_ls

upvoted 5 times

andyto 2 years, 8 months ago

B - S3 File Gateway.

C - this is wrong answer because data migration is scheduled (this is not continuous task), so condition "near-real time" is not fulfilled

upvoted 4 times

Thief 2 years, 8 months ago

C is the best ans

upvoted 1 times

lizzard812 2 years, 8 months ago

Why not A? There is no scheduled job?

upvoted 2 times

A company is storing petabytes of data in Amazon S3 Standard. The data is stored in multiple S3 buckets and is accessed with varying frequency. The company does not know access patterns for all the data. The company needs to implement a solution for each S3 bucket to optimize the cost of S3 usage.

Which solution will meet these requirements with the MOST operational efficiency?

A. Create an S3 Lifecycle configuration with a rule to transition the objects in the S3 bucket to S3 Intelligent-Tiering.

Most Voted

B. Use the S3 storage class analysis tool to determine the correct tier for each object in the S3 bucket. Move each object to the identified storage tier.

C. Create an S3 Lifecycle configuration with a rule to transition the objects in the S3 bucket to S3 Glacier Instant Retrieval.

D. Create an S3 Lifecycle configuration with a rule to transition the objects in the S3 bucket to S3 One Zone-Infrequent Access (S3 One Zone-IA).

Correct Answer: A

Community vote distribution

A (100%)

Comments

TariqKipkemei **Highly Voted** 2 years ago

Selected Answer: A

Unknown access patterns for the data = S3 Intelligent-Tiering
upvoted 8 times

channn **Highly Voted** 2 years, 2 months ago

Selected Answer: A

Key words: 'The company does not know access patterns for all the data', so A.
upvoted 5 times

Guru4Cloud **Most Recent** 1 year, 9 months ago

Selected Answer: A

Create an S3 Lifecycle configuration with a rule to transition the objects in the S3 bucket to S3 Intelligent-Tiering.
upvoted 3 times

Buruguduystunstugudunstuy 2 years, 2 months ago

Selected Answer: A

The correct answer is A.

Creating an S3 Lifecycle configuration with a rule to transition the objects in the S3 bucket to S3 Intelligent-Tiering would be the most efficient solution to optimize the cost of S3 usage. S3 Intelligent-Tiering is a storage class that automatically moves objects between two access tiers (frequent and infrequent) based on changing access patterns. It is a cost-effective solution that does not require any manual intervention to move data to different storage classes, unlike the other options.

upvoted 5 times

Buruguduystunstugudunstuy 2 years, 2 months ago

Answer B, Using the S3 storage class analysis tool to determine the correct tier for each object and manually moving objects to the identified storage tier would be time-consuming and require more operational overhead.

Answer C, Transitioning objects to S3 Glacier Instant Retrieval would be appropriate for data that is accessed less frequently and does not require immediate access.

Answer D, S3 One Zone-IA would be appropriate for data that can be recreated if lost and does not require the durability of S3 Standard or S3 Standard-IA.

upvoted 3 times

COTIT 2 years, 2 months ago

Selected Answer: A

For me is A. Create an S3 Lifecycle configuration with a rule to transition the objects in the S3 bucket to S3 Intelligent-Tiering.

Why?

"S3 Intelligent-Tiering is the ideal storage class for data with unknown, changing, or unpredictable access patterns"

<https://aws.amazon.com/s3/storage-classes/intelligent-tiering/>

upvoted 3 times

Bofi 2 years, 2 months ago

Selected Answer: A

Once the data traffic is unpredictable, Intelligent-Tiering is the best option

upvoted 3 times

NIL8891 2 years, 2 months ago

Selected Answer: A

Create an S3 Lifecycle configuration with a rule to transition the objects in the S3 bucket to S3 Intelligent-Tiering.

upvoted 2 times

Maximus007 2 years, 2 months ago

Selected Answer: A

A: as exact pattern is not clear

upvoted 3 times

A rapidly growing global ecommerce company is hosting its web application on AWS. The web application includes static content and dynamic content. The website stores online transaction processing (OLTP) data in an Amazon RDS database. The website's users are experiencing slow page loads.

Which combination of actions should a solutions architect take to resolve this issue? (Choose two.)

- A. Configure an Amazon Redshift cluster.
- B. Set up an Amazon CloudFront distribution. **Most Voted**
- C. Host the dynamic web content in Amazon S3.
- D. Create a read replica for the RDS DB instance. **Most Voted**
- E. Configure a Multi-AZ deployment for the RDS DB instance.

Correct Answer: BD

Community vote distribution

BD (90%)

8%

Comments

Buruguduystunstugudunstuy **Highly Voted** 3 years, 2 months ago

Selected Answer: BD

To resolve the issue of slow page loads for a rapidly growing e-commerce website hosted on AWS, a solutions architect can take the following two actions:

1. Set up an Amazon CloudFront distribution
2. Create a read replica for the RDS DB instance

Configuring an Amazon Redshift cluster is not relevant to this issue since Redshift is a data warehousing service and is typically used for the analytical processing of large amounts of data.

Hosting the dynamic web content in Amazon S3 may not necessarily improve performance since S3 is an object storage service, not a web application server. While S3 can be used to host static web content, it may not be suitable for hosting dynamic web content since S3 doesn't support server-side scripting or processing.

Configuring a Multi-AZ deployment for the RDS DB instance will improve high availability but may not necessarily improve performance.

upvoted 16 times

acts268 **Highly Voted** 3 years, 2 months ago

Selected Answer: BD

Cloud Front and Read Replica

upvoted 5 times

robst84 **Most Recent** 5 months, 2 weeks ago

Selected Answer: BD

Probably the answer is B and D. I hate how AWS is getting so sloppy with the questions. We have to answer by eliminating the ones that seem farthest from the best solution. Creating a replica for the RDS DB Instance is not going to do help, unless you change your application to start using it for reading operations. For some sloppy questions like this one, you have to assume that the rest of the answer is implicit, however, for some other questions they are very strict and you can not assume a comma...

upvoted 1 times

pentium75 2 years, 5 months ago

Selected Answer: BD

A - Redshift is for OLAP, not OLTP

B - Caching, reduces page load time and server load

C - S3 can't host dynamic (!) content

D - Read Replica is meant for increasing DB performance
E - Multi-AZ is meant for HA (not asked here)

upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: BD

The two options that will best help resolve the slow page loads are:

B) Set up an Amazon CloudFront distribution

and

E) Configure a Multi-AZ deployment for the RDS DB instance

Explanation:

CloudFront can cache static content globally and improve latency for static content delivery.
Multi-AZ RDS improves performance and availability of the database driving dynamic content.

upvoted 4 times

MatAlves 1 year, 9 months ago

Wrong. Multi-AZ is meant to "enhance availability by deploying a standby instance in a second AZ, and achieve fault tolerance in the event of an AZ or database instance failure."

It does not improve performance.

upvoted 1 times

antropaws 3 years ago

Selected Answer: BD

BD is correct.

upvoted 4 times

TariqKipkemei 3 years ago

Selected Answer: BD

Resolve latency = Amazon CloudFront distribution and read replica for the RDS DB

upvoted 5 times

SamDouk 3 years, 2 months ago

Selected Answer: BD

B and D

upvoted 3 times

klayytech 3 years, 2 months ago

Selected Answer: BD

The website's users are experiencing slow page loads.

To resolve this issue, a solutions architect should take the following two actions:

Create a read replica for the RDS DB instance. This will help to offload read traffic from the primary database instance and improve performance.

upvoted 3 times

zooba72 3 years, 2 months ago

Selected Answer: BD

Question asked about performance improvements, not HA. Cloudfront & Read Replica

upvoted 3 times

thaotnt 3 years, 2 months ago

Selected Answer: BD

slow page loads. >>> D

upvoted 3 times

andyto 3 years, 2 months ago

Selected Answer: BD

Read Replica will speed up Reads on RDS DB.

E is wrong. It brings HA but doesn't contribute to speed which is impacted in this case. Multi-AZ is Active-Standby solution.

upvoted 1 times

COTIT 3 years, 2 months ago

Selected Answer: BE

I agree with B & E.

B. Set up an Amazon CloudFront distribution. (Amazon CloudFront is a content delivery network (CDN) service)

E. Configure a Multi-AZ deployment for the RDS DB instance. (Good idea for loadbalance the DB workflow)

upvoted 2 times

pentium75 2 years, 5 months ago

Multi-AZ for HA, Read Replica for Scalability

https://aws.amazon.com/rds/features/read-replicas/?nc1=h_ls

upvoted 2 times

Santosh43 3 years, 2 months ago

B and E (as there is nothing mention about read transactions)

upvoted 1 times

awsgeek75 2 years, 4 months ago

Why E? There is nothing mentioned about High Availability also. E is wrong because Multi AZ won't help with scaling

upvoted 2 times

Akademik6 3 years, 2 months ago

Selected Answer: BD

Cloudfront and Read Replica. We don't need HA here.

upvoted 4 times

Bofi 3 years, 2 months ago

Selected Answer: BE

Amazon CloudFront can handle both static and Dynamic contents hence there is not need for option C I.e hosting the static data on Amazon S3. RDS read replica will reduce the amount of reads on the RDS hence leading a better performance. Multi-AZ is for disaster Recovery , which means D is also out.

upvoted 1 times

Thief 3 years, 2 months ago

Selected Answer: BC

CloudFont with S3

upvoted 1 times

pentium75 2 years, 5 months ago

S3 can't host "dynamic content"

upvoted 3 times

A company uses Amazon EC2 instances and AWS Lambda functions to run its application. The company has VPCs with public subnets and private subnets in its AWS account. The EC2 instances run in a private subnet in one of the VPCs. The Lambda functions need direct network access to the EC2 instances for the application to work.

The application will run for at least 1 year. The company expects the number of Lambda functions that the application uses to increase during that time. The company wants to maximize its savings on all application resources and to keep network latency between the services low.

Which solution will meet these requirements?

- A. Purchase an EC2 Instance Savings Plan. Optimize the Lambda functions' duration and memory usage and the number of invocations. Connect the Lambda functions to the private subnet that contains the EC2 instances.
- B. Purchase an EC2 Instance Savings Plan. Optimize the Lambda functions' duration and memory usage, the number of invocations, and the amount of data that is transferred. Connect the Lambda functions to a public subnet in the same VPC where the EC2 instances run.
- C. Purchase a Compute Savings Plan. Optimize the Lambda functions' duration and memory usage, the number of invocations, and the amount of data that is transferred. Connect the Lambda functions to the private subnet that contains the EC2 instances. **Most Voted**
- D. Purchase a Compute Savings Plan. Optimize the Lambda functions' duration and memory usage, the number of invocations, and the amount of data that is transferred. Keep the Lambda functions in the Lambda service VPC.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 3 years, 2 months ago

Selected Answer: C

Answer C is the best solution that meets the company's requirements.

By purchasing a Compute Savings Plan, the company can save on the costs of running both EC2 instances and Lambda functions. The Lambda functions can be connected to the private subnet that contains the EC2 instances through a VPC endpoint for AWS services or a VPC peering connection. This provides direct network access to the EC2 instances while keeping the traffic within the private network, which helps to minimize network latency.

Optimizing the Lambda functions' duration, memory usage, number of invocations, and amount of data transferred can help to further minimize costs and improve performance. Additionally, using a private subnet helps to ensure that the EC2 instances are not directly accessible from the public internet, which is a security best practice.

upvoted 20 times

Buruguduystunstugudunstuy 3 years, 2 months ago

Answer A is not the best solution because connecting the Lambda functions directly to the private subnet that contains the EC2 instances may not be scalable as the number of Lambda functions increases. Additionally, using an EC2 Instance Savings Plan may not provide savings on the costs of running Lambda functions.

Answer B is not the best solution because connecting the Lambda functions to a public subnet may not be as secure as connecting them to a private subnet. Also, keeping the EC2 instances in a private subnet helps to ensure that they are not directly accessible from the public internet.

Answer D is not the best solution because keeping the Lambda functions in the Lambda service VPC may not provide direct network access to the EC2 instances, which may impact the performance of the application.

upvoted 9 times

TariqKipkemei Highly Voted 2 years, 7 months ago

Selected Answer: C

Implement Compute Savings Plan because it applies to Lambda usage as well, then connect the Lambda functions to the private subnet that contain the EC2 instances

upvoted 8 times

MatAlves Most Recent 1 year, 9 months ago

"Savings Plans are a flexible pricing model that offer low prices on Amazon EC2, AWS Lambda, and AWS Fargate usage, in exchange for a commitment to a consistent amount of usage (measured in \$/hour) for a 1 or 3 year term."

- That already excludes A and B.

The question requires to "keep network latency between the services low", which can be achieved by connecting the Lambda functions to the private subnet that contains the EC2 instances.

C is the answer.

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: C

A Compute Savings Plan covers both EC2 and Lambda and allows maximizing savings on all resources.

Optimizing Lambda configuration reduces costs.

Connecting the Lambda functions to the private subnet with the EC2 instances provides direct network access between them, keeping latency low.

The Lambda functions are isolated in the private subnet rather than public, improving security.

upvoted 4 times

jaehoon090 2 years, 10 months ago

CCCCCCCCCCCCCCCCCCCC

upvoted 2 times

elearningtakai 3 years, 2 months ago

Selected Answer: C

Connect Lambda to Private Subnet contains EC2

upvoted 2 times

zooba72 3 years, 2 months ago

Selected Answer: C

Compute savings plan covers both EC2 & Lambda

upvoted 6 times

Zox42 3 years, 2 months ago

C. I would go with C, because Compute savings plans cover Lambda as well.

upvoted 5 times

andyto 3 years, 2 months ago

A. I would go with A. Saving and low network latency are required.

EC2 instance savings plans offer savings of up to 72%

Compute savings plans offer savings of up to 66%

Placing Lambda on the same private network with EC2 instances provides the lowest latency.

upvoted 1 times

abitwrong 3 years, 2 months ago

EC2 Instance Savings Plans apply to EC2 usage only. Compute Savings Plans apply to usage across Amazon EC2, AWS Lambda, and AWS Fargate. (<https://aws.amazon.com/savingsplans/faq/>)

Lambda functions need direct network access to the EC2 instances for the application to work and these EC2 instances are in the private subnet. So the correct answer is C.

upvoted 3 times

A solutions architect needs to allow team members to access Amazon S3 buckets in two different AWS accounts: a development account and a production account. The team currently has access to S3 buckets in the development account by using unique IAM users that are assigned to an IAM group that has appropriate permissions in the account.

The solutions architect has created an IAM role in the production account. The role has a policy that grants access to an S3 bucket in the production account.

Which solution will meet these requirements while complying with the principle of least privilege?

- A. Attach the Administrator Access policy to the development account users.
- B. Add the development account as a principal in the trust policy of the role in the production account. **Most Voted**
- C. Turn off the S3 Block Public Access feature on the S3 bucket in the production account.
- D. Create a user in the production account with unique credentials for each team member.

Correct Answer: B

Community vote distribution

B (100%)

Comments

kels1 **Highly Voted** 2 years, 1 month ago

well, if you made it this far, it means you are persistent :) Good luck with your exam!
upvoted 83 times

SkyZeroZx 2 years, 1 month ago

Thanks good luck for all
upvoted 10 times

Kimnesh 1 year, 9 months ago

thank you!
upvoted 6 times

gpt_test **Highly Voted** 2 years, 2 months ago

Selected Answer: B

By adding the development account as a principal in the trust policy of the IAM role in the production account, you are allowing users from the development account to assume the role in the production account. This allows the team members to access the S3 bucket in the production account without granting them unnecessary privileges.
upvoted 10 times

digital_eye **Most Recent** 1 year, 2 months ago

Selected Answer: B

Good luck everyone
upvoted 2 times

TariqKipkemei 1 year, 7 months ago

Selected Answer: B

Add the development account as a principal in the trust policy of the role in the production account
upvoted 3 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: B

The best solution is B) Add the development account as a principal in the trust policy of the role in the production account.

This allows cross-account access to the S3 bucket in the production account by assuming the IAM role. The development account users can assume the role to gain temporary access to the production bucket.

upvoted 5 times

nilandd44gg 1 year, 11 months ago

Selected Answer: B

<https://aws.amazon.com/blogs/security/how-to-use-trust-policies-with-iam-roles/>

An AWS account accesses another AWS account – This use case is commonly referred to as a cross-account role pattern. It allows human or machine IAM principals from one AWS account to assume this role and act on resources within a second AWS account. A role is assumed to enable this behavior when the resource in the target account doesn't have a resource-based policy that could be used to grant cross-account access.

upvoted 3 times

elearningtakai 2 years, 2 months ago

Selected Answer: B

About Trust policy – The trust policy defines which principals can assume the role, and under which conditions. A trust policy is a specific type of resource-based policy for IAM roles.

Answer A: overhead permission Admin to development.

Answer C: Block public access is a security best practice and seems not relevant to this scenario.

Answer D: difficult to manage and scale

upvoted 3 times

Buruguduystunstugudunstuy 2 years, 2 months ago

Selected Answer: B

Answer A, attaching the Administrator Access policy to development account users, provides too many permissions and violates the principle of least privilege. This would give users more access than they need, which could lead to security issues if their credentials are compromised.

Answer C, turning off the S3 Block Public Access feature, is not a recommended solution as it is a security best practice to enable S3 Block Public Access to prevent accidental public access to S3 buckets.

Answer D, creating a user in the production account with unique credentials for each team member, is also not a recommended solution as it can be difficult to manage and scale for large teams. It is also less secure, as individual user credentials can be more easily compromised.

upvoted 3 times

klayytech 2 years, 2 months ago

Selected Answer: B

The solution that will meet these requirements while complying with the principle of least privilege is to add the development account as a principal in the trust policy of the role in the production account. This will allow team members to access Amazon S3 buckets in two different AWS accounts while complying with the principle of least privilege.

Option A is not recommended because it grants too much access to development account users. Option C is not relevant to this scenario. Option D is not recommended because it does not comply with the principle of least privilege.

upvoted 2 times

Akademik6 2 years, 2 months ago

Selected Answer: B

B is the correct answer

upvoted 3 times

A company uses AWS Organizations with all features enabled and runs multiple Amazon EC2 workloads in the ap-southeast-2 Region. The company has a service control policy (SCP) that prevents any resources from being created in any other Region. A security policy requires the company to encrypt all data at rest.

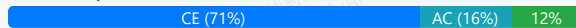
An audit discovers that employees have created Amazon Elastic Block Store (Amazon EBS) volumes for EC2 instances without encrypting the volumes. The company wants any new EC2 instances that any IAM user or root user launches in ap-southeast-2 to use encrypted EBS volumes. The company wants a solution that will have minimal effect on employees who create EBS volumes.

Which combination of steps will meet these requirements? (Choose two.)

- A. In the Amazon EC2 console, select the EBS encryption account attribute and define a default encryption key.
- B. Create an IAM permission boundary. Attach the permission boundary to the root organizational unit (OU). Define the boundary to deny the ec2:CreateVolume action when the ec2:Encrypted condition equals false.
- C. Create an SCP. Attach the SCP to the root organizational unit (OU). Define the SCP to deny the ec2:CreateVolume action when the ec2:Encrypted condition equals false. **Most Voted**
- D. Update the IAM policies for each account to deny the ec2:CreateVolume action when the ec2:Encrypted condition equals false.
- E. In the Organizations management account, specify the Default EBS volume encryption setting. **Most Voted**

Correct Answer: CE

Community vote distribution



Comments

Guru4Cloud **Highly Voted** 2 years, 3 months ago

Selected Answer: CE

The correct answer is (C) and (E).

Option (C): Creating an SCP and attaching it to the root organizational unit (OU) will deny the ec2:CreateVolume action when the ec2:Encrypted condition equals false. This means that any IAM user or root user in any account in the organization will not be able to create an EBS volume without encrypting it.

Option (E): Specifying the Default EBS volume encryption setting in the Organizations management account will ensure that all new EBS volumes created in any account in the organization are encrypted by default.

upvoted 12 times

Axaus **Highly Voted** 2 years, 6 months ago

Selected Answer: CE

CE

Prevent future issues by creating a SCP and set a default encryption.

upvoted 10 times

network_enthusiast **Most Recent** 1 year, 1 month ago

Selected Answer: AC

Answer will be A&C.

AWS does not currently provide a direct setting at the organization level to enforce default EBS encryption across all member accounts in AWS Organizations. However, you can achieve this goal by implementing default encryption at the account level and enforcing compliance using an SCP (Service Control Policy).

upvoted 2 times

Andreew883 1 year, 1 month ago

Selected Answer: CE

C and E as shaded by guru4cloud

upvoted 1 times

Jazz888 1 year, 6 months ago

The problem here is we don't know in which account the workload is on. The account in ap-xx-is that the management account or it's a member account?? That will decide to select either A or E. C is certainly correct

upvoted 2 times

NSA_Poker 1 year, 6 months ago

Selected Answer: CE

(A) is incorrect bc absent of SCP or the Organizations management account, the scope of the EC2 console is too narrow to be applied to 'any IAM user or root user'.

upvoted 2 times

venutadi 1 year, 7 months ago

Selected Answer: AC

<https://repost.aws/knowledge-center/ebs-automatic-encryption>

Newly created Amazon EBS volumes aren't encrypted by default. However, you can turn on default encryption for new EBS volumes and snapshot copies that are created within a specified Region. To turn on encryption by default, use the Amazon Elastic Compute Cloud (Amazon EC2) console.

upvoted 2 times

1rob 1 year, 11 months ago

Selected Answer: AC

A: will enforce automatic encryption in a account. This will have no effect on employees. Do this in every account.

B: permission boundary is not appropriate here.

C: an SCP will force employees to create encrypted volumes in every account.

D: This would work but is too much maintenance.

E: Setting EBS volume encryption in the Organizations management account will only have impact on volumes in that account, not on other accounts.

upvoted 4 times

pentium75 1 year, 11 months ago

Selected Answer: AE

The solution should "have minimal effect on employees who create EBS volumes". Thus new volumes should automatically be encrypted. Options B, C and D do NOT automatically encrypt volumes, they simply cause requests to create non-encrypted volumes to fail.

upvoted 3 times

dkw2342 1 year, 9 months ago

IMO the correct solution is AC:

In the Amazon EC2 console, select the EBS encryption account attribute and define a default encryption key.

-> This has to be done in every AWS account separately.

Create an SCP. Attach the SCP to the root organizational unit (OU). Define the SCP to deny the ec2:CreateVolume action when the ec2:Encrypted condition equals false.

-> This will just act as a safeguard in case an admin would disable default encryption in the member account, so it should not have any effect on employees who create EBS volumes.

I think an updated question would offer options A and an updated C:

Create an SCP. Attach the SCP to the root organizational unit (OU). Define the SCP to deny the ec2:DisableEbsEncryptionByDefault action.

-> This will prevent disabling default encryption once it has been enabled.

upvoted 2 times

Valder21 2 years, 3 months ago

Wondering if just C would be sufficient?

upvoted 1 times

bjexamprep 2 years, 3 months ago

Seems many people selected E as part of the correct answer. But I didn't find so called Organization level EBS default setting in my Organization management account. I tried setting default EBS encryption setting in my Organization management account, and it didn't apply to the member account. If E cannot guarantee default encryption in all other account, E has no advantage over A. Anyone can explain why E is better than A?

upvoted 4 times

novelai_me 2 years, 5 months ago

Selected Answer: AE

Option A: By default, EBS encryption is not enabled for EC2 instances. However, you can set an EBS encryption by default in your AWS account in the Amazon EC2 console. This ensures that every new EBS volume that is created is encrypted.

Option E: With AWS Organizations, you can centrally set the default EBS encryption for your organization's accounts. This helps in enforcing a consistent encryption policy across your organization.

Option B, C and D are not correct because while you can use IAM policies or SCPs to restrict the creation of unencrypted EBS volumes, this could

potentially impact employees' ability to create necessary resources if not properly configured. They might require additional permissions management, which is not mentioned in the requirements. By setting the EBS encryption by default at the account or organization level (Options A and E), you can ensure all new volumes are encrypted without affecting the ability of employees to create resources.

upvoted 3 times

Buruguduystunstugudunstuy 2 years, 6 months ago

Selected Answer: CE

SCPs are a great way to enforce policies across an entire AWS Organization, preventing users from creating resources that do not comply with the se policies.

In AWS Management Console, one can go to EC2 dashboard -> Settings -> Data encryption -> Check "Always encrypt new EBS volumes" and choose a default KMS key. This ensures that every new EBS volume created will be encrypted by default, regardless of how it is created.

upvoted 3 times

PRASAD180 2 years, 6 months ago

1000% CE crt

upvoted 1 times

hiroohiroo 2 years, 6 months ago

Selected Answer: CE

CとEが正しいと考える。

upvoted 3 times

Efren 2 years, 6 months ago

Selected Answer: CE

CE for me as well

upvoted 2 times

nosense 2 years, 6 months ago

Selected Answer: CE

SCP that denies the ec2:CreateVolume action when the ec2:Encrypted condition equals false. This will prevent users and service accounts in member accounts from creating unencrypted EBS volumes in the ap-southeast-2 Region.

upvoted 2 times

Efren 2 years, 6 months ago

agreed

upvoted 1 times

pentium75 1 year, 11 months ago

Wouldn't this have "effect on employees who create EBS volumes", which we are asked to minimize?

upvoted 1 times

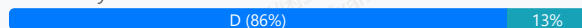
A company wants to use an Amazon RDS for PostgreSQL DB cluster to simplify time-consuming database administrative tasks for production database workloads. The company wants to ensure that its database is highly available and will provide automatic failover support in most scenarios in less than 40 seconds. The company wants to offload reads off of the primary instance and keep costs as low as possible.

Which solution will meet these requirements?

- A. Use an Amazon RDS Multi-AZ DB instance deployment. Create one read replica and point the read workload to the read replica.
- B. Use an Amazon RDS Multi-AZ DB cluster deployment. Create two read replicas and point the read workload to the read replicas.
- C. Use an Amazon RDS Multi-AZ DB instance deployment. Point the read workload to the secondary instances in the Multi-AZ pair.
- D. Use an Amazon RDS Multi-AZ DB cluster deployment. Point the read workload to the reader endpoint. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

ogerber **Highly Voted** 3 years ago

Selected Answer: D

A - multi-az instance : failover takes between 60-120 sec

D - multi-az cluster: failover around 35 sec

upvoted 20 times

kelmryan1 2 years, 1 month ago

They want to keep cost low as possible, A is the right answer

upvoted 2 times

MatAlves 1 year, 9 months ago

No, the question clearly says "failover support in most scenarios in less than 40 seconds."

D is the only possible answer.

upvoted 4 times

Buruguduystunstugudunstuy **Highly Voted** 2 years, 12 months ago

Selected Answer: D

The correct answer is:

D. Use an Amazon RDS Multi-AZ DB cluster deployment. Point the read workload to the reader endpoint.

Explanation:

The company wants high availability, automatic failover support in less than 40 seconds, read offloading from the primary instance, and cost-effectiveness.

Answer D is the best choice for several reasons:

1. Amazon RDS Multi-AZ deployments provide high availability and automatic failover support.
2. In a Multi-AZ DB cluster, Amazon RDS automatically provisions and maintains a standby in a different Availability Zone. If a failure occurs, Amazon RDS performs an automatic failover to the standby, minimizing downtime.
3. The "Reader endpoint" for an Amazon RDS DB cluster provides load-balancing support for read-only connections to the DB cluster. Directing read traffic to the reader endpoint helps in offloading read operations from the primary instance.

upvoted 14 times

Kiki_Pass 2 years, 10 months ago

Sorry I'm a bit confused... I thought only Aurora DB Cluster has reader endpoint. Do you by any chance has the link to the doc for RDS Reader Endpoint?

upvoted 4 times

lemur88 2 years, 9 months ago

<https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/multi-az-db-clusters-concepts-connection-management.html#multi-az-db-clusters-concepts-connection-management-endpoints-reader>

upvoted 6 times

zikou **Most Recent** 1 year, 9 months ago

to offload read we use read replicas also there is no such thing as reader endpoint in rds, it is only on aurora

upvoted 2 times

rondellidell 2 years, 2 months ago

Selected Answer: B

<https://aws.amazon.com/rds/features/multi-az/>
Amazon RDS Multi-AZ with two readable standbys

upvoted 1 times

hro 2 years, 2 months ago

I think the cluster is over-kill - but the company 'wants to use an Amazon RDS ... DB cluster'.

upvoted 3 times

pentium75 2 years, 5 months ago

Selected Answer: D

A would be cheapest but "failover times are typically 60–120 seconds" which does not meet our requirements. We need Multi-AZ DB cluster (not instance). This has a reader endpoint by default, thus no need for additional read replicas (to "keep costs as low as possible").

<https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/Concepts.MultiAZSingleStandby.html>

upvoted 7 times

master9 2 years, 5 months ago

Selected Answer: A

in question, it has mentioned that "keep costs as low as possible"

In a Multi-AZ configuration, the DB instances and EBS storage volumes are deployed across two Availability Zones.

It provides high availability and failover support for DB instances.

This setup is primarily for disaster recovery.

It involves a primary DB instance and a standby replica, which is a copy of the primary DB instance.

The standby replica is not accessible directly; instead, it serves as a failover target in case the primary instance fails.

upvoted 1 times

potomac 2 years, 7 months ago

Selected Answer: D

It is D.

A is not correct. Multi-AZ DB instance deployment, which creates a primary instance and a standby instance to provide failover support. However, the standby instance does not serve traffic.

upvoted 2 times

maudsha 2 years, 7 months ago

Selected Answer: D

<https://aws.amazon.com/blogs/database/choose-the-right-amazon-rds-deployment-option-single-az-instance-multi-az-instance-or-multi-az-database-cluster/#:~:text=Unlike%20Multi%2DAZ%20instance%20deployment,different%20AZs%20serving%20read%20traffic.>

According to this the answer is D

"Unlike Multi-AZ instance deployment, where the secondary instance can't be accessed for read or writes, Multi-AZ DB cluster deployment consists of primary instance running in one AZ serving read-write traffic and two other standby running in two different AZs serving read traffic."

You don't have to create read replicas with cluster deployment so B is out.

upvoted 2 times

kwang312 2 years, 9 months ago

D

Fail-over on Multi-AZ DB instance is 60-120s

On Cluster, the time under 35s

upvoted 5 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

D. Use an Amazon RDS Multi-AZ DB cluster deployment. Point the read workload to the reader endpoint
upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

Use an Amazon RDS Multi-AZ DB cluster deployment Point the read workload to the reader endpoint.
upvoted 2 times

Eminenza22 2 years, 9 months ago

Selected Answer: A

The solutions architect should use an Amazon RDS Multi-AZ DB instance deployment. The company can create one read replica and point the read workload to the read replica. Amazon RDS provides high availability and failover support for DB instances using Multi-AZ deployments.
upvoted 1 times

Gooniegoogoo 2 years, 11 months ago

and d..

Multi-AZ DB clusters typically have lower write latency when compared to Multi-AZ DB instance deployments. They also allow read-only workloads to run on reader DB instances.
upvoted 2 times

TariqKipkemei 3 years ago

Selected Answer: D

This is as case where both option A and D can work, but option D gives 2 DB instances for read compared to only 1 given by option A. Costwise they are the same as both options use 3 DB instances.
upvoted 2 times

Henrytml 3 years ago

Selected Answer: A

lowest cost option, and effective with read replica
upvoted 3 times

antropaws 3 years ago

Selected Answer: D

It's D. Read well: "A company wants to use an Amazon RDS for PostgreSQL DB CLUSTER".
upvoted 4 times

A company runs a highly available SFTP service. The SFTP service uses two Amazon EC2 Linux instances that run with elastic IP addresses to accept traffic from trusted IP sources on the internet. The SFTP service is backed by shared storage that is attached to the instances. User accounts are created and managed as Linux users in the SFTP servers.

The company wants a serverless option that provides high IOPS performance and highly configurable security. The company also wants to maintain control over user permissions.

Which solution will meet these requirements?

A. Create an encrypted Amazon Elastic Block Store (Amazon EBS) volume. Create an AWS Transfer Family SFTP service with a public endpoint that allows only trusted IP addresses. Attach the EBS volume to the SFTP service endpoint. Grant users access to the SFTP service.

B. Create an encrypted Amazon Elastic File System (Amazon EFS) volume. Create an AWS Transfer Family SFTP service with elastic IP addresses and a VPC endpoint that has internet-facing access. Attach a security group to the endpoint that allows only trusted IP addresses. Attach the EFS volume to the SFTP service endpoint. Grant users access to the SFTP service.

Most Voted

C. Create an Amazon S3 bucket with default encryption enabled. Create an AWS Transfer Family SFTP service with a public endpoint that allows only trusted IP addresses. Attach the S3 bucket to the SFTP service endpoint. Grant users access to the SFTP service.

D. Create an Amazon S3 bucket with default encryption enabled. Create an AWS Transfer Family SFTP service with a VPC endpoint that has internal access in a private subnet. Attach a security group that allows only trusted IP addresses. Attach the S3 bucket to the SFTP service endpoint. Grant users access to the SFTP service.

Correct Answer: B

Community vote distribution

B (85%)

Other

Comments

pentium75 **Highly Voted** 2 years, 5 months ago

Selected Answer: B

Not A - Transfer Family can't use EBS

B - Possible and meets requirement

Not C - S3 doesn't guarantee "high IOPS performance"; also there is no "public endpoint that allows only trusted IP addresses" (you can assign a Security Group to a public endpoint but that is not mentioned here)

Not D - Endpoint would be in private subnet, not accessible from Internet at all

upvoted 10 times

alexandercamachop **Highly Voted** 3 years ago

Selected Answer: B

First Serverless - EFS

Second it says it is attached to the Linux instances at the same time, only EFS can do that.

upvoted 8 times

bora4motion 1 year ago

dude, if you have linux instances that's no longer serverless =)))

upvoted 1 times

bora4motion **Most Recent** 1 year ago

Selected Answer: C

I'm going with C. EFS needs to be attached to something, used with a server/instance.

upvoted 1 times

FlyingHawk 1 year, 6 months ago

Selected Answer: B

D, the description "VPC endpoint that has internal access in a private subnet" is technically incorrect and doesn't make sense for AWS Transfer Family. AWS Transfer Family actually supports:

Public endpoints (internet-facing)

VPC endpoints (private network access)

The VPC endpoint doesn't have "internal access" - it's about how the SFTP service is configured to interact with your VPC network resources.

upvoted 1 times

FlyingHawk 1 year, 6 months ago

A more accurate description for a VPC-based AWS Transfer Family SFTP service would be:

Create an AWS Transfer Family SFTP service with a VPC endpoint

Place the endpoint in a private subnet

Configure security groups to allow only trusted IP sources

Attach the S3 bucket to the SFTP service endpoint

So while the overall strategy in option D (using S3 with VPC endpoint and security group restrictions) could be valid, the specific wording about "internal access" is incorrect.

This technical inaccuracy in the description would make D an incorrect answer in a certification or technical assessment.

upvoted 1 times

FlyingHawk 1 year, 6 months ago

C public endpoint that allows only trusted IP addresses does not sound technical correct as only the security group can restrict the IP address. the correct description should " Create an AWS Transfer Family SFTP service in public subnet with security group allows only trusted IP address, create the vpc endpoint (gateway endpoint here for saving the cost) of S3 bucket to allow the SFTP service can access it privately, grant user access to the SFTP service via IAM policies and grant the access of S3 to SFTP service via IAM role

upvoted 1 times

FlyingHawk 1 year, 6 months ago

Selected Answer: C

Amazon S3 provides high throughput and performance suitable for many use cases, including those requiring high IOPS (Input/Output Operations Per Second). S3 is not a block storage solution (like EBS) or a file system (like EFS). While its performance is exceptional for its intended use case (object storage), it may not match the millisecond latency or consistent high IOPS required for transactional databases or other ultra-low latency applications.

For applications where sub-millisecond latency or extremely high random IOPS (e.g., 64,000 IOPS) is required, solutions like EBS or EFS would be better.

upvoted 1 times

FlyingHawk 1 year, 6 months ago

After read the description of C, I feel the description of "an AWS Transfer Family SFTP service with a public endpoint that allows only trusted IP addresses" does not sound correct, the endpoint itself cannot restrict the IP address from internet, only the security group. the correct description should be: Create an AWS Transfer Family SFTP service with a public endpoint and security group that allows only trusted IP addresses. create the vpc endpoint (gateway endpoint here for saving the cost) of S3 bucket to allow the SFTP service can access it privately, grant user access to the SFTP service via IAM policies and grant the access of S3 to SFTP service via IAM role

upvoted 1 times

JA2018 1 year, 6 months ago

Selected Answer: B

Actually AWS Transfer Family can use S3, so it's a toss-up between Options B & C but I tend to favour Option B for the following reasons (based on the keys in STEM):

company runs a highly available SFTP service. The SFTP service uses two Amazon EC2 Linux instances that run with elastic IP addresses to accept traffic from trusted IP sources on the internet. The SFTP service is backed by shared storage that is attached to the instances. User accounts are created and managed as Linux users in the SFTP servers.

1. Requires a serverless option that provides high IOPS performance and highly configurable security.

2. User also wants to maintain control over user permissions

upvoted 1 times

523db89 1 year, 9 months ago

Option B best meets the company's requirements by leveraging AWS Transfer Family with an EFS volume, ensuring high availability, security, and performance.

upvoted 2 times

NickGordon 2 years, 7 months ago

Selected Answer: B

A is incorrect as EBS is not an option
C is incorrect as when I select public accessible, I don't see an option I can set up trusted IP address
D is incorrect as it is internal.

B, followed the steps and I can set up a sftp in this way
upvoted 4 times

potomac 2 years, 7 months ago

Selected Answer: B

B
EFS has lower latency and higher throughput than S3 when accessed from within the same availability zone.
upvoted 3 times

thanhnv142 2 years, 7 months ago

C: Because it is server-less. definitely not A or B because it utilizes server.
upvoted 1 times

warp 2 years, 7 months ago

Amazon Elastic File System - Serverless, fully elastic file storage:
<https://aws.amazon.com/efs/>
upvoted 5 times

bsbs1234 2 years, 8 months ago

B,
A), transfer family does not support EBS
C,D), S3 has lower IOPS than EFS
upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

Create an encrypted Amazon Elastic File System (Amazon EFS) volume. Create an AWS Transfer Family SFTP service with elastic IP addresses and a VPC endpoint that has internet-facing access. Attach a security group to the endpoint that allows only trusted IP addresses. Attach the EFS volume to the SFTP service endpoint. Grant users access to the SFTP service.
upvoted 2 times

Axeashes 2 years, 12 months ago

<https://aws.amazon.com/blogs/storage/use-ip-whitelisting-to-secure-your-aws-transfer-for-sftp-servers/>
upvoted 2 times

TariqKipkemei 3 years ago

Selected Answer: B

EFS is best to serve this purpose.
upvoted 2 times

envest 3 years ago

Answer C (from abylead.com)
Transfer Family offers fully managed serverless support for B2B file transfers via SFTP, AS2, FTPS, & FTP directly in & out of S3 or EFS. For a controlled internet access you can use internet-facing endpoints with Transfer SFTP servers & restrict trusted internet sources with VPC's default Sgrp. In addition, S3 Access Points aliases allows you to use S3 bucket names for a unique access control policy on shared S3 datasets.
Transfer SFTP & S3: <https://aws.amazon.com/blogs/apn/how-to-use-aws-transfer-family-to-replace-and-scale-sftp-servers/>

A) Transfer SFTP doesn't support EBS, not for share data, & not serverless: infeasible.
B) EFS mounts via ENIs not endpoints: infeasible.
D) public endpoint for internet access is missing: infeasible.
upvoted 4 times

omoakin 3 years ago

BBBBBBBBBBBBBBB
upvoted 1 times

vesen22 3 years ago

Selected Answer: B

EFS all day
upvoted 2 times

A company is developing a new machine learning (ML) model solution on AWS. The models are developed as independent microservices that fetch approximately 1 GB of model data from Amazon S3 at startup and load the data into memory. Users access the models through an asynchronous API. Users can send a request or a batch of requests and specify where the results should be sent.

The company provides models to hundreds of users. The usage patterns for the models are irregular. Some models could be unused for days or weeks. Other models could receive batches of thousands of requests at a time.

Which design should a solutions architect recommend to meet these requirements?

- A. Direct the requests from the API to a Network Load Balancer (NLB). Deploy the models as AWS Lambda functions that are invoked by the NLB.
- B. Direct the requests from the API to an Application Load Balancer (ALB). Deploy the models as Amazon Elastic Container Service (Amazon ECS) services that read from an Amazon Simple Queue Service (Amazon SQS) queue. Use AWS App Mesh to scale the instances of the ECS cluster based on the SQS queue size.
- C. Direct the requests from the API into an Amazon Simple Queue Service (Amazon SQS) queue. Deploy the models as AWS Lambda functions that are invoked by SQS events. Use AWS Auto Scaling to increase the number of vCPUs for the Lambda functions based on the SQS queue size.
- D. Direct the requests from the API into an Amazon Simple Queue Service (Amazon SQS) queue. Deploy the models as Amazon Elastic Container Service (Amazon ECS) services that read from the queue. Enable AWS Auto Scaling on Amazon ECS for both the cluster and copies of the service based on the queue size. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

examtopictempacc **Highly Voted** 2 years, 6 months ago
asynchronous=SQS, microservices=ECS.
Use AWS Auto Scaling to adjust the number of ECS services.
upvoted 17 times

TariqKipkemei 2 years, 6 months ago
good breakdown :)
upvoted 4 times

TariqKipkemei **Highly Voted** 2 years, 6 months ago
Selected Answer: D
For once examtopic answer is correct :) haha...

Batch requests/async = Amazon SQS
Microservices = Amazon ECS
Workload variations = AWS Auto Scaling on Amazon ECS
upvoted 11 times

wizcloudifa **Most Recent** 1 year, 7 months ago
Selected Answer: D

ALB is mentioned in other options to distract you, you don't need ALB for scaling here, we would need ECS autoscaling, they play with that idea in option B a bit however D gets it in a completely optimized way.... A and C both have lambda which for Machine learning models with workloads on heavy side, will not fly

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: D

I go with everyone D.

upvoted 3 times

alexandercamachop 2 years, 6 months ago

Selected Answer: D

D, no need for an App Load balancer like C says, no where in the text.

SQS is needed to ensure all request gets routed properly in a Microservices architecture and also that it waits until its picked up.

ECS with Autoscaling, will scale based on the unknown pattern of usage as mentioned.

upvoted 2 times

anibinaadi 2 years, 6 months ago

It is D

Refer <https://aws.amazon.com/blogs/containers/amazon-elastic-container-service-ecs-auto-scaling-using-custom-metrics/> for additional information/knowledge.

upvoted 2 times

nosense 2 years, 6 months ago

Selected Answer: D

because it is scalable, reliable, and efficient.

C does not scale the models automatically

upvoted 4 times

deechean 2 years, 3 months ago

why C doesn't scale the model? Application Auto Scaling can apply to lambda.

upvoted 1 times

pentium75 1 year, 11 months ago

How would you "use Auto Scaling (!) to increase the number of vCPUs (!) for the Lambda functions"?

upvoted 3 times

NSA_Poker 1 year, 6 months ago

Auto Scaling doesn't apply to Lambda. As your functions receive more requests, Lambda automatically handles scaling the number of execution environments until you reach your account's concurrency limit.

upvoted 2 times

A solutions architect wants to use the following JSON text as an identity-based policy to grant specific permissions:

```
{  "Statement": [    {      "Action": [        "asm:ListDocuments",        "asm:GetDocument"      ],      "Effect": "Allow",      "Resource": "*",      "Sid": ""    }  ],  "Version": "2012-10-17"}
```

Which IAM principals can the solutions architect attach this policy to? (Choose two.)

A. Role **Most Voted**

B. Group **Most Voted**

C. Organization

D. Amazon Elastic Container Service (Amazon ECS) resource

E. Amazon EC2 resource

Correct Answer: AB

Community vote distribution

AB (100%)

Comments

nosense **Highly Voted** 2 years, 6 months ago

Selected Answer: AB

identity-based policy used for role and group
upvoted 20 times

pentium75 **Highly Voted** 1 year, 11 months ago

Selected Answer: AB

Isn't the content of the policy completely irrelevant? IAM policies are applied to users, groups or roles ...
upvoted 8 times

dkw2342 **Most Recent** 1 year, 9 months ago

AB is correct, but the question is misleading because, according to the AWS IAM documentation, groups are not considered principals:
[https://docs.aws.amazon.com/IAM/latest/UserGuide/intro-structure.html#intro-structure-principal.](https://docs.aws.amazon.com/IAM/latest/UserGuide/intro-structure.html#intro-structure-principal)
upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: AB

A. Role
B. Group
upvoted 3 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: AB

Role or group
upvoted 3 times

A company is running a custom application on Amazon EC2 On-Demand Instances. The application has frontend nodes that need to run 24 hours a day, 7 days a week and backend nodes that need to run only for a short time based on workload. The number of backend nodes varies during the day.

The company needs to scale out and scale in more instances based on workload.

Which solution will meet these requirements MOST cost-effectively?

- A. Use Reserved Instances for the frontend nodes. Use AWS Fargate for the backend nodes.
- B. Use Reserved Instances for the frontend nodes. Use Spot Instances for the backend nodes. **Most Voted**
- C. Use Spot Instances for the frontend nodes. Use Reserved Instances for the backend nodes.
- D. Use Spot Instances for the frontend nodes. Use AWS Fargate for the backend nodes.

Correct Answer: B

Community vote distribution

B (62%)

A (38%)

Comments

Ramdi1 **Highly Voted** 2 years, 8 months ago

Selected Answer: A

Has to be A, It can scale down if required and you will be charged for what you use with fargate. Secondly they have not said the backend can have timeouts or can be down for a little period of time or something. So it has to rule out any spot instances even though they are cheaper.

upvoted 18 times

awsgeek75 2 years, 5 months ago

Fargate is serverless EKS so it cannot manage EC2 nodes

upvoted 7 times

nosense **Highly Voted** 3 years ago

Selected Answer: B

Reserved+ spot .

Fargate for serverless

upvoted 16 times

JosephKY **Most Recent** 7 months ago

Selected Answer: B

If I was building a solution in the real world and given these options, I would choose A every time. If I don't know the fault tolerance of the application logic, I have to assume it cannot be interrupted. If I'm taking an AWS exam, however, I know to answer B.

upvoted 1 times

Andreew883 1 year, 1 month ago

Selected Answer: B

The question is about instances so container and fargate is not applicable and we need to choose response B

upvoted 3 times

zdi561 1 year, 4 months ago

Selected Answer: A

I think in A it should be Fargate spot, B means standard spot. The former uses auto scaling and the latter does not

upvoted 2 times

zdi561 1 year, 4 months ago

Selected Answer: A

The requirement is to have an autoscaling , and fargate can do that. Spot instance only works when the work load is fault tolerant. Furthermore it is told to use on-demand instance.

upvoted 2 times

Mayur_B 1 year, 8 months ago

I will go with A.

Where does the question mentions that the workload interruptions are accepted.

upvoted 3 times

mussaha 2 years, 3 months ago

Selected Answer: B

B) because firegate is container

upvoted 3 times

noircesar25 2 years, 3 months ago

so what ive make up from this scenario is: the key word right here is "backend nodes" you cant use a serverless compute service with nodes and you need to use EC2s

so if we had ECS EC2 lunch type or on-demand EC2s as an options for the backend, they would be true?

upvoted 2 times

mwwt2022 2 years, 5 months ago

Selected Answer: B

24-7 usage for fe -> reserved instance

irregular workload for be -> spot instance

upvoted 4 times

pentium75 2 years, 5 months ago

Selected Answer: B

Not A because Fargate runs containers, not EC2 instances. But we have no indication that the workload would be containerized; it runs "on EC2 instances".

Not C and D because frontend must run 24/7, can't use Spot.

Thus B, yes, Spot instances are risky, but as they need to run "only for a short time" it seems acceptable.

Technically ideal option would be Reserved Instances for frontend nodes and On-demand instances for backend nodes, but that is not an option here.

upvoted 9 times

AwsZora 2 years, 6 months ago

Selected Answer: A

it is safe

upvoted 1 times

awsgeek75 2 years, 4 months ago

Fargate = containers

A is wrong

upvoted 2 times

meowruki 2 years, 6 months ago

Selected Answer: B

Reserved Instances (RIs) for Frontend Nodes: Since the frontend nodes need to run continuously (24/7), using Reserved Instances for them makes sense. RIs provide significant cost savings compared to On-Demand Instances for steady-state workloads.

Spot Instances for Backend Nodes: Spot Instances are suitable for short-duration workloads and can be significantly cheaper than On-Demand Instances. Since the number of backend nodes varies during the day, Spot Instances can help you take advantage of spare capacity at a lower cost. Keep in mind that Spot Instances may be interrupted if the capacity is needed elsewhere, so they are best suited for stateless and fault-tolerant workloads.

upvoted 2 times

meowruki 2 years, 6 months ago

Option A (Use Reserved Instances for the frontend nodes. Use AWS Fargate for the backend nodes): AWS Fargate is a serverless compute engine for containers, and it may not be the best fit for the described backend workload, especially if the number of backend nodes varies during the day.

upvoted 1 times

Goutham4981 2 years, 6 months ago

Selected Answer: B

AWS Fargate is a serverless compute engine for containers that allows you to run containers without having to manage the underlying infrastructure. It simplifies the process of deploying and managing containerized applications by abstracting away the complexities of server management, scaling,

and cluster orchestration.

No containerized application requirements are mentioned in the question. Plain EC2 instances. So Fargate is not actually an option

upvoted 3 times

thanhnv142 2 years, 7 months ago

A is fargate, which is none sense. B seems more OK (though none-sense)

upvoted 4 times

dilaaziz 2 years, 8 months ago

Selected Answer: A

Fargate for backend node

upvoted 2 times

awsgeek75 2 years, 4 months ago

Fargate is for containers not EC2 so A is wrong

upvoted 2 times

Wayne23Fang 2 years, 8 months ago

Selected Answer: A

(B) would take chance, though unlikely (A) is server-less auto-scaling. In case backend is idle, it might scale down, save money but no need to worry for interruption by Spot instance.

upvoted 3 times

A company uses high block storage capacity to run its workloads on premises. The company's daily peak input and output transactions per second are not more than 15,000 IOPS. The company wants to migrate the workloads to Amazon EC2 and to provision disk performance independent of storage capacity.

Which Amazon Elastic Block Store (Amazon EBS) volume type will meet these requirements MOST cost-effectively?

- A. GP2 volume type
- B. io2 volume type
- C. GP3 volume type **Most Voted**
- D. io1 volume type

Correct Answer: C

Community vote distribution

C (96%)

4%

Comments

nosense **Highly Voted** 3 years ago

Selected Answer: C

Gp3 \$ 0.08 usd per gb

Gp2 \$ 0.10 usd per gb

upvoted 15 times

Yadav_Sanjay **Highly Voted** 3 years ago

Selected Answer: C

Both GP2 and GP3 has max IOPS 16000 but GP3 is cost effective.

<https://aws.amazon.com/blogs/storage/migrate-your-amazon-ebs-volumes-from-gp2-to-gp3-and-save-up-to-20-on-costs/>

upvoted 12 times

LeonSauveterre **Most Recent** 1 year, 6 months ago

Selected Answer: C

OK, and is this something I should know to prepare for the exam...

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: C

C. GP3 volume type

upvoted 4 times

james2033 2 years, 10 months ago

Selected Answer: C

Quote "customers can scale up to 16,000 IOPS and" at <https://aws.amazon.com/about-aws/whats-new/2020/12/introducing-new-amazon-ebs-general-purpose-volumes-gp3/>

upvoted 3 times

alexandercamachop 3 years ago

Selected Answer: C

The GP3 (General Purpose SSD) volume type in Amazon Elastic Block Store (EBS) is the most cost-effective option for the given requirements. GP3 volumes offer a balance of price and performance and are suitable for a wide range of workloads, including those with moderate I/O needs.

GP3 volumes allow you to provision performance independently from storage capacity, which means you can adjust the baseline performance (measured in IOPS) and throughput (measured in MiB/s) separately from the volume size. This flexibility allows you to optimize your costs while meeting the workload requirements.

In this case, since the company's daily peak input and output transactions per second are not more than 15,000 IOPS, GP3 volumes provide a suitable and cost-effective option for their workloads.

upvoted 2 times

maver144 3 years ago

Selected Answer: B

It is not C pals. The company wants to migrate the workloads to Amazon EC2 and to provision disk performance independent of storage capacity. With GP3 we have to increase storage capacity to increase IOPS over baseline.

You can only chose IOPS independetly with IO family and IO2 is in general better then IO1.

upvoted 2 times

somsundar 2 years, 11 months ago

@maver144 - That's the case with GP2 volumes. With GP3 we can define IOPS independent of storage capacity.

upvoted 4 times

Joselucho38 3 years ago

Selected Answer: C

Therefore, the most suitable and cost-effective option in this scenario is the GP3 volume type (option C).

upvoted 2 times

Efren 3 years ago

Selected Answer: C

GPS3 allows 16000 IOPS

upvoted 4 times

A company needs to store data from its healthcare application. The application's data frequently changes. A new regulation requires audit access at all levels of the stored data.

The company hosts the application on an on-premises infrastructure that is running out of storage capacity. A solutions architect must securely migrate the existing data to AWS while satisfying the new regulation.

Which solution will meet these requirements?

- A. Use AWS DataSync to move the existing data to Amazon S3. Use AWS CloudTrail to log data events. **Most Voted**
- B. Use AWS Snowcone to move the existing data to Amazon S3. Use AWS CloudTrail to log management events.
- C. Use Amazon S3 Transfer Acceleration to move the existing data to Amazon S3. Use AWS CloudTrail to log data events.
- D. Use AWS Storage Gateway to move the existing data to Amazon S3. Use AWS CloudTrail to log management events.

Correct Answer: A

Community vote distribution

A (69%)

D (31%)

Comments

thanhnv142 **Highly Voted** 2 years, 7 months ago

A is better because:

- Data sync is used for migrate. Storage gw is used to connect on-prem to AWS.
- dataevents is to log for access, management events is for config or management

upvoted 15 times

pentium75 **Highly Voted** 2 years, 5 months ago

Selected Answer: A

We need to log "access at all levels" aka "data events", thus B and D are out (logging only "management events" like granting permissions or changing the access tier).

C, S3 Transfer Acceleration is to increase upload performance from widespread sources or over unreliable networks, but it just provides an endpoint, it does not upload anything itself.

upvoted 11 times

hpirnaj **Most Recent** 1 year, 5 months ago

Selected Answer: A

data management shows management operations that are performed on resources in your AWS account so, D is wrong
<https://repost.aws/knowledge-center/cloudtrail-data-management-events>

upvoted 2 times

JA2018 1 year, 6 months ago

Selected Answer: A

Keys in STEM:

A company needs to store data from its healthcare application. The application's data frequently changes. A new regulation requires audit access at all levels of the stored data.

1. Must securely migrate the existing data to AWS while satisfying the new regulation.

2. Logs access at all data levels

upvoted 2 times

1e22522 1 year, 10 months ago

Selected Answer: A

Typical DataSync scenario me thinks!

upvoted 3 times

osmk 2 years, 3 months ago

Selected Answer: A

Use AWS DataSync to migrate existing data to Amazon S3 <https://aws.amazon.com/datasync/faqs/>

upvoted 4 times

NayeraB 2 years, 3 months ago

Selected Answer: A

It's DataSync for me

upvoted 3 times

frmrkc 2 years, 4 months ago

Selected Answer: D

Storage Gateway integration with CloudTrail :

<https://docs.aws.amazon.com/filegateway/latest/filefsxw/logging-using-cloudtrail.html>

whereas DataSync can be monitored with Amazon CloudWatch:

<https://docs.aws.amazon.com/datasync/latest/userguide/monitor-datasync.html>

upvoted 2 times

frmrkc 2 years, 4 months ago

and here are all Storage Gateway actions monitored by CloudTrail:

https://docs.aws.amazon.com/storagegateway/latest/APIReference/API_Operations.html

upvoted 1 times

awsgeek75 2 years, 5 months ago

Selected Answer: A

B and C don't solve the problem

A is extending the data and management events are for administrative actions only (tracking account creation, user security actions etc.).

C uses DataSync to move all the data and logs data events which include S3 file uploads and downloads.

Management events: User logs into an EC2 instance, creates an S3 IAM role

Data events: User uploads a file to S3

upvoted 4 times

benacert 2 years, 5 months ago

A- DataSync secure fast data transfer

upvoted 2 times

ZZZ_Sleep 2 years, 5 months ago

Selected Answer: D

Keyword of this question = running out of storage capacity

AWS Storage Gateway = extend the on-premises storage

AWS DataSync = copy data between on-premises storage

So, the answer should be D (AWS Storage Gateway)

upvoted 6 times

pentium75 2 years, 5 months ago

"Securely migrate the existing data to AWS" -> move data away from on-premises storage to AWS. Plus, D logs only management events, not "access at all levels".

upvoted 6 times

aws94 2 years, 6 months ago

Selected Answer: D

AWS DataSync is designed for fast, simple, and secure data transfer, but it focuses more on data synchronization rather than on-premises migration.

upvoted 1 times

pentium75 2 years, 5 months ago

Thus it is wrong, but more because of the incorrect logging option in this answer.

upvoted 1 times

meowruki 2 years, 6 months ago

Selected Answer: A

AWS DataSync is suitable for data transfer and synchronization

Option D (Use AWS Storage Gateway to move the existing data to Amazon S3. Use AWS CloudTrail to log management events): AWS Storage

Gateway is typically used for hybrid cloud storage solutions and may introduce additional complexity for a one-time data migration task. It might not be as straightforward as using AWS Snowcone for this specific scenario.

upvoted 2 times

chikuwan 2 years, 6 months ago

Selected Answer: A

both DataSync and Storage Gateway are fine to sync data...but to "audit access at all levels of the stored data" ...it should be data events(data plane operation)..management event is some account level things.

So answer should be A

upvoted 3 times

bogobob 2 years, 6 months ago

Selected Answer: D

While both DataSync and Storage Gateway allow syncing of data between on-premise and cloud, DataSync is built for rapid shifting of data into a cloud environment, not specifically for continued use in on-premise servers.

upvoted 2 times

potomac 2 years, 7 months ago

Selected Answer: A

AWS DataSync is an online data transfer service that simplifies, automates, and accelerates the process of copying large amounts of data to and from AWS storage services over the Internet or over AWS Direct Connect.

upvoted 2 times

pentium75 2 years, 5 months ago

What about logging?

upvoted 2 times

JA2018 1 year, 6 months ago

This is where AWS CloudTrail comes into play.

upvoted 1 times

canonlycontainletters1 2 years, 7 months ago

Selected Answer: A

A seems to be more convincing to me.

upvoted 2 times

A solutions architect is implementing a complex Java application with a MySQL database. The Java application must be deployed on Apache Tomcat and must be highly available.

What should the solutions architect do to meet these requirements?

- A. Deploy the application in AWS Lambda. Configure an Amazon API Gateway API to connect with the Lambda functions.
- B. Deploy the application by using AWS Elastic Beanstalk. Configure a load-balanced environment and a rolling deployment policy. **Most Voted**
- C. Migrate the database to Amazon ElastiCache. Configure the ElastiCache security group to allow access from the application.
- D. Launch an Amazon EC2 instance. Install a MySQL server on the EC2 instance. Configure the application on the server. Create an AMI. Use the AMI to create a launch template with an Auto Scaling group.

Correct Answer: B

Community vote distribution

B (100%)

Comments

cloudehustiaist **Highly Voted** 3 years ago

B

AWS Elastic Beanstalk provides an easy and quick way to deploy, manage, and scale applications. It supports a variety of platforms, including Java and Apache Tomcat. By using Elastic Beanstalk, the solutions architect can upload the Java application and configure the environment to run Apache Tomcat.

upvoted 11 times

KennethNg923 **Most Recent** 1 year, 12 months ago

Selected Answer: B

Beanstalk for sure

upvoted 2 times

zinabu 2 years, 1 month ago

By using Elastic Beanstalk, the solutions architect can upload the Java application and configure the environment to run Apache Tomcat.

upvoted 3 times

wizcloudifa 2 years, 1 month ago

Selected Answer: B

The key word here from the question if you notice is "The Java application must be DEPLOYED..." hence Elastic Beanstalk, it is a serverless deployment service and supports a variety of platforms (Apache Tomcat in our situation), and it will scale automatically with less operational overhead (unlike option D with a lot of operation overhead)

upvoted 3 times

awsgeek75 2 years, 4 months ago

Selected Answer: B

<https://aws.amazon.com/elasticbeanstalk/details/>

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

B. Deploy the application by using AWS Elastic Beanstalk. Configure a load-balanced environment and a rolling deployment policy.

upvoted 4 times

james2033 2 years, 10 months ago

Selected Answer: B

Keyword "AWS Elastic Beanstalk" for re-architecture from Java web-app inside Apache Tomcat to AWS Cloud.

upvoted 3 times

TariqKipkemei 3 years ago

Selected Answer: B

Definitely B

upvoted 2 times

antropaws 3 years ago

Selected Answer: B

Clearly B.

upvoted 2 times

nosense 3 years ago

Selected Answer: B

Easy deploy, management and scale

upvoted 3 times

greyrose 3 years ago

Selected Answer: B

BB

upvoted 1 times

A serverless application uses Amazon API Gateway, AWS Lambda, and Amazon DynamoDB. The Lambda function needs permissions to read and write to the DynamoDB table.

Which solution will give the Lambda function access to the DynamoDB table MOST securely?

A. Create an IAM user with programmatic access to the Lambda function. Attach a policy to the user that allows read and write access to the DynamoDB table. Store the `access_key_id` and `secret_access_key` parameters as part of the Lambda environment variables. Ensure that other AWS users do not have read and write access to the Lambda function configuration.

B. Create an IAM role that includes Lambda as a trusted service. Attach a policy to the role that allows read and write access to the DynamoDB table. Update the configuration of the Lambda function to use the new role as the execution role. **Most Voted**

C. Create an IAM user with programmatic access to the Lambda function. Attach a policy to the user that allows read and write access to the DynamoDB table. Store the `access_key_id` and `secret_access_key` parameters in AWS Systems Manager Parameter Store as secure string parameters. Update the Lambda function code to retrieve the secure string parameters before connecting to the DynamoDB table.

D. Create an IAM role that includes DynamoDB as a trusted service. Attach a policy to the role that allows read and write access from the Lambda function. Update the code of the Lambda function to attach to the new role as an execution role.

Correct Answer: B

Community vote distribution

B (100%)

Comments

awsgeek75 **Highly Voted** 2 years, 5 months ago

Selected Answer: B

DynamoDB needs to trust Lambda. NOT the other way around. So Lambda must be configured as a trusted service. Role for service which gives B and D options. D is setting up (somehow?) to allow Lambda to trust DynamoDB... or the wording makes no sense.

upvoted 8 times

james2033 **Highly Voted** 2 years, 10 months ago

Selected Answer: B

Keyword B. "IAM role that includes Lambda as a trusted service", not "IAM role that includes DynamoDB as a trusted service" in D. It is IAM role, not IAM user.

upvoted 6 times

KennethNg923 **Most Recent** 1 year, 12 months ago

Selected Answer: B

IAM Role for access to DynamoDB, not for access Lambda

upvoted 2 times

antropaws 3 years ago

Selected Answer: B

B sounds better.

upvoted 3 times

omoakin 3 years ago

BBBBBBBBBB

upvoted 2 times

alvinnguyennexcel 3 years ago

Selected Answer: B

vote B

upvoted 2 times

cloudenthusiast 3 years ago

B

Option B suggests creating an IAM role that includes Lambda as a trusted service, meaning the role is specifically designed for Lambda functions. The role should have a policy attached to it that grants the required read and write access to the DynamoDB table.

upvoted 4 times

nosense 3 years ago

Selected Answer: B

B is right

Role key word and trusted service lambda

upvoted 5 times

The following IAM policy is attached to an IAM group. This is the only policy applied to the group.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "1",
      "Effect": "Allow",
      "Action": "ec2:*",
      "Resource": "*",
      "Condition": {
        "StringEquals": {
          "ec2:Region": "us-east-1"
        }
      }
    },
    {
      "Sid": "2",
      "Effect": "Deny",
      "Action": [
        "ec2:StopInstances",
        "ec2:TerminateInstances"
      ],
      "Resource": "*",
      "Condition": {
        "BoolIfExists": {"aws:MultiFactorAuthPresent": false}
      }
    }
  ]
}
```

What are the effective IAM permissions of this policy for group members?

- A. Group members are permitted any Amazon EC2 action within the us-east-1 Region. Statements after the Allow permission are not applied.
- B. Group members are denied any Amazon EC2 permissions in the us-east-1 Region unless they are logged in with multi-factor authentication (MFA).
- C. Group members are allowed the ec2:StopInstances and ec2:TerminateInstances permissions for all Regions when logged in with multi-factor authentication (MFA). Group members are permitted any other Amazon EC2 action.
- D. Group members are allowed the ec2:StopInstances and ec2:TerminateInstances permissions for the us-east-1 Region only when logged in with multi-factor authentication (MFA). Group members are permitted any other Amazon EC2 action within the us-east-1 Region. **Most Voted**

Correct Answer: D

Community vote distribution

D (90%)

10%

Comments

jack79 **Highly Voted** 2 years, 12 months ago
came in exam today
upvoted 14 times

KennethNg923 **Most Recent** 1 year, 12 months ago

Selected Answer: D

for the us-east-1 Region only, not for all region
upvoted 2 times

wizcloudifa 2 years, 1 month ago

Selected Answer: D

One of the few situations when actual answer is same as the most voted answer lol
upvoted 2 times

pdragon1981 2 years, 5 months ago

Selected Answer: C

Not sure why everyone vote D, I think that the valid option as to be C as the second condition regarding MFA there is point that only refer to a specific region, so basically this means that is for all the regions
upvoted 2 times

pdragon1981 2 years, 5 months ago

Ok ignore D is right as the first condition is what gives permission to make anything for EC2 but is restricted to us-east-1 region
upvoted 6 times

yodelin 2 years, 8 months ago

the json is describing a lot of things apparently, so I go with the longest answer lol
upvoted 1 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

D. Group members are allowed the ec2:StopInstances and ec2:TerminateInstances permissions for the us-east-1 Region only when logged in with multi-factor authentication (MFA). Group members are permitted any other Amazon EC2 action within the us-east-1 Region
upvoted 3 times

james2033 2 years, 10 months ago

Selected Answer: D

A. "Statements after the Allow permission are not applied." --> Wrong.
B. "denied any Amazon EC2 permissions in the us-east-1 Region" --> Wrong. Just deny 2 items.
C. "allowed the ec2:StopInstances and ec2:TerminateInstances permissions for all Regions" --> Wrong. Just region us-east-1.
D. ok.
upvoted 2 times

TariqKipkemei 3 years ago

Selected Answer: D

Only D makes sense
upvoted 2 times

antropaws 3 years ago

Selected Answer: D

D sounds about right.
upvoted 2 times

alvinnguyennexcel 3 years ago

Selected Answer: D

D is correct
upvoted 3 times

omoakin 3 years ago

D is correct
upvoted 2 times

nosense 3 years ago

Selected Answer: D

D is right
upvoted 3 times

A manufacturing company has machine sensors that upload .csv files to an Amazon S3 bucket. These .csv files must be converted into images and must be made available as soon as possible for the automatic generation of graphical reports.

The images become irrelevant after 1 month, but the .csv files must be kept to train machine learning (ML) models twice a year. The ML trainings and audits are planned weeks in advance.

Which combination of steps will meet these requirements MOST cost-effectively? (Choose two.)

A. Launch an Amazon EC2 Spot Instance that downloads the .csv files every hour, generates the image files, and uploads the images to the S3 bucket.

B. Design an AWS Lambda function that converts the .csv files into images and stores the images in the S3 bucket. Invoke the Lambda function when a .csv file is uploaded. **Most Voted**

C. Create S3 Lifecycle rules for .csv files and image files in the S3 bucket. Transition the .csv files from S3 Standard to S3 Glacier 1 day after they are uploaded. Expire the image files after 30 days. **Most Voted**

D. Create S3 Lifecycle rules for .csv files and image files in the S3 bucket. Transition the .csv files from S3 Standard to S3 One Zone-Infrequent Access (S3 One Zone-IA) 1 day after they are uploaded. Expire the image files after 30 days.

E. Create S3 Lifecycle rules for .csv files and image files in the S3 bucket. Transition the .csv files from S3 Standard to S3 Standard-Infrequent Access (S3 Standard-IA) 1 day after they are uploaded. Keep the image files in Reduced Redundancy Storage (RRS).

Correct Answer: BC

Community vote distribution

BC (90%)

10%

Comments

vesen22 **Highly Voted** 2 years, 6 months ago

Selected Answer: BC

<https://docs.aws.amazon.com/amazonglacier/latest/dev/introduction.html>

upvoted 5 times

awsgeek75 **Most Recent** 1 year, 11 months ago

Selected Answer: BC

B for processing the images via Lambda as it's more cost efficient than EC2 spot instances

C for expiring images after 30 days and because the ML trainings are planned weeks in advance so S3 glacier is ideal for slow retrieval and cheap storage.

D and E uses S3 infrequent access which is more expensive than glacier

upvoted 3 times

pentium75 1 year, 11 months ago

Selected Answer: BC

Not A, we need the images "as soon as possible", A runs every hour

"ML trainings and audits are planned weeks in advance" thus Glacier (C) is ok.

upvoted 3 times

Xin123 2 years, 2 months ago

Selected Answer: BC

Answer is B&C. For D, you must store data for 30 days in s3 standard before move to IA tiers, glacier is fine

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/lifecycle-transition-general->

considerations.html#:~:text=Before%20you%20transition%20objects%20to%20S3%20Standard%2DIA%20or%20S3%20One%20Zone%2DIA%2C%20you%20must%20store%20them%20for%20at%20least%2030%20days%20in%20Amazon%20S3

upvoted 4 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: BC

Definitely B & C

upvoted 3 times

jayce5 2 years, 4 months ago

Selected Answer: BC

A. Wrong, the .csv files must be processed asap.

D and E are incorrect since Glacier is the most cost-effective option, and plans for using .csv files are known weeks in advance.

upvoted 2 times

james2033 2 years, 4 months ago

Why need "These .csv files must be converted into images"?

upvoted 1 times

awsgeek75 1 year, 10 months ago

Because they are being used in some graphical reports (probably fancy powerpoint presentations!)

upvoted 2 times

smartegnine 2 years, 5 months ago

Selected Answer: BC

the key word is Weeks in advance, even you save data in S3 Gracia will also OK to take couples days to retrieve the data

upvoted 3 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: BC

Definitely B & C

upvoted 2 times

Abrar2022 2 years, 6 months ago

Selected Answer: BC

A. Wrong because Lifecycle rule is not mentioned.

B. CORRECT

C. CORRECT

D. Why Store on S3 One Zone-Infrequent Access (S3 One Zone-IA) when the files are going to irrelevant after 1 month? (Availability 99.99% - consider cost)

E. again, Why use Reduced Redundancy Storage (RRS) when the files are irrelevant after 1 month? (Availability 99.99% - consider cost)

upvoted 4 times

RoroJ 2 years, 6 months ago

Selected Answer: BE

B: Serverless and fast responding

E: will keep .csv file for a year, C and D expires the file after 30 days.

upvoted 4 times

RoroJ 2 years, 6 months ago

B&C, misread the question, expires the image files after 30 days.

upvoted 3 times

hiroohiroo 2 years, 6 months ago

Selected Answer: BC

<https://aws.amazon.com/jp/about-aws/whats-new/2021/11/amazon-s3-glacier-storage-class-amazon-s3-glacier-flexible-retrieval/>

upvoted 3 times

nosense 2 years, 6 months ago

Selected Answer: BC

B severless and cost effective

C corrtl rule to store

upvoted 3 times

A company has developed a new video game as a web application. The application is in a three-tier architecture in a VPC with Amazon RDS for MySQL in the database layer. Several players will compete concurrently online. The game's developers want to display a top-10 scoreboard in near-real time and offer the ability to stop and restore the game while preserving the current scores.

What should a solutions architect do to meet these requirements?

- A. Set up an Amazon ElastiCache for Memcached cluster to cache the scores for the web application to display.
- B. Set up an Amazon ElastiCache for Redis cluster to compute and cache the scores for the web application to display.
- C. Place an Amazon CloudFront distribution in front of the web application to cache the scoreboard in a section of the application.
- D. Create a read replica on Amazon RDS for MySQL to run queries to compute the scoreboard and serve the read traffic to the web application.

Most Voted

Correct Answer: B

Community vote distribution

B (92%)

8%

Comments

Guru4Cloud **Highly Voted** 2 years, 3 months ago

Selected Answer: B

Redis provides fast in-memory data storage and processing. It can compute the top 10 scores and update the cache in milliseconds. ElastiCache Redis supports sorting and ranking operations needed for the top 10 leaderboard. The cached leaderboard can be retrieved from Redis vs hitting the MySQL database for every read. This reduces load on the database. Redis supports persistence, so scores are preserved if the cache stops/restarts

upvoted 10 times

TariqKipkemei **Highly Voted** 2 years, 1 month ago

Selected Answer: B

Real-time gaming leaderboards are easy to create with Amazon ElastiCache for Redis. Just use the Redis Sorted Set data structure, which provides uniqueness of elements while maintaining the list sorted by their scores. Creating a real-time ranked list is as simple as updating a user's score each time it changes. You can also use Sorted Sets to handle time series data by using timestamps as the score.

<https://aws.amazon.com/elasticache/redis/#:~:text=ElastiCache%20for%20Redis,-Gaming,-Leaderboards>

upvoted 8 times

potomac **Most Recent** 2 years, 1 month ago

Selected Answer: B

ElastiCache for Redis sorts and ranks datasets

upvoted 5 times

5ab5e39 2 years, 3 months ago

<https://aws.amazon.com/blogs/database/building-a-real-time-gaming-leaderboard-with-amazon-elasticache-for-redis/>

upvoted 4 times

ukivanlamipi 2 years, 3 months ago

Selected Answer: A

concurrently = memcached

upvoted 3 times

awsgeek75 1 year, 11 months ago

Amazon ElastiCache for Memcached is non-persistent so start/stop of game will lose the score." and offer the ability to stop and restore the game while preserving the current scores."

upvoted 3 times

james2033 2 years, 4 months ago

Selected Answer: B

See case study of leaderboard with Redis at <https://redis.io/docs/data-types/sorted-sets/> , it is feature "sorted sets". See comparison between Redis and Memcached at <https://docs.aws.amazon.com/AmazonElastiCache/latest/mem-ug/SelectEngine.html> , the different at feature "Sorted sets"

upvoted 4 times

live_reply_developers 2 years, 5 months ago

Selected Answer: B

advanced data structures, complex querying, pub/sub messaging, or persistence, Redis may be a better fit.

upvoted 2 times

haoAWS 2 years, 5 months ago

B is correct

upvoted 2 times

jf_topics 2 years, 6 months ago

B correct.

upvoted 2 times

hiroohiroo 2 years, 6 months ago

Selected Answer: B

<https://aws.amazon.com/jp/blogs/news/building-a-real-time-gaming-leaderboard-with-amazon-elasticache-for-redis/>

upvoted 4 times

cloudenthusiast 2 years, 6 months ago

Amazon ElastiCache for Redis is a highly scalable and fully managed in-memory data store. It can be used to store and compute the scores in real time for the top-10 scoreboard. Redis supports sorted sets, which can be used to store the scores as well as perform efficient queries to retrieve the top scores. By utilizing ElastiCache for Redis, the web application can quickly retrieve the current scores without the need to perform complex and potentially resource-intensive database queries.

upvoted 3 times

nosense 2 years, 6 months ago

Selected Answer: B

B is right

upvoted 2 times

Efren 2 years, 6 months ago

More questions!!!

upvoted 5 times

An ecommerce company wants to use machine learning (ML) algorithms to build and train models. The company will use the models to visualize complex scenarios and to detect trends in customer data. The architecture team wants to integrate its ML models with a reporting platform to analyze the augmented data and use the data directly in its business intelligence dashboards.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Use AWS Glue to create an ML transform to build and train models. Use Amazon OpenSearch Service to visualize the data.
- B. Use Amazon SageMaker to build and train models. Use Amazon QuickSight to visualize the data. **Most Voted**
- C. Use a pre-built ML Amazon Machine Image (AMI) from the AWS Marketplace to build and train models. Use Amazon OpenSearch Service to visualize the data.
- D. Use Amazon QuickSight to build and train models by using calculated fields. Use Amazon QuickSight to visualize the data.

Correct Answer: B

Community vote distribution

B (100%)

Comments

vincent2023 1 year, 4 months ago

Selected Answer: B

Sage Maker - ML
upvoted 1 times

67db0ed 1 year, 10 months ago

<https://docs.aws.amazon.com/quicksight/latest/user/sagemaker-integration.html>
upvoted 2 times

awsgeek75 2 years, 5 months ago

Selected Answer: B

Machine Learning = Sage Maker so B for least operational overhead
A and D are not right technologies.
C is possible but with more overhead of using AMI even if you can get OpenSearch to visualize the data somehow which I don't think is possible without massive overhead
upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

Use Amazon SageMaker to build and train models. Use Amazon QuickSight to visualize the data.
upvoted 3 times

james2033 2 years, 10 months ago

Selected Answer: B

Question keyword "machine learning", answer keyword "Amazon SageMaker". Choose B. Use Amazon QuickSight for visualization. See "Gaining insights with machine learning (ML) in Amazon QuickSight" at <https://docs.aws.amazon.com/quicksight/latest/user/making-data-driven-decisions-with-ml-in-quicksight.html>
upvoted 3 times

VellaDevil 2 years, 11 months ago

Selected Answer: B

Sagemaker.
upvoted 2 times

TariqKipkemei 3 years ago

Selected Answer: B

Business intelligence, visualiations = AmazonQuicksight

ML = Amazon SageMaker

upvoted 3 times

antropaws 3 years ago

Selected Answer: B

Most likely B.

upvoted 2 times

omoakin 3 years ago

Amazon SageMaker is a fully managed service that provides every developer and data scientist with the ability to build, train, and deploy ML models quickly.

upvoted 2 times

cloudenthusiast 3 years ago

Amazon SageMaker is a fully managed service that provides a complete set of tools and capabilities for building, training, and deploying ML models. It simplifies the end-to-end ML workflow and reduces operational overhead by handling infrastructure provisioning, model training, and deployment. To visualize the data and integrate it into business intelligence dashboards, Amazon QuickSight can be used. QuickSight is a cloud-native business intelligence service that allows users to easily create interactive visualizations, reports, and dashboards from various data sources, including the augmented data generated by the ML models.

upvoted 4 times

Efren 3 years ago

Selected Answer: B

ML== SageMaker

upvoted 2 times

nosense 3 years ago

Selected Answer: B

B sagemaker provide deploy ml models

upvoted 2 times

A company is running its production and nonproduction environment workloads in multiple AWS accounts. The accounts are in an organization in AWS Organizations. The company needs to design a solution that will prevent the modification of cost usage tags.

Which solution will meet these requirements?

- A. Create a custom AWS Config rule to prevent tag modification except by authorized principals.
- B. Create a custom trail in AWS CloudTrail to prevent tag modification.
- C. Create a service control policy (SCP) to prevent tag modification except by authorized principals. **Most Voted**
- D. Create custom Amazon CloudWatch logs to prevent tag modification.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Guru4Cloud **Highly Voted** 2 years, 3 months ago

Selected Answer: C

Tip: AWS Organizaton + service control policy (SCP) - This for any questions, you see both together. then you tell me

C. Create a service control policy (SCP) to prevent tag modification except by authorized principals.

upvoted 7 times

awsgeek75 **Most Recent** 1 year, 11 months ago

Selected Answer: C

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_scps.html

AWS example for this question/use case:

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_scps_examples_tagging.html#example-require-restrict-tag-mods-to-admin

upvoted 3 times

james2033 2 years, 4 months ago

Selected Answer: C

D "Amazon CloudWatch" just for logging, not for prevent tag modification

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_tag-policies-cwe.html

Amazon Organizaton has "Service Control Policy (SCP)" with "tag policy"

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_tag-policies.html . Choose C.

AWS Config for technical stuff, not for tag policies. Not A.

upvoted 4 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: C

Service control policies (SCPs) are a type of organization policy that you can use to manage permissions in your organization.

upvoted 2 times

alexandercamachop 2 years, 6 months ago

Selected Answer: C

Anytime we need to restrict anything in an AWS Organization, it is SCP Policies.

upvoted 3 times

Abrar2022 2 years, 6 months ago

AWS Config is for tracking configuration changes

upvoted 2 times

Abrar2022 2 years, 6 months ago

so it's wrong. Right answer is C

upvoted 3 times

antropaws 2 years, 6 months ago

Selected Answer: C

I'd say C.

upvoted 3 times

hiroohiroo 2 years, 6 months ago

Selected Answer: C

https://docs.aws.amazon.com/ja_jp/organizations/latest/userguide/orgs_manage_policies_scps_examples_tagging.html

upvoted 4 times

nosense 2 years, 6 months ago

Selected Answer: C

Denies tag: modify

upvoted 3 times

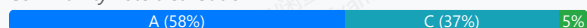
A company hosts its application in the AWS Cloud. The application runs on Amazon EC2 instances behind an Elastic Load Balancer in an Auto Scaling group and with an Amazon DynamoDB table. The company wants to ensure the application can be made available in another AWS Region with minimal downtime.

What should a solutions architect do to meet these requirements with the LEAST amount of downtime?

- A. Create an Auto Scaling group and a load balancer in the disaster recovery Region. Configure the DynamoDB table as a global table. Configure DNS failover to point to the new disaster recovery Region's load balancer. **Most Voted**
- B. Create an AWS CloudFormation template to create EC2 instances, load balancers, and DynamoDB tables to be launched when needed. Configure DNS failover to point to the new disaster recovery Region's load balancer.
- C. Create an AWS CloudFormation template to create EC2 instances and a load balancer to be launched when needed. Configure the DynamoDB table as a global table. Configure DNS failover to point to the new disaster recovery Region's load balancer.
- D. Create an Auto Scaling group and load balancer in the disaster recovery Region. Configure the DynamoDB table as a global table. Create an Amazon CloudWatch alarm to trigger an AWS Lambda function that updates Amazon Route 53 pointing to the disaster recovery load balancer.

Correct Answer: A

Community vote distribution



Comments

lucdt4 **Highly Voted** 3 years ago

Selected Answer: A

A and D is correct.
But Route 53 has a feature DNS failover when instances down so we don't need use Cloudwatch and lambda to trigger
-> A correct
upvoted 17 times

Wablo 2 years, 11 months ago

Yes it does but you configure it. It's not automated anymore. D is the best answer!
upvoted 2 times

Kp88 2 years, 10 months ago

What are you talking about configuring? Yes you have to configure everything at some point
<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/dns-failover-configuring.html>
upvoted 2 times

JA2018 1 year, 6 months ago

to be launched when needed => requires manual human intervention

Certainly will not help with requirement for the least amount of downtime.
upvoted 1 times

smartegnine 2 years, 11 months ago

Did not see Route 53 in this question right? So my opinion is D
upvoted 1 times

pentium75 **Highly Voted** 2 years, 5 months ago

Selected Answer: C

They are not asking for automatic failover, they want to "ensure the application can (!) be made available in another AWS Region with minimal downtime". This works with C; they would just execute the template and it would be available in short time.

A would create a DR environment that IS already available, which is not what the question asks for.
D is like A, just abusing Lambda to update the DNS record (which doesn't make sense).
B would create a separate, empty database

upvoted 10 times

bora4motion 1 year, 1 month ago

I am going with A. There's no downtime with A, maybe a blip because of DNS.

upvoted 1 times

Charlesvg Most Recent 11 months, 2 weeks ago

Selected Answer: A

obviously A, as launching a cloudformation template is not the LEAST Amounts of time compare to A

upvoted 1 times

anikolov 2 years, 4 months ago

Selected Answer: A

With the LEAST amount of downtime = A

Cost effective = C, but risky some of EC2 types/capacity not available in Region at the time, when need to switch to DR

upvoted 8 times

awsgeek75 2 years, 5 months ago

Selected Answer: C

There are 2 parts. DB and application. Dynamo DB recovery in another region is not possible without global table so option B is out.

A will make the infra available in 2 regions which is not required. The question is about DR, not scaling.

D Use Lambda to modify R53 to point to new region. This is going to cause delays but is possible and it will also be running a scaled EC2 instances in a passive region.

C Make a CF template which can launch the infra when needed. DB is global table so it will be available.

upvoted 4 times

LeonSauveterre 1 year, 6 months ago

Option C relies on launching EC2 instances and load balancers only when needed, which may increase downtime. On contrary, I think A is better because it ensures resources are pre-configured and ready-to-go.

It's true nobody mentioned anything about infra availability in 2 regions... but it couldn't hurt since we simply need the least amount of downtime.

upvoted 1 times

bora4motion 1 year, 1 month ago

you need minimum downtime, nobody asks about what's required or not apart of minimal downtime. I'm going with a.

upvoted 1 times

meowruki 2 years, 6 months ago

Selected Answer: C

AWS CloudFormation Template: Use CloudFormation to define the infrastructure components (EC2 instances, load balancer, etc.) in a template. This allows for consistent and repeatable infrastructure deployment.

EC2 Instances and Load Balancer: Launch the EC2 instances and load balancer in the disaster recovery (DR) Region using the CloudFormation template. This enables the deployment of the application in the DR Region when needed.

DynamoDB Global Table: Configure the DynamoDB table as a global table. DynamoDB Global Tables provide automatic multi-region, multi-master replication, ensuring that the data is available in both the primary and DR Regions.

DNS Failover: Configure DNS failover to point to the new DR Region's load balancer. This allows for seamless failover of traffic to the DR Region when needed.

Option A is close, but it introduces an Auto Scaling group in the disaster recovery Region, which might introduce unnecessary complexity and potential scaling delays. Option D introduces a Lambda function triggered by CloudWatch alarms, which might add latency and complexity compared to the more direct approach in Option C.

upvoted 1 times

bogobob 2 years, 6 months ago

Selected Answer: A

Assuming they're using Route53 as a DNS then A <https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/dns-failover.html>

upvoted 3 times

EEK2k 2 years, 7 months ago

Selected Answer: C

Only B and C take care of EC2 instances. But since B does not take care of Data in the Dynamo DB, C is the only correct Answer.

upvoted 1 times

potomac 2 years, 7 months ago

Selected Answer: A

Route 53 has a feature DNS failover when instances down
upvoted 1 times

thanhnv142 2 years, 7 months ago

C is the best choice here
upvoted 1 times

Wayne23Fang 2 years, 7 months ago

Selected Answer: C

I think CloudFormation is easier than manual provision of Auto Scaling group and load balancer in DR region.
upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: A

Creating Auto Scaling group and load balancer in DR region allows fast launch of capacity when needed.
Configuring DynamoDB as a global table provides continuous data replication.
Using DNS failover via Route 53 to point to the DR region's load balancer enables rapid traffic shifting.
upvoted 2 times

Wablo 2 years, 11 months ago

Both Option A and Option D include the necessary steps of setting up an Auto Scaling group and load balancer in the disaster recovery Region, configuring the DynamoDB table as a global table, and updating DNS records. However, Option D provides a more detailed approach by explicitly mentioning the use of an Amazon CloudWatch alarm and AWS Lambda function to automate the DNS update process.

By leveraging an Amazon CloudWatch alarm, Option D allows for an automated failover mechanism. When triggered, the CloudWatch alarm can execute an AWS Lambda function, which in turn can update the DNS records in Amazon Route 53 to redirect traffic to the disaster recovery load balancer in the new Region. This automation helps reduce the potential for human error and further minimizes downtime.

Answer is D

upvoted 2 times

Kp88 2 years, 10 months ago

Failover policy takes care of DNS record update so no need for cloud watch/lambda
upvoted 1 times

TariqKipkemei 2 years, 12 months ago

Selected Answer: C

The company wants to ensure the application 'CAN' be made available in another AWS Region with minimal downtime. Meaning they want to be able to launch infra on need basis.

Best answer is C.

upvoted 2 times

Wablo 2 years, 11 months ago

minimal downtime not minimal effort!

D

upvoted 1 times

dajform 2 years, 11 months ago

B, C are not OK because "launching resources when needed", which will increase the time to recover "DR"

upvoted 2 times

AshishRocks 3 years ago

I feel it is A

Configure DNS failover: Use DNS failover to point the application's DNS record to the load balancer in the disaster recovery Region. DNS failover allows you to route traffic to the disaster recovery Region in case of a failure in the primary Region.

upvoted 3 times

Wablo 2 years, 11 months ago

Once you configure manually the DNS, it's no more automated like Lambda does.

upvoted 2 times

Yadav_Sanjay 3 years ago

Selected Answer: C

C suits best

upvoted 3 times

hiroohiroo 3 years ago

Selected Answer: A

AがDNS フェイルオーバー
upvoted 1 times

A company needs to migrate a MySQL database from its on-premises data center to AWS within 2 weeks. The database is 20 TB in size. The company wants to complete the migration with minimal downtime.

Which solution will migrate the database MOST cost-effectively?

- A. Order an AWS Snowball Edge Storage Optimized device. Use AWS Database Migration Service (AWS DMS) with AWS Schema Conversion Tool (AWS SCT) to migrate the database with replication of ongoing changes. Send the Snowball Edge device to AWS to finish the migration and continue the ongoing replication. **Most Voted**
- B. Order an AWS Snowmobile vehicle. Use AWS Database Migration Service (AWS DMS) with AWS Schema Conversion Tool (AWS SCT) to migrate the database with ongoing changes. Send the Snowmobile vehicle back to AWS to finish the migration and continue the ongoing replication.
- C. Order an AWS Snowball Edge Compute Optimized with GPU device. Use AWS Database Migration Service (AWS DMS) with AWS Schema Conversion Tool (AWS SCT) to migrate the database with ongoing changes. Send the Snowball device to AWS to finish the migration and continue the ongoing replication
- D. Order a 1 GB dedicated AWS Direct Connect connection to establish a connection with the data center. Use AWS Database Migration Service (AWS DMS) with AWS Schema Conversion Tool (AWS SCT) to migrate the database with replication of ongoing changes.

Correct Answer: A

Community vote distribution

A (89%)

11%

Comments

nosense **Highly Voted** 2 years, 6 months ago

Selected Answer: A

A) 300 first 10 days. 150 shipping

D) 750 for 2 weeks

upvoted 14 times

Efren 2 years, 6 months ago

Thanks, i was checking the speed more than price. Thanks for the clarification

upvoted 3 times

Goutham4981 **Highly Voted** 2 years ago

Selected Answer: A

Direct Connect takes at least 1 month to setup - D is invalid

AWS Snowmobile is used for transferring large amounts of data (petabytes) from remote locations where establishing a connection to the cloud is impossible - B is invalid

AWS Snowball Edge Compute Optimized provides higher vCPU performance and lower storage as compared to Snowball storage optimized. As our need is solely data transfer, high vCPU performance is not required but high storage is - C is invalid

upvoted 7 times

zinabu **Most Recent** 1 year, 7 months ago

Selected Answer: A

But I didn't understand why we are using a schema conversion tool, because AWS have already a managed service engine for MySQL Db, (RDS for MySQL or Aurora for my SQL is on the table)

upvoted 7 times

EEK2k 2 years, 1 month ago

Selected Answer: D

To calculate the time it would take to transfer 20TB of data over a 1 GB dedicated AWS Direct Connect, we can use the formula:

time = data size / data transfer rate

Here, the data size is 20TB, which is equivalent to 20,000 GB or 20,000,000 MB. The data transfer rate is 1 GB/s.

Converting the data size to MB, we get:

20,000,000 MB / 1 GB/s = 20,000 seconds

Therefore, it would take approximately 20,000 seconds or 5.56 hours to transfer 20TB of data over a 1 GB dedicated AWS Direct Connect.
upvoted 3 times

Murtadhaceit 2 years ago

It takes more way more than 2 weeks to setup Direct Connect. Therefore, D is not valid since we have to do the transfer within 2 weeks.
upvoted 4 times

potomac 2 years, 1 month ago

Selected Answer: A

C is wrong, GPU is not needed
upvoted 4 times

Ramdi1 2 years, 2 months ago

Selected Answer: A

Has to be A. the option for D would only work if they said they have like 6 Months plus. It would take too long to set up.
upvoted 3 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: A

I agreed with A.
Why not D?
When you initiate the process by requesting an AWS Direct Connect connection, it typically starts with the AWS Direct Connect provider. This provider may need to coordinate with AWS to allocate the necessary resources. This initial setup phase can take anywhere from a few days to a couple of weeks.
Couple of weeks? No Good
upvoted 4 times

Guru4Cloud 2 years, 3 months ago

When you create a Snowball job in the AWS console, it will estimate the delivery date based on your location. Being near a facility shows 1-2 day estimated delivery.
For extremely urgent requests, you can contact AWS Support and inquire about expedited Snowball delivery. If inventory is available, they may be able to ship same day or next day.
upvoted 3 times

james2033 2 years, 4 months ago

Selected Answer: A

Keyword "20 TB", choose "AWS Snowball", there are A or C. C has word "GPU" what is not related, therefore choose A.
upvoted 3 times

Zox42 2 years, 5 months ago

Selected Answer: A

Answer A
upvoted 2 times

MrAWSAssociate 2 years, 5 months ago

Selected Answer: D

D is correct
upvoted 1 times

pentium75 1 year, 11 months ago

No, takes months, not weeks
upvoted 2 times

DrWatson 2 years, 6 months ago

Selected Answer: A

https://docs.aws.amazon.com/dms/latest/userguide/CHAP_LargeDBs.Process.html
upvoted 3 times

RoroJ 2 years, 6 months ago

Selected Answer: A

D Direct Connection will need a long time to setup plus need to deal with Network and Security changes with existing environment. Ad then plus the Data trans time... No way can be done in 2 weeks.

upvoted 5 times

Joselucho38 2 years, 6 months ago

Selected Answer: D

Overall, option D combines the reliability and cost-effectiveness of AWS Direct Connect, AWS DMS, and AWS SCT to migrate the database efficiently and minimize downtime.

upvoted 2 times

Abhineet9148232 2 years, 6 months ago

Selected Answer: A

D - Direct Connect takes atleast a month to setup! Requirement is for within 2 weeks.

upvoted 5 times

Rob1L 2 years, 6 months ago

Selected Answer: D

AWS Snowball Edge Storage Optimized device is used for large-scale data transfers, but the lead time for delivery, data transfer, and return shipping would likely exceed the 2-week time frame. Also, ongoing database changes wouldn't be replicated while the device is in transit.

upvoted 1 times

Rob1L 2 years, 6 months ago

Change to A because "Most cost effective"

upvoted 3 times

hiroohiroo 2 years, 6 months ago

Selected Answer: A

https://docs.aws.amazon.com/ja_jp/snowball/latest/developer-guide/device-differences.html#device-options

Aです。

upvoted 4 times

norris81 2 years, 6 months ago

Selected Answer: A

How long does direct connect take to provision ?

upvoted 2 times

examtopictempacc 2 years, 6 months ago

At least one month and expensive.

upvoted 1 times

A company moved its on-premises PostgreSQL database to an Amazon RDS for PostgreSQL DB instance. The company successfully launched a new product. The workload on the database has increased. The company wants to accommodate the larger workload without adding infrastructure.

Which solution will meet these requirements MOST cost-effectively?

- A. Buy reserved DB instances for the total workload. Make the Amazon RDS for PostgreSQL DB instance larger. **Most Voted**
- B. Make the Amazon RDS for PostgreSQL DB instance a Multi-AZ DB instance.
- C. Buy reserved DB instances for the total workload. Add another Amazon RDS for PostgreSQL DB instance.
- D. Make the Amazon RDS for PostgreSQL DB instance an on-demand DB instance.

Correct Answer: A

Community vote distribution

A (84%)

Other

Comments

elmogy **Highly Voted** 3 years ago

Selected Answer: A

A.
"without adding infrastructure" means scaling vertically and choosing larger instance.
"MOST cost-effectively" reserved instances
upvoted 17 times

wsdasdasdqwda 2 years, 7 months ago

"MOST cost-effectively" doesn't mean reserved instances. Only in this case it is but not in general.
upvoted 6 times

zdi561 **Most Recent** 1 year, 4 months ago

Selected Answer: B

without adding infrastructure means without adding more capacity per instance, multi-az adding read replicas solve the problem.
upvoted 1 times

awsgeek75 2 years, 5 months ago

Selected Answer: A

accommodate the larger workload without adding infrastructure. = Reserved DB instance
upvoted 3 times

awsgeek75 2 years, 5 months ago

+ make the instance larger so most cost effective is to reserve a large instance suitable for workload which is A
upvoted 3 times

pentium75 2 years, 5 months ago

Selected Answer: A

B - Multi-AZ is for HA, does not help 'accommodating the larger workload'
C - Adding "another instance" will not help, we can't split the workload between two instances
D - On-demand instance is a good choice for unknown workload, but here we know the workload, it's just higher than before
upvoted 3 times

Goutham4981 2 years, 6 months ago

Selected Answer: A

Cannot add more infrastructure - C is invalid
Multi AZ DB instance is for high availability and failure mitigation, does not increase performance, higher workload support - B is invalid
On demand instances are costlier than Reserved instances - D is invalid

upvoted 2 times

bogobob 2 years, 6 months ago

Selected Answer: D

Not A : "launched a new product", reserved instances are for known workloads, a new product doesn't have known workload.
Not B : "accommodate the larger workload", while Multi-AZ can help with larger workloads, they are more for higher availability.
Not C : "without adding infrastructure", adding a PostGresQL instance is new infrastructure.

upvoted 3 times

pentium75 2 years, 5 months ago

Question says nothing about unknown load. New product -> more total products -> load has increased.

upvoted 2 times

JA2018 1 year, 6 months ago

Please refer to following Keys in Stem:

1. The company successfully launched a new product.
2. The workload on the database has increased.
3. The company wants to accommodate the larger workload without adding infrastructure.

upvoted 1 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

B is the best approach in this scenario overall:

Making the RDS PostgreSQL instance Multi-AZ adds a standby replica to handle larger workloads and provides high availability. Even though it adds infrastructure, the cost is less than doubling the infrastructure with a separate DB instance. It provides better performance, availability, and disaster recovery than a single larger instance.

upvoted 2 times

BillyBlunts 2 years, 8 months ago

Agreed the answer is B

Multi-AZ deployments are cost-effective because they leverage the standby instance without incurring additional charges. You only pay for the primary instance's regular usage costs.

upvoted 1 times

pentium75 2 years, 5 months ago

Multi-AZ is for HA, does not add performance. Meaning, will not help 'accommodating the larger workload'.

upvoted 3 times

james2033 2 years, 10 months ago

Selected Answer: A

Buy larger instance.

upvoted 2 times

james2033 2 years, 10 months ago

Selected Answer: A

Keyword "Amazon RDS for PostgreSQL instance large" . See list of size of instance at <https://aws.amazon.com/rds/instance-types/>

upvoted 2 times

examtopictempacc 3 years ago

Selected Answer: A

A.

Not C: without adding infrastructure

upvoted 3 times

EA100 3 years ago

Answer - C

Option B, making the Amazon RDS for PostgreSQL DB instance a Multi-AZ DB instance, would provide high availability and fault tolerance but may not directly address the need for increased capacity to handle the larger workload.

Therefore, the recommended solution is Option C: Buy reserved DB instances for the workload and add another Amazon RDS for PostgreSQL DB instance to accommodate the increased workload in a cost-effective manner.

upvoted 1 times

cloudenthusiast 3 years ago

C

Option C: buying reserved DB instances for the total workload and adding another Amazon RDS for PostgreSQL DB instance seems to be the most

appropriate choice. It allows for workload distribution across multiple instances, providing scalability and potential performance improvements. Additionally, reserved instances can provide cost savings in the long term.

upvoted 1 times

nosense 3 years ago

A for me, because without adding additional infrastructure

upvoted 4 times

th3k33n 3 years ago

Should be C

upvoted 1 times

Efren 3 years ago

That would add more infrastructure. A would increase the size, keeping the number of instances, i think

upvoted 2 times

cloudenthusiast 3 years ago

Option A involves making the existing Amazon RDS for PostgreSQL DB instance larger. While this can improve performance, it may not be sufficient to handle a significantly increased workload. It also doesn't distribute the workload or provide scalability.

upvoted 1 times

nosense 3 years ago

The main not HA, cost-effectively and without adding infrastructure

upvoted 2 times

omoakin 3 years ago

A is the best

upvoted 2 times

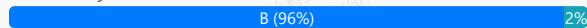
A company operates an ecommerce website on Amazon EC2 instances behind an Application Load Balancer (ALB) in an Auto Scaling group. The site is experiencing performance issues related to a high request rate from illegitimate external systems with changing IP addresses. The security team is worried about potential DDoS attacks against the website. The company must block the illegitimate incoming requests in a way that has a minimal impact on legitimate users.

What should a solutions architect recommend?

- A. Deploy Amazon Inspector and associate it with the ALB.
- B. Deploy AWS WAF, associate it with the ALB, and configure a rate-limiting rule. **Most Voted**
- C. Deploy rules to the network ACLs associated with the ALB to block the incoming traffic.
- D. Deploy Amazon GuardDuty and enable rate-limiting protection when configuring GuardDuty.

Correct Answer: B

Community vote distribution



Comments

samehpalass **Highly Voted** 2 years, 5 months ago

Selected Answer: B

As no shield protect here so WAF rate limit
upvoted 11 times

hydro143 2 years, 2 months ago

Where's your Shield Advanced now, in your hour of need he has abandoned you
upvoted 12 times

pentium75 **Highly Voted** 1 year, 11 months ago

Selected Answer: B

Best solution Shield Advanced, not listed here, thus second-best solution, WAF with rate limiting
upvoted 8 times

awsgeek75 **Most Recent** 1 year, 11 months ago

Selected Answer: B

A. Amazon Inspector = Software vulnerabilities like OS patches etc. Not fit for purpose.
C. Changing IP from DDoS so don't know the incoming traffic for configuration (even if it was possible)
D. GardDuty is for workload and AWS account monitoring so it can't help with DDoS.

B is correct as AWS WAF + ALB can configure rate limiting even if source IP changes.
upvoted 6 times

jAtlas7 1 year, 11 months ago

Selected Answer: B

according to some google searches... to protect against DDOS attack:
* AWS WAF(Web Application Firewall) provides protection on the application layer (I think Application Load Balancer belongs to this level)
* AWS Shield protects the infrastructure layers of the OSI mode (I think AWS Network Load Balancer belongs to this level)
upvoted 3 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: A

This case is A
upvoted 1 times

pentium75 1 year, 11 months ago

Inspector is for detecting vulnerabilities, has nothing to do with the requirement.
upvoted 3 times

vincent2023 1 year, 4 months ago

Inspector wouldn't help in this case as it's used to detect vulnerabilities.
upvoted 1 times

james2033 2 years, 4 months ago

Selected Answer: B

AWS Web Application Firewall (WAF) + ALB (Application Load Balancer) See image at <https://aws.amazon.com/waf/> .
<https://docs.aws.amazon.com/waf/latest/developerguide/ddos-responding.html> .

Question keyword "high request rate", answer keyword "rate-limiting rule" <https://docs.aws.amazon.com/waf/latest/developerguide/waf-rate-based-example-limit-login-page-keys.html>

Amazon GuardDuty for threat detection <https://aws.amazon.com/guardduty/> , not for DDoS.
upvoted 3 times

TariqKipkemei 2 years, 5 months ago

Selected Answer: B

B in swahili 'ba' :)
external systems, incoming requests = AWS WAF
upvoted 2 times

Axeashes 2 years, 5 months ago

Selected Answer: B

layer 7 DDoS protection with WAF
<https://docs.aws.amazon.com/waf/latest/developerguide/ddos-get-started-web-acl-rbr.html>
upvoted 2 times

antropaws 2 years, 6 months ago

Selected Answer: B

B no doubt.
upvoted 1 times

Joselucho38 2 years, 6 months ago

Selected Answer: B

AWS WAF (Web Application Firewall) is a service that provides protection for web applications against common web exploits. By associating AWS WAF with the Application Load Balancer (ALB), you can inspect incoming traffic and define rules to allow or block requests based on various criteria.
upvoted 5 times

cloudenthusiast 2 years, 6 months ago

B

AWS Web Application Firewall (WAF) is a service that helps protect web applications from common web exploits and provides advanced security features. By deploying AWS WAF and associating it with the ALB, the company can set up rules to filter and block incoming requests based on specific criteria, such as IP addresses.

In this scenario, the company is facing performance issues due to a high request rate from illegitimate external systems with changing IP addresses. By configuring a rate-limiting rule in AWS WAF, the company can restrict the number of requests coming from each IP address, preventing excessive traffic from overwhelming the website. This will help mitigate the impact of potential DDoS attacks and ensure that legitimate users can access the site without interruption.

upvoted 5 times

Efren 2 years, 6 months ago

Selected Answer: B

If not AWS Shield, then WAF
upvoted 4 times

nosense 2 years, 6 months ago

Selected Answer: B

B obv for this
upvoted 4 times

Efren 2 years, 6 months ago

My mind slipped with AWS Shield. GuardDuty can be working along with WAF for DDOS attack, but ultimately would be WAF

<https://aws.amazon.com/blogs/security/how-to-use-amazon-guardduty-and-aws-web-application-firewall-to-automatically-block-suspicious-hosts/>

upvoted 2 times

Mia2009687 2 years, 5 months ago

Same here, I was looking for AWS Shield

upvoted 1 times

Efren 2 years, 6 months ago

Selected Answer: D

D, Guard Duty for me

upvoted 1 times

pentium75 1 year, 11 months ago

Guard Duty detects threats, has nothing to do with rate-limiting.

upvoted 2 times

A company wants to share accounting data with an external auditor. The data is stored in an Amazon RDS DB instance that resides in a private subnet. The auditor has its own AWS account and requires its own copy of the database.

What is the MOST secure way for the company to share the database with the auditor?

- A. Create a read replica of the database. Configure IAM standard database authentication to grant the auditor access.
- B. Export the database contents to text files. Store the files in an Amazon S3 bucket. Create a new IAM user for the auditor. Grant the user access to the S3 bucket.
- C. Copy a snapshot of the database to an Amazon S3 bucket. Create an IAM user. Share the user's keys with the auditor to grant access to the object in the S3 bucket.
- D. Create an encrypted snapshot of the database. Share the snapshot with the auditor. Allow access to the AWS Key Management Service (AWS KMS) encryption key. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

alexandercamachop **Highly Voted** 3 years ago

Selected Answer: D

The most secure way for the company to share the database with the auditor is option D: Create an encrypted snapshot of the database, share the snapshot with the auditor, and allow access to the AWS Key Management Service (AWS KMS) encryption key.

By creating an encrypted snapshot, the company ensures that the database data is protected at rest. Sharing the encrypted snapshot with the auditor allows them to have their own copy of the database securely.

In addition, granting access to the AWS KMS encryption key ensures that the auditor has the necessary permissions to decrypt and access the encrypted snapshot. This allows the auditor to restore the snapshot and access the data securely.

This approach provides both data protection and access control, ensuring that the database is securely shared with the auditor while maintaining the confidentiality and integrity of the data.

upvoted 24 times

TariqKipkemei 2 years, 12 months ago

best explanation ever

upvoted 4 times

24b2e9e **Highly Voted** 1 year, 11 months ago

why not A ?

upvoted 5 times

LeonSauveterre 1 year, 6 months ago

I think it's simply because it introduces more risks than sharing just a snapshot. External entities don't need that many privileges, so it's not the MOST secure way

upvoted 1 times

KennethNg923 **Most Recent** 1 year, 12 months ago

Selected Answer: D

Encrypted snapshot must be most secure compare others

upvoted 2 times

awsgeek75 2 years, 5 months ago

Selected Answer: D

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_ShareSnapshot.html

With AWS RDS, you can share snapshots across accounts so no need to go through S3 or replication. Option D allows more secure way by using encryption and sharing encryption key.

upvoted 3 times

potomac 2 years, 7 months ago

Selected Answer: D

MOST secure way

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

Key word: "Secure way"

The snapshot contents are encrypted using KMS keys for data security.

Sharing the snapshot directly removes risks of extracting/transferring data.

The auditor can restore the snapshot into their own RDS instance.

Access is controlled through sharing the encrypted snapshot and KMS key.

upvoted 5 times

antropaws 3 years ago

Selected Answer: D

Most likely D.

upvoted 4 times

cloudenthusiast 3 years ago

Option D (Creating an encrypted snapshot of the database, sharing the snapshot, and allowing access to the AWS Key Management Service encryption key) is generally considered a better option for sharing the database with the auditor in terms of security and control.

upvoted 3 times

nosense 3 years ago

Selected Answer: D

D for me

upvoted 3 times

A solutions architect configured a VPC that has a small range of IP addresses. The number of Amazon EC2 instances that are in the VPC is increasing, and there is an insufficient number of IP addresses for future workloads.

Which solution resolves this issue with the LEAST operational overhead?

- A. Add an additional IPv4 CIDR block to increase the number of IP addresses and create additional subnets in the VPC. Create new resources in the new subnets by using the new CIDR. **Most Voted**
- B. Create a second VPC with additional subnets. Use a peering connection to connect the second VPC with the first VPC. Update the routes and create new resources in the subnets of the second VPC.
- C. Use AWS Transit Gateway to add a transit gateway and connect a second VPC with the first VPC. Update the routes of the transit gateway and VPCs. Create new resources in the subnets of the second VPC.
- D. Create a second VPC. Create a Site-to-Site VPN connection between the first VPC and the second VPC by using a VPN-hosted solution on Amazon EC2 and a virtual private gateway. Update the route between VPCs to the traffic through the VPN. Create new resources in the subnets of the second VPC.

Correct Answer: A

Community vote distribution

A (100%)

Comments

antropaws **Highly Voted** 3 years ago

Selected Answer: A

A is correct: You assign a single CIDR IP address range as the primary CIDR block when you create a VPC and can add up to four secondary CIDR blocks after creation of the VPC.

upvoted 8 times

Guru4Cloud **Highly Voted** 2 years, 9 months ago

Selected Answer: A

the architect just needs to:

Add the CIDR using the AWS console or CLI
Create new subnets in the VPC using the new CIDR
Launch resources in the new subnets

upvoted 5 times

Tendu **Most Recent** 1 year, 3 months ago

Selected Answer: A

A - is the correct answer

upvoted 1 times

rosanna 1 year, 6 months ago

Selected Answer: A

The CIDR block of a VPC can NOT be changed but can be extended.
Thus, for the LEAST operational overhead; extend the existing one.

upvoted 1 times

f2e2419 2 years, 5 months ago

Selected Answer: A

best option

upvoted 3 times

awsgeek75 2 years, 5 months ago

Selected Answer: A

A: LEAST operational overhead is by creating a new CIDR block in existing VPC.
All other options require additional overhead of gateway or second VPC

upvoted 4 times

potomac 2 years, 7 months ago

Selected Answer: A

After you've created your VPC, you can associate additional IPv4 CIDR blocks with the VPC

upvoted 4 times

TariqKipkemei 2 years, 12 months ago

Selected Answer: A

A is best

upvoted 2 times

Yadav_Sanjay 3 years ago

Selected Answer: A

Add additional CIDR of bigger range

upvoted 3 times

Efren 3 years ago

Selected Answer: A

Add new bigger subnets

upvoted 3 times

nosense 3 years ago

Selected Answer: A

A valid

upvoted 2 times

A company used an Amazon RDS for MySQL DB instance during application testing. Before terminating the DB instance at the end of the test cycle, a solutions architect created two backups. The solutions architect created the first backup by using the mysqldump utility to create a database dump. The solutions architect created the second backup by enabling the final DB snapshot option on RDS termination.

The company is now planning for a new test cycle and wants to create a new DB instance from the most recent backup. The company has chosen a MySQL-compatible edition of Amazon Aurora to host the DB instance.

Which solutions will create the new DB instance? (Choose two.)

- A. Import the RDS snapshot directly into Aurora. **Most Voted**
- B. Upload the RDS snapshot to Amazon S3. Then import the RDS snapshot into Aurora.
- C. Upload the database dump to Amazon S3. Then import the database dump into Aurora. **Most Voted**
- D. Use AWS Database Migration Service (AWS DMS) to import the RDS snapshot into Aurora.
- E. Upload the database dump to Amazon S3. Then use AWS Database Migration Service (AWS DMS) to import the database dump into Aurora.

Correct Answer: AC

Community vote distribution

AC (76%)

CE (15%)

4%

Comments

Axaus **Highly Voted** 3 years ago

Selected Answer: AC

A,C

A because the snapshot is already stored in AWS.

C because you don't need a migration tool going from MySQL to MySQL. You would use the MySQL utility.

upvoted 13 times

oras2023 **Highly Voted** 3 years ago

Selected Answer: AC

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/AuroraMySQL.Migrating.RDSMySQL.Import.html>

upvoted 8 times

bora4motion **Most Recent** 1 year, 1 month ago

Selected Answer: CE

c + e is correct

upvoted 2 times

tch 1 year, 2 months ago

Selected Answer: AC

Answer B is out, why upload to S3?

This is to call the most recent backup, answer D & E are out to use AWS Database Migration Service (AWS DMS).

upvoted 1 times

JackyCCK 2 years, 2 months ago

If you can use option A - "Import the RDS snapshot directly into Aurora", why go S3 in option C?

Non sense, A and C cannot co-exist

upvoted 2 times

pentium75 2 years, 5 months ago

Selected Answer: AC

A per <https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/AuroraMySQL.Migrating.RDSMySQL.Snapshot.html>

C per <https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/AuroraMySQL.Migrating.ExtMySQL.html>
upvoted 7 times

aws94 2 years, 6 months ago

Selected Answer: AB

A and B

upvoted 1 times

meowruki 2 years, 6 months ago

Selected Answer: AC

Similar : <https://repost.aws/knowledge-center/aurora-postgresql-migrate-from-rds>
upvoted 3 times

potomac 2 years, 7 months ago

Selected Answer: AD

A and C

upvoted 1 times

TariqKipkemei 2 years, 7 months ago

Selected Answer: AC

Either import the RDS snapshot directly into Aurora or upload the database dump to Amazon S3, then import the database dump into Aurora.
upvoted 3 times

thanhnv142 2 years, 7 months ago

AC:

- store dump in s3 then upload to aurora
- no need to store snapshot in s3 because is in AWS already

upvoted 5 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: CE

C and E are the solutions that can restore the backups into Amazon Aurora.

The RDS DB snapshot contains backup data in a proprietary format that cannot be directly imported into Aurora.
The mysqldump database dump contains SQL statements that can be imported into Aurora after uploading to S3.
AWS DMS can migrate the dump file from S3 into Aurora.

upvoted 6 times

james2033 2 years, 10 months ago

Selected Answer: AC

Amazon RDS for MySQL --> Amazon Aurora MySQL-compatible.

* mysqldump, database dump --> (C) Upload to Amazon S3, Import dump to Aurora.

* DB snapshot --> (A) Import RDS Snapshot directly Aurora. The correct word should be "migration". "Use console to migrate the DB snapshot and create an Aurora MySQL DB cluster with the same databases as the original MySQL DB instance."

Exclude B, because no need upload DB snapshot to Amazon S3. Exclude D, because no need Migration service. Exclude E, because no need Migration service. Use exclusion method is more easy for this question.

Related links:

- Amazon RDS create database snapshot https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_CreateSnapshot.html
- <https://aws.amazon.com/rds/aurora/>

upvoted 3 times

marufxplorer 2 years, 11 months ago

CE

Since the backup created by the solutions architect was a database dump using the mysqldump utility, it cannot be directly imported into Aurora using RDS snapshots. Amazon Aurora has its own specific backup format that is different from RDS snapshots

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

C and E are the solutions that can restore the backups into Amazon Aurora.

The RDS DB snapshot contains backup data in a proprietary format that cannot be directly imported into Aurora.
The mysqldump database dump contains SQL statements that can be imported into Aurora after uploading to S3.
AWS DMS can migrate the dump file from S3 into Aurora.

upvoted 3 times

antropaws 3 years ago

Selected Answer: AC

Migrating data from MySQL by using an Amazon S3 bucket

You can copy the full and incremental backup files from your source MySQL version 5.7 database to an Amazon S3 bucket, and then restore an Amazon Aurora MySQL DB cluster from those files.

This option can be considerably faster than migrating data using mysqldump, because using mysqldump replays all of the commands to recreate the schema and data from your source database in your new Aurora MySQL DB cluster.

By copying your source MySQL data files, Aurora MySQL can immediately use those files as the data for an Aurora MySQL DB cluster.

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/AuroraMySQL.Migrating.ExtMySQL.html>

upvoted 4 times

Efren 3 years ago

Selected Answer: BC

Id say B and C

You can create a dump of your data using the mysqldump utility, and then import that data into an existing Amazon Aurora MySQL DB cluster.

c> - Because Amazon Aurora MySQL is a MySQL-compatible database, you can use the mysqldump utility to copy data from your MySQL or MariaDB database to an existing Amazon Aurora MySQL DB cluster.

B.- You can copy the source files from your source MySQL version 5.5, 5.6, or 5.7 database to an Amazon S3 bucket, and then restore an Amazon Aurora MySQL DB cluster from those files.

upvoted 2 times

nosense 3 years ago

Selected Answer: BE

Rds required upload to s3

upvoted 1 times

nosense 3 years ago

If too be honestly can't decide between be and bc...

upvoted 1 times

Guru4Cloud 2 years, 9 months ago

using the mysqldump database dump provide valid solutions to restore into Aurora. Options A, B, and D using the RDS snapshot cannot directly restore into Aurora.

upvoted 1 times

nosense 3 years ago

in the end, apparently the A and C.

a) because it creates a new DB

b) no sense to load in s3. can directly

c) yes, creates a new inst

d and e migration

upvoted 2 times

A company hosts a multi-tier web application on Amazon Linux Amazon EC2 instances behind an Application Load Balancer. The instances run in an Auto Scaling group across multiple Availability Zones. The company observes that the Auto Scaling group launches more On-Demand Instances when the application's end users access high volumes of static web content. The company wants to optimize cost.

What should a solutions architect do to redesign the application MOST cost-effectively?

- A. Update the Auto Scaling group to use Reserved Instances instead of On-Demand Instances.
- B. Update the Auto Scaling group to scale by launching Spot Instances instead of On-Demand Instances.
- C. Create an Amazon CloudFront distribution to host the static web contents from an Amazon S3 bucket. **Most Voted**
- D. Create an AWS Lambda function behind an Amazon API Gateway API to host the static website contents.

Correct Answer: C

Community vote distribution

C (100%)

Comments

awsgeek75 1 year, 11 months ago

Selected Answer: C

C: Cost effective static content scaling = CloudFront

A and B scale instances so not the best use of money for static content

D Probably most expensive way of service static content at scale as you'll be charged for Lambda execution also

upvoted 4 times

mwwt2022 1 year, 11 months ago

Selected Answer: C

static content -> CloudFront

upvoted 4 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: C

implementing CloudFront to serve static content is the most cost-optimal architectural change for this use case.

upvoted 4 times

james2033 2 years, 4 months ago

Selected Answer: C

Keyword "Amazon CloudFront", "high volumes of static web content", choose C.

upvoted 3 times

TariqKipkemei 2 years, 5 months ago

Selected Answer: C

static web content = Amazon CloudFront

upvoted 3 times

alexandercamachop 2 years, 6 months ago

Selected Answer: C

Static Web Content = S3 Always.

CloudFront = Closer to the users locations since it will cache in the Edge nodes.

upvoted 4 times

cloudenthusiast 2 years, 6 months ago

By leveraging Amazon CloudFront, you can cache and serve the static web content from edge locations worldwide, reducing the load on your EC2 instances. This can help lower the number of On-Demand Instances required to handle high volumes of static web content requests. Storing the static content in an Amazon S3 bucket and using CloudFront as a content delivery network (CDN) improves performance and reduces costs by reducing the load on your EC2 instances.

upvoted 4 times

Efren 2 years, 6 months ago

Selected Answer: C

Static content, cloudFront plus S3

upvoted 4 times

nosense 2 years, 6 months ago

Selected Answer: C

c for me

upvoted 2 times

A company stores several petabytes of data across multiple AWS accounts. The company uses AWS Lake Formation to manage its data lake. The company's data science team wants to securely share selective data from its accounts with the company's engineering team for analytical purposes.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Copy the required data to a common account. Create an IAM access role in that account. Grant access by specifying a permission policy that includes users from the engineering team accounts as trusted entities.
- B. Use the Lake Formation permissions Grant command in each account where the data is stored to allow the required engineering team users to access the data.
- C. Use AWS Data Exchange to privately publish the required data to the required engineering team accounts.
- D. Use Lake Formation tag-based access control to authorize and grant cross-account permissions for the required data to the engineering team accounts. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

cloudenthusiast **Highly Voted** 2 years, 6 months ago

Selected Answer: D

By utilizing Lake Formation's tag-based access control, you can define tags and tag-based policies to grant selective access to the required data for the engineering team accounts. This approach allows you to control access at a granular level without the need to copy or move the data to a common account or manage permissions individually in each account. It provides a centralized and scalable solution for securely sharing data across accounts with minimal operational overhead.

upvoted 18 times

NSA_Poker **Most Recent** 1 year, 6 months ago

Selected Answer: D

(B) uses the CLI command that has many options: principal, TableName, ColumnNames, LFTag etc providing a way to manage granular access permissions for different users at the table and column level. That way you don't give full access to the all the data. The problem with (B) is to implement this in each account has a lot more operational overhead than (D).

upvoted 2 times

awsgeek75 1 year, 11 months ago

Selected Answer: D

D: Selective data = tagging

A and B gives full access to all the data

C is possible but with complex operational overhead as you have to publish your data to the Data Exchange. (this is based on my limited knowledge so happy to be corrected)

upvoted 4 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: D

D is the correct option with the least operational overhead.

Using Lake Formation tag-based access control allows granting cross-account permissions to access data in other accounts based on tags, without having to copy data or configure individual permissions in each account.

This provides a centralized, tag-based way to share selective data across accounts to authorized users with least operational overhead.

upvoted 4 times

luisgu 2 years, 6 months ago

Selected Answer: D

<https://aws.amazon.com/blogs/big-data/securely-share-your-data-across-aws-accounts-using-aws-lake-formation/>
upvoted 4 times

A company wants to host a scalable web application on AWS. The application will be accessed by users from different geographic regions of the world. Application users will be able to download and upload unique data up to gigabytes in size. The development team wants a cost-effective solution to minimize upload and download latency and maximize performance.

What should a solutions architect do to accomplish this?

- A. Use Amazon S3 with Transfer Acceleration to host the application. **Most Voted**
- B. Use Amazon S3 with CacheControl headers to host the application.
- C. Use Amazon EC2 with Auto Scaling and Amazon CloudFront to host the application.
- D. Use Amazon EC2 with Auto Scaling and Amazon ElastiCache to host the application.

Correct Answer: A

Community vote distribution

A (73%)

C (27%)

Comments

pentium75 **Highly Voted** 2 years, 5 months ago

Selected Answer: A

The question asks for "a cost-effective solution [ONLY TO] to minimize upload and download latency and maximize performance", not for the actual application. And the 'cost-effective solution to minimize upload and download latency and maximize performance' is S3 Transfer Acceleration. Obviously there is more required to host the app, but that is not asked for.

upvoted 15 times

chris0975 **Highly Voted** 2 years, 7 months ago

Selected Answer: A

The question is focused on large downloads and uploads. S3 Transfer Acceleration is what fits. CloudFront is for caching which cannot be used when the data is unique. They aren't as concerned with regular web traffic.

Amazon S3 Transfer Acceleration can speed up content transfers to and from Amazon S3 by as much as 50-500% for long-distance transfer of larger objects.

upvoted 7 times

e10c971 **Most Recent** 7 months ago

Selected Answer: C

Global Access & Performance

CloudFront reduces the load on your EC2 instances by serving cached content, which can lower compute costs.

Auto Scaling helps optimize costs by only running the necessary number of instances.

upvoted 1 times

ln2718 11 months ago

Selected Answer: C

Could be answer C... "users from different geographic regions of the world". And, I agree, absolutely poorly written question.

upvoted 1 times

allanm 10 months ago

Cloudfront is a CDN that helps to show dynamic content to users around the world. It's not a two way system so C is out.

upvoted 1 times

bora4motion 1 year, 1 month ago

Selected Answer: A

voting for A.

upvoted 1 times

Ben_88 1 year, 7 months ago

Selected Answer: C

S3 can't "host an application", so even tho S3 Transfer acceleration would optimize the performances of the download/upload, still it doesn't host the app. the question is poorly written i guess

upvoted 2 times

vincent2023 1 year, 4 months ago

I agree with your point !!

upvoted 2 times

ChymKuBoy 1 year, 9 months ago

Selected Answer: A

A for sure

upvoted 3 times

awsgeek75 2 years, 5 months ago

Selected Answer: A

Not C, D No requirements to scale the application itself so EC2 is not applicable.

B is for caching so not sure how/if that helps the upload speed for global users

A is correct as Transfer Accelerator is best for uploading and downloading unique items near the user's region/location

upvoted 6 times

tosuccess 2 years, 5 months ago

Selected Answer: A

for datas greater than 1 GB, s3 transfer acceleration is the best

upvoted 3 times

Cyberkayu 2 years, 5 months ago

Selected Answer: A

Application users will be able to download and upload UNIQUE data up to gigabytes in size

Thus all caching related solution dont work.

upvoted 5 times

Goutham4981 2 years, 6 months ago

Selected Answer: A

Downloading data upto gigabytes in size - Cloudfront is a content delivery service that acts as an edge caching layer for images and other data. Not a service that minimizes upload and download latency.

upvoted 2 times

potomac 2 years, 7 months ago

Selected Answer: A

The question is focused on large downloads and uploads. S3 Transfer Acceleration is what fits. CloudFront is for caching which cannot be used when the data is unique. They aren't as concerned with regular web traffic.

C didn't mention S3. Where the data is stored?

upvoted 4 times

pentium75 2 years, 5 months ago

A doesn't mention EC2 or EKS or ECS or Elastic Beanstalk or Lambda. Where does the "scalable web application" run?

upvoted 2 times

JA2018 1 year, 6 months ago

hmm... could be a red herring?

upvoted 1 times

beast2091 2 years, 7 months ago

It is A.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/transfer-acceleration.html>

upvoted 2 times

danielmakita 2 years, 7 months ago

It is A as the Transfer Acceleration will minimize upload and download latency.

If you choose C, where would the files be stored? There is no mention of any S3. Will it be stored inside the EC2? That's why I didn't go for C

upvoted 5 times

Sindokuhlep 2 years, 7 months ago

Selected Answer: C

Amazon S3 with Transfer Acceleration (option A) is designed for speeding up uploads to Amazon S3, and it's not used for hosting scalable web applications. It doesn't mention using EC2 instances for hosting the application.

upvoted 4 times

canonlycontainletters1 2 years, 7 months ago

Selected Answer: C

My answer is C

upvoted 1 times

thanhnv142 2 years, 7 months ago

C because A is for upload data to S3, not for web app

upvoted 1 times

A company has hired a solutions architect to design a reliable architecture for its application. The application consists of one Amazon RDS DB instance and two manually provisioned Amazon EC2 instances that run web servers. The EC2 instances are located in a single Availability Zone.

An employee recently deleted the DB instance, and the application was unavailable for 24 hours as a result. The company is concerned with the overall reliability of its environment.

What should the solutions architect do to maximize reliability of the application's infrastructure?

A. Delete one EC2 instance and enable termination protection on the other EC2 instance. Update the DB instance to be Multi-AZ, and enable deletion protection.

B. Update the DB instance to be Multi-AZ, and enable deletion protection. Place the EC2 instances behind an Application Load Balancer, and run them in an EC2 Auto Scaling group across multiple Availability Zones. **Most Voted**

C. Create an additional DB instance along with an Amazon API Gateway and an AWS Lambda function. Configure the application to invoke the Lambda function through API Gateway. Have the Lambda function write the data to the two DB instances.

D. Place the EC2 instances in an EC2 Auto Scaling group that has multiple subnets located in multiple Availability Zones. Use Spot Instances instead of On-Demand Instances. Set up Amazon CloudWatch alarms to monitor the health of the instances. Update the DB instance to be Multi-AZ, and enable deletion protection.

Correct Answer: B

Community vote distribution

B (100%)

Comments

awsgeek75 **Highly Voted** 1 year, 11 months ago

Option E: Sack the employee who did this :)

upvoted 11 times

nosense **Highly Voted** 2 years, 6 months ago

Selected Answer: B

B is correct. HA ensured by DB in Multi-AZ and EC2 in AG

upvoted 7 times

wizcloudifa **Most Recent** 1 year, 7 months ago

Selected Answer: B

A: delete one instance, why?. Although takes care of reliability of DB instance however not EC2.

B. seems perfect as takes care of reliability of both EC2 as well as DB

C. DB instance's reliability is not taken care of

D. seems to be trying to address cost alongside reliability of EC2 and DB.

upvoted 3 times

awsgeek75 1 year, 11 months ago

Selected Answer: B

A: Deleting one EC2 instance makes no sense. Why would you do that?

C: API Gateway, Lambda etc are all nice but they don't solve the problem of DB instance deletion

D: EC2 subnet blah, what? The problem is reliability, not networking!

B is correct as it solves the DB deletion issue and increases reliability by Multi AZ scaling of EC2 instances

upvoted 5 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: B

The key points:

- ° RDS Multi-AZ and deletion protection provide high availability for the database.
- ° The load balancer and Auto Scaling group across AZs give high availability for EC2.
- ° Options A, C, D have limitations that would reduce reliability vs option B.

upvoted 3 times

TariqKipkemei 2 years, 5 months ago

Selected Answer: B

Update the DB instance to be Multi-AZ, and enable deletion protection. Place the EC2 instances behind an Application Load Balancer, and run them in an EC2 Auto Scaling group across multiple Availability Zones

upvoted 2 times

antropaws 2 years, 6 months ago

Selected Answer: B

B for sure.

upvoted 2 times

alexandercamachop 2 years, 6 months ago

Selected Answer: B

It is the only one with High Availability.

Amazon RDS with Multi AZ

EC2 with Auto Scaling Group in Multi Az

upvoted 2 times

omoakin 2 years, 6 months ago

same question from

<https://www.examttopics.com/exams/amazon/aws-certified-solutions-architect-associate-saa-c02/>

long time ago and still same option B

upvoted 3 times

A company is storing 700 terabytes of data on a large network-attached storage (NAS) system in its corporate data center. The company has a hybrid environment with a 10 Gbps AWS Direct Connect connection.

After an audit from a regulator, the company has 90 days to move the data to the cloud. The company needs to move the data efficiently and without disruption. The company still needs to be able to access and update the data during the transfer window.

Which solution will meet these requirements?

A. Create an AWS DataSync agent in the corporate data center. Create a data transfer task Start the transfer to an Amazon S3 bucket. **Most Voted**

B. Back up the data to AWS Snowball Edge Storage Optimized devices. Ship the devices to an AWS data center. Mount a target Amazon S3 bucket on the on-premises file system.

C. Use rsync to copy the data directly from local storage to a designated Amazon S3 bucket over the Direct Connect connection.

D. Back up the data on tapes. Ship the tapes to an AWS data center. Mount a target Amazon S3 bucket on the on-premises file system.

Correct Answer: A

Community vote distribution

A (100%)

Comments

wRhIH **Highly Voted** 2 years, 12 months ago

For those who wonders why not B. Snowball Edge Storage Optimized device for data transfer is up to 100TB
<https://docs.aws.amazon.com/snowball/latest/developer-guide/device-differences.html>

upvoted 13 times

smartegnine 2 years, 11 months ago

10GBs * 24*60*60 = 864,000 GB estimate around 864 TB a day, 2 days will transfer all data. But for snowball at least 4 days for delivery to the data center.

upvoted 1 times

siGma182 2 years, 10 months ago

This account is wrong but I get your point. It is wrong cause 10Gb/s is not the same as 10GB/s (Gigabits vs Gigabytes). However, the correct count is 864Tb/8 = 108TB per day. In one week you should've transferred all the data.

upvoted 8 times

Maru86 2 years, 2 months ago

That's right, 1 GB = 8 Gb. Essentially we have a speed of 1.25GB/s.

upvoted 2 times

JA2018 1 year, 6 months ago

folks, you had to consider the typical TCP overheads for such over-the-wire data transfers.... it could be as high as 70% of the total available bandwidth due to a wide variety of factors

upvoted 1 times

Maru86 2 years, 2 months ago

The question explicitly mentioned "devices", also Snowball Edge Storage Optimized is 80TB HDD. So it is possible, but the answer is A because we can transfer with DataSync in 6.5 days.

upvoted 4 times

hsinchang **Highly Voted** 2 years, 10 months ago

Selected Answer: A

Access during the transfer window -> DataSync
upvoted 11 times

surajkrishnamurthy **Most Recent** 1 year, 4 months ago

Selected Answer: A

700TB with a Bandwidth of 10Gbps takes approximately 7 days, 3 Hours, 2 Minutes, 7 Seconds

As per the below link>>

<https://expedient.com/knowledgebase/tools-and-calculators/file-transfer-time-calculator/>

So Option A is Correct & using AWS Datasync facilitates further access to data still present in the OnPrem to access and S3 provides with unlimited data storage capacity

upvoted 1 times

Danilus 1 year, 7 months ago

A- Porque La compañía todavía necesita poder acceder y actualizar los datos durante la ventana de transferencia

B-Aunque esta podría ser una opción porque dice de enviar los dispositivos al centro de datos que serían más o menos entre 9 y 10 dispositivos ya que el almacenamiento de storage optimized es de 80 TB el problema está en que con Snowball Edge Storage Optimized no se puede acceder y actualizar a los datos durante la transferencia

upvoted 2 times

JA2018 1 year, 6 months ago

From Google Translate:

A- Because The company still needs to be able to access and update the data during the transfer window

B-Although this could be an option because it says that sending the devices to the data center would be more or less between 9 and 10 devices since the storage optimized storage is 80 TB, the problem is that with Snowball Edge Storage Optimized it does not Data can be accessed and updated during transfer

upvoted 3 times

Danilus 1 year, 7 months ago

Selected Answer: A

A- Porque La compañía todavía necesita poder acceder y actualizar los datos durante la ventana de transferencia

B-Aunque esta podría ser una opción porque dice de enviar los dispositivos al centro de datos que serían más o menos entre 9 y 10 dispositivos ya que el almacenamiento de storage optimized es de 80 TB el problema está en que con Snowball Edge Storage Optimized no se puede acceder y actualizar a los datos durante la transferencia

upvoted 1 times

MatAlves 1 year, 9 months ago

Selected Answer: A

Finally, a company with good bandwidth.

upvoted 5 times

NSA_Poker 2 years ago

Selected Answer: A

(B) is incorrect bc although Mountpoint for S3 is possible for on-premises NAS, this is not as efficient as AWS DataSync. Data updates made during the transfer window would have to be resolved later.

upvoted 2 times

Burrito69 2 years, 2 months ago

I will put simple calculation for you guys to just store in your head to quickly answer:

for 10GBPS its 1.25GBPS because its bits to bytes.

for one minute its 75GBPS

for one hour its 4500 GBPS

for one day its 10.8 TB

so you can calculate easily if you just store these numbers in your head. let's say if question is 1GBPS DirectConnect that means everything above should be divide by 8. cool

upvoted 1 times

JA2018 1 year, 6 months ago

folks, you had to consider the typical TCP overheads for such over-the-wire data transfers.... it could be as high as 70% of the total available bandwidth due to a wide variety of factors

in a worst case scenario, we could be looking @ 3.24 TB/ day

upvoted 1 times

awsgeek75 2 years, 5 months ago

Selected Answer: A

Critical requirement: "The company needs to move the data efficiently and without disruption."

B: Causes disruption

C: I don't think that is possible without a gateway kind of thing

D: Tape backups? "Mount a target Amazon S3 bucket on the on-premises file system"? This requires some gateway which is not mentioned

A is the answer as DataSync allows transfer without disruption and with 10Gbps, it can be done in 90 days.

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: A

AWS DataSync can efficiently transfer large datasets from on-premises NAS to Amazon S3 over Direct Connect.

DataSync allows accessing and updating the data continuously during the transfer process.

upvoted 5 times

TariqKipkemei 2 years, 12 months ago

Selected Answer: A

AWS DataSync is a secure, online service that automates and accelerates moving data between on premises and AWS Storage services.

upvoted 3 times

omoakin 3 years ago

A

<https://www.examttopics.com/discussions/amazon/view/46492-exam-aws-certified-solutions-architect-associate-saa-c02/#:~:text=Exam%20question%20from,Question%20%23%3A%20385>

upvoted 2 times

cloudenthusiast 3 years ago

Selected Answer: A

By leveraging AWS DataSync in combination with AWS Direct Connect, the company can efficiently and securely transfer its 700 terabytes of data to an Amazon S3 bucket without disruption. The solution allows continued access and updates to the data during the transfer window, ensuring business continuity throughout the migration process.

upvoted 4 times

nosense 3 years ago

Selected Answer: A

A for me, bcs egde storage up to 100tb

upvoted 5 times

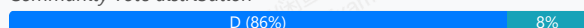
A company stores data in PDF format in an Amazon S3 bucket. The company must follow a legal requirement to retain all new and existing data in Amazon S3 for 7 years.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Turn on the S3 Versioning feature for the S3 bucket. Configure S3 Lifecycle to delete the data after 7 years. Configure multi-factor authentication (MFA) delete for all S3 objects.
- B. Turn on S3 Object Lock with governance retention mode for the S3 bucket. Set the retention period to expire after 7 years. Recopy all existing objects to bring the existing data into compliance.
- C. Turn on S3 Object Lock with compliance retention mode for the S3 bucket. Set the retention period to expire after 7 years. Recopy all existing objects to bring the existing data into compliance.
- D. Turn on S3 Object Lock with compliance retention mode for the S3 bucket. Set the retention period to expire after 7 years. Use S3 Batch Operations to bring the existing data into compliance. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

omarshaban **Highly Voted** 2 years, 4 months ago

THIS WAS IN MY EXAM

upvoted 13 times

awsgeek75 2 years, 4 months ago

Did you pass?

upvoted 7 times

iapps369 **Highly Voted** 2 years, 5 months ago

D

as S3 batch operations reduce risk and manual copy/paste overhead.

upvoted 6 times

Lin878 **Most Recent** 1 year, 12 months ago

Selected Answer: D

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock.html>

upvoted 3 times

awsgeek75 2 years, 5 months ago

Selected Answer: D

A: Versioning, not relevant

B: Governance, it won't enforce object lock

C: Recopy existing objects may work but lots of operational overhead (see link)

D: Compliance on existing objects with batch operations is least operational overhead

<https://repost.aws/questions/QUGKrl8XRLTEeulzUHq0Ikew/s3-object-lock-on-existing-s3-objects>

upvoted 5 times

awsgeek75 2 years, 5 months ago

With option C, you have to copy the object for it to be complaint and then delete the original as only the new copy will be compliant. So D is the only option

upvoted 2 times

mr123dd 2 years, 5 months ago

Selected Answer: A

To enable Object Lock on an Amazon S3 bucket, you must first enable versioning on that bucket. other 3 option did not enable versioning first
upvoted 2 times

allanm 10 months ago

A does not bring existing data into compliance and MFA is not required for this use case. Answer is D
upvoted 1 times

fb4afde 2 years, 6 months ago

Selected Answer: D

Recopying offers more control but requires users to manage the process. S3 Batch Operations automates the process at scale but with less granular control - LEAST operational overhead
upvoted 3 times

moonster 2 years, 7 months ago

Its C because you only need to recopy all existing objects one time, so why use S3 batch operations if new datas going to be in compliance retention mode? I can see why its C although my initial gut answer was D.
upvoted 2 times

pentium75 2 years, 5 months ago

What if I don't have the original files anymore? Where should I copy them from?
upvoted 3 times

kwang312 2 years, 8 months ago

You can only enable Object Lock for new buckets. If you want to turn on Object Lock for an existing bucket, contact AWS Support.
upvoted 2 times

pentium75 2 years, 5 months ago

You need a token from AWS Support, but you CAN enable Object Lock for an existing bucket.
upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

Turn on S3 Object Lock with compliance retention mode for the S3 bucket. Set the retention period to expire after 7 years. Use S3 Batch Operations to bring the existing data into compliance.
upvoted 2 times

MrAWSAssociate 2 years, 11 months ago

Selected Answer: D

To replicate existing object/data in S3 Bucket to bring them to compliance, optionally we use "S3 Batch Replication", so option D is the most appropriate, especially if we have big data in S3.
upvoted 2 times

TariqKipkemei 2 years, 12 months ago

Selected Answer: D

For minimum ops D is best
upvoted 2 times

DrWatson 3 years ago

Selected Answer: D

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/batch-ops-retention-date.html>
upvoted 6 times

antropaws 3 years ago

Selected Answer: C

Batch operations will add operational overhead.
upvoted 3 times

pentium75 2 years, 5 months ago

And gathering all the files for copying them again does not?
upvoted 2 times

Abrar2022 3 years ago

Use Object Lock in Compliance mode. Then Use Batch operation.
WRONG>> manual work and not automated>>>Recopy all existing objects to bring the existing data into compliance.
upvoted 1 times

pentium75 2 years, 5 months ago

Batch IS automated. You just need to create the batch which is a one-time operation.
"Recopy all existing objects" is not operational overhead?

upvoted 2 times

omoakin 3 years ago

C

When an object is locked in compliance mode, its retention mode can't be changed, and its retention period can't be shortened. Compliance mode helps ensure that an object version can't be overwritten or deleted for the duration of the retention period.

upvoted 3 times

omoakin 3 years ago

error i meant to type D

i wont do recopy

upvoted 2 times

lucdt4 3 years ago

No, D for me because the requirement is LEAST operational overhead

So RECOpy is the manual operation -> C is wrong

D is correct

upvoted 3 times

cloudenthusiast 3 years ago

Recopying vs. S3 Batch Operations: In Option C, the recommendation is to recopy all existing objects to ensure they have the appropriate retention settings. This can be done using simple S3 copy operations. On the other hand, Option D suggests using S3 Batch Operations, which is a more advanced feature and may require additional configuration and management. S3 Batch Operations can be beneficial if you have a massive number of objects and need to perform complex operations, but it might introduce more overhead for this specific use case.

Operational complexity: Option C has a straightforward process of recopying existing objects. It is a well-known operation in S3 and doesn't require additional setup or management. Option D introduces the need to set up and configure S3 Batch Operations, which can involve creating job definitions, specifying job parameters, and monitoring the progress of batch operations. This additional complexity may increase the operational overhead.

upvoted 2 times

Efren 3 years ago

Selected Answer: D

You need AWS Batch to re-apply certain config to files that were already in S3, like encryption

upvoted 5 times

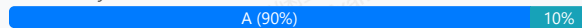
A company has a stateless web application that runs on AWS Lambda functions that are invoked by Amazon API Gateway. The company wants to deploy the application across multiple AWS Regions to provide Regional failover capabilities.

What should a solutions architect do to route traffic to multiple Regions?

- A. Create Amazon Route 53 health checks for each Region. Use an active-active failover configuration. **Most Voted**
- B. Create an Amazon CloudFront distribution with an origin for each Region. Use CloudFront health checks to route traffic.
- C. Create a transit gateway. Attach the transit gateway to the API Gateway endpoint in each Region. Configure the transit gateway to route requests.
- D. Create an Application Load Balancer in the primary Region. Set the target group to point to the API Gateway endpoint hostnames in each Region.

Correct Answer: A

Community vote distribution



Comments

TariqKipkemei **Highly Voted** 2 years, 12 months ago

Selected Answer: A

Global, Reduce latency, health checks, no failover = Amazon CloudFront
 Global, Reduce latency, health checks, failover, Route traffic = Amazon Route 53
 option A has more weight.
 upvoted 35 times

Anmol_1010 2 years, 7 months ago

nicley explained
 upvoted 3 times

ManikRoy 2 years, 1 month ago

Cloud front does have failover capabilities.
https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/high_availability_origin_failover.html#:~:text=the%20secondary%20origin-,Note,Choose%20Create%20origin%20group.
 upvoted 4 times

examtopictempacc **Highly Voted** 3 years ago

Selected Answer: A

A. I'm not an expert in this area, but I still want to express my opinion. After carefully reviewing the question and thinking about it for a long time, I actually don't know the reason. As I mentioned at the beginning, I'm not an expert in this field.
 upvoted 20 times

awsgeek75 2 years, 4 months ago

All the explanation you need for this question and option A is in this article:
<https://aws.amazon.com/blogs/compute/building-a-multi-region-serverless-application-with-amazon-api-gateway-and-aws-lambda/>
 upvoted 3 times

MatAlves **Most Recent** 1 year, 9 months ago

Selected Answer: A

Correct me if I'm wrong but CloudFront DOES NOT have health check capabilities out of the box. Route 53 and Global Accelerator do.
 upvoted 3 times

ChymKuBoy 1 year, 9 months ago

Selected Answer: A

A for sure

upvoted 2 times

awsgeek75 2 years, 5 months ago

Selected Answer: A

B: Caching solution. Not ideal for failover although it will work. Would have been a correct answer if A wasn't an option

C: Transit gateway is for VPC connectivity not AWS API or Lambda

D: Even if it was possible, there is a primary region dependency of ALB

A: correct because R53 health checks can failover across regions

Good explanation here:

<https://aws.amazon.com/blogs/compute/building-a-multi-region-serverless-application-with-amazon-api-gateway-and-aws-lambda/>

upvoted 4 times

awsgeek75 2 years, 4 months ago

The article also explains why you cannot use a CloudFront distribution for API Gateway, Lambda for failover

upvoted 2 times

tosuccess 2 years, 5 months ago

Selected Answer: B

we can set primary and secondary regions in cloud front for failover.

upvoted 2 times

pentium75 2 years, 5 months ago

Selected Answer: A

Application is serverless, it doesn't matter where it runs, so can be active-active setup and run wherever the request comes in. Route 53 with health checks will route to a healthy region.

B, could work too, but CloudFront is for caching which does not seem to help with an API. The goal here is "failover capabilities", not caching/performance/latency etc.

upvoted 3 times

Goutham4981 2 years, 6 months ago

Selected Answer: A

In active active failover config, route53 continuously monitors its endpoints and if one of them is unhealthy, it excludes the region/endpoint from its valid traffic route - Only Sensible option

Cloudfront is a content delivery network - not used to route traffic

Transit gateway for traffic routing - aws devs will hit us with a stick on hearing this option

You cant use a load balancer for cross region load balancing - invalid

upvoted 2 times

potomac 2 years, 7 months ago

Selected Answer: A

Global ,Reduce latency, health checks, failover, Route traffic = Amazon Route 53

upvoted 2 times

youdelin 2 years, 8 months ago

"What the?" yeah I know right

upvoted 1 times

jrestrepob 2 years, 9 months ago

Selected Answer: B

"Stateless applications provide one service or function and use content delivery network (CDN), web, or print servers to process these short-term requests.

<https://docs.aws.amazon.com/architecture-diagrams/latest/multi-region-api-gateway-with-cloudfront/multi-region-api-gateway-with-cloudfront.html>

upvoted 1 times

deechean 2 years, 9 months ago

its not static content, actually they deployed a API Gateway backed by lambda

upvoted 3 times

MrAWSAssociate 2 years, 11 months ago

Selected Answer: A

A option does make sense.

upvoted 2 times

Sangsation 2 years, 12 months ago

Selected Answer: B